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Author(s)	Smyth, Coleman Andrew
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Diet and Renal Outcomes

A thesis submitted to the School of Medicine, National University of Ireland, Galway
in fulfilment of the requirements for the degree of Doctor of Philosophy

By

Coleman Andrew Smyth M.B., B.Ch., B.A.O., M. Med. Sc., M.R.C.P.I.

HRB Clinical Research Facility Galway
National University of Ireland Galway

Supervisor: Professor Martin O'Donnell

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Abstract

Chronic Kidney Disease (CKD) is associated with significant morbidity, increased cardiovascular risk and premature mortality and advanced CKD places a significant burden on patients and health systems. There is a need to reduce the global CKD burden; dietary modification may be one approach. In this work, I examine the association between diet and renal outcomes, using multiple methodologies.

First, I completed a systematic review of dietary sodium intake and renal outcomes and report an association between high intake (>4.6g/day) and adverse outcomes, but no difference between low (<2.3g/day) and moderate (2.3-4.6g/day) intakes. Second, I completed post-hoc analyses of two large clinical trials (ONTARGET and TRANSCNED) exploring the association between sodium and potassium intakes and surrogate outcomes (n=28,879). This study showed no overall association between sodium intake and surrogate renal outcomes (although trends towards adverse outcomes with high intake). High potassium intake was associated with reduced odds of outcomes (there was loss of statistical significance with eGFR<45 or macroalbuminuria) but an increased risk of hyperkalemia. Third, I completed the first renal analyses of a large prospective cohort study (n=544,635) exploring the association between diet quality, sodium and potassium intake and hard outcomes (dialysis or death from a renal cause). This study showed higher sodium intake was associated with an increased risk of the outcome but higher potassium intake and diet quality were associated with reduced risks. Fourth, I completed a cross-sectional analysis (n=3,499) that showed CKD was associated with an increased risk of functional impairment (patient-important outcome). Finally, I developed a clinical trial protocol testing the hypothesis that low sodium (as recommended by guidelines) improves renal outcomes. This trial, funded by the Health Research Board, will be one of the longest trials of sodium restriction in patients with CKD.

This thesis provides novel observational evidence on the association between diet and renal outcomes (including two of the largest prospective cohort studies) and identified considerable uncertainty about the benefits of low sodium intake in patients with CKD, which will be explored in an upcoming clinical trial.

Declaration

This thesis is submitted to National University of Ireland, Galway in accordance with the requirements for the degree of Doctor of Philosophy (PhD) in the School of Medicine.

I declare that this thesis is a record of my own work and has not been submitted for any other academic award in this University, or elsewhere. All information sources have been fully acknowledged and referenced.

Parts of this work have appeared in peer-reviewed publications and presentations (Appendix 1).

Coleman Andrew Smyth

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Finally, I would like to thank my parents, my sisters, family and friends for their support and encouragement.

Dedication

I would like to dedicate this PhD thesis to my parents, Joe and Anne Smyth.

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List of Abbreviations

24UNa = 24 hour urine collection for sodium

AARP = American Association of Retired Persons

ACE = Angiotensin converting enzyme inhibitor

ACR = Albumin creatinine ratio

ADL = Activities of daily living

AHA = American Heart Association

AHEI = Alternate Healthy Eating Index

AKI = Acute kidney injury

ARB = Angiotensin II receptor blocker

BADL = Basic activities of daily living

BMI = Body mass index

BNP = Brain natriuretic peptide

BP = Blood pressure

BRA = British Renal Association

CABG = Coronary artery bypass grafting

CARI = Caring for Australasians with renal impairment

CKD = Chronic kidney disease

CKD-EPI = Chronic Kidney Disease Epidemiology Collaboration

CLARITY = Cardiovascular Multimorbidity in primary care study

Co-I = Co-Investigator

CrCl = Creatinine clearance

CRF = Case report form

CSN = Canadian Society of Nephrology

CV = Cardiovascular

CVD = Cardiovascular disease

DASH = Dietary Approaches to Stop Hypertension

DSMB = Data Safety & Monitoring Board

ECG = Electrocardiogram

eCRF = Electronic case report form

EDTNA-ERA = European Dialysis & Transplantation Nurses Association and European Renal Care Association

eGFR = Estimated glomerular filtration rate

ESRD = End-stage renal disease

FDA = Food and Drug Administration

FFQ = Food frequency questionnaire

FSAI = Food Safety Authority of Ireland

GFR = Glomerular filtration rate

GMS = General medical service

GUH = Galway University Hospitals

GWAS = Genome-wide association studies

HEI = Healthy Eating Index

HERS = Heart and Estrogen/Progestin Replacement Study

HR = Dazard ratio (95% confidence interval)

HRB = Health Research Board

HRB-CRFG = Health Research Board Clinical Research Facility Galway

IADL = Instrumental activities of daily living

ICD = International Classification of Diseases

ICGP = Irish College of General Practitioners

ICH-GCP = International Conference of Harmonisation Guidelines for Good Clinical Practice

IDMS = Isotope dilution mass spectrometry

IOM = Institute of Medicine

IQR = interquartile range

KDIGO = Kidney Disease Improving Global Outcomes

KDOQI = Kidney Disease Outcomes Quality Initiative

LVEF = Left ventricular ejection fraction

MDRD = Modification of Diet in Renal Disease

MDS = Mediterranean Diet Score

MUFA = Monounsaturated fat

NDI = National Death Index

NHANES = National Health And Nutrition Examination Survey

NIH = National Institutes of Health

NKCC = Sodium-potassium-chloride transporter

NKF = National Kidney Foundation

NSAID = Non-steroidal anti-inflammatory drugs

ONTARGET = Ongoing telmisartan alone and in combination with ramipril global endpoint trial

OR = Odds ratio (95% confidence interval)

PABA = P-aminobenzoic acid

PCI = Percutaneous coronary intervention

PCR = Protein creatinine ratio

PI = Principal investigator

PICOS = Participants, intervention, comparison, study design

PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PTCA = Percutaneous transluminal coronary angioplasty

PUFA = Polyunsaturated fat

PURE = Prospective Urban Rural Epidemiology study

RAAS = Renin-angiotensin-aldosterone system

RCT = Randomised controlled trials

REC = Research ethics committee

RFS = Recommended Food Score

ROMK = Renal outer medullary potassium channel

RRT = Renal replacement therapy

SBP = Systolic blood pressure

SD = Standard deviation

SFA = Saturated fat

SHR = Sub Hazard Ratio (95% confidence interval)

SSA DMF = Social Security Administration Death Master File

STICK = Sodium Intake in Chronic Kidney Disease

TOHP = Trials of Hypertension Prevention

TONE = Trial Of Non-pharmacologic interventions in the Elderly

TRANSCEND = Telmisartan randomised assessment study in ACE intolerant subjects with cardiovascular disease

UAE = Urinary albumin excretion

USRDS = United States Renal Data System

WestREN = Western Research and Education Network

Chapter 1: Introduction

1.1 Importance of Chronic Kidney Disease

Chronic kidney disease (CKD), characterized by proteinuria and/or reduction in glomerular filtration rate (GFR), affects up to one in eight adults[1] and is associated with increased morbidity and premature mortality [23-6]. CKD is estimated to affect 14.0% of the population, including approximately 300,000 people in Ireland[7]. Over 1.8 million people worldwide receive renal replacement therapy (RRT) for ESRD[8], the final endpoint of chronic kidney disease, including 5,000 people in Ireland[7]. As CKD is often under-recognised and frequently under-treated, the global prevalence of ESRD is rising. In addition, the cost of dialysis is currently prohibitive for many countries and may challenge even the most affluent economies[9].

1.2 Definition of Chronic Kidney Disease

Renal function is most frequently measured by GFR, most frequently estimated from serum creatinine using prediction equations (eGFR). Multiple equations exist, including Cockcroft-Gault[10], Modification of Diet in Renal Disease (MDRD)[11] and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI)[12] equations. Previous studies report an association between GFR and increased risk of death and cardiovascular events[13].

The Cockcroft-Gault equation[10], estimates creatinine clearance (ml/min), using age, weight, gender and serum creatinine. It is readily calculable, but routine reporting on laboratory reports is limited, as patient weight is rarely provided to the laboratory. The four variable MDRD equation[14] reports estimated GFR (eGFR), indexed to body surface area (ml/min/1.73m^2), using age, race, gender and serum creatinine. The original MDRD equation required six variables[11], also including blood urea nitrogen and albumin levels, but is less frequently used. The MDRD formula has been validated in patients with CKD[15,16] but is less accurate when GFR is >60ml/min/1.73m^2. As a result of this issue, the CKD-EPI formula[12] was developed using pooled data from multiple studies, using the same four variables (age, race, gender and serum creatinine) to estimate GFR. The introduction of isotope dilution mass spectrometry (IDMS) calibration for the measurement of serum creatinine requires an additional adjustment factor of 0.94086, as creatinine estimates using this methodology are approximately 6% lower than without this methodology[17].

Chapter 1: Introduction

In 2002, the National Kidney Foundation (NFK) Kidney Disease Outcomes Quality Initiative (KDOQI introduced the first definition of CKD (independent of underlying cause), defined as GFR<60ml/min/1.73m^2. Subsequent guidelines further categorized CKD into five stages (1-5) based on GFR level and proteinuria and added suffixes for transplant (T) and dialysis (D). The most recent Kidney Disease: Improving Global Outcomes (KDIGO) defined CKD as “abnormalities of kidney structure or function, present for > 3months, with implications for health and CKD is classified based on cause, GFR category and albuminuria category”, as well as adding a heat map to express risk of progression (Table 1.1).

Table 1.1 Chronic Kidney Disease Nomenclature

Risk categories for prognosis of Chronic Kidney Disease. Adapted from KDIGO[18]

				Persistent Albuminuria		
				A1	**A2**	**A3**
				Normal to mildly ↑	Moderately ↑	Severely ↑
				<30mg/g <3mg/mmol	30-300mg/g 3-30mg/mmol	>300mg/g >30mg/mmol
GFR Categories (ml/min/1.73m^2)	**G1**	Normal or ↑	≥90	Low Risk	Moderate Risk	High Risk
	G2	Mildly ↓	60-89	Low Risk	Moderate Risk	High Risk
	G3a	Mildly to moderately ↓	45-59	Moderate Risk	High Risk	Very High Risk
	G3b	Moderately to severely ↓	30-44	High Risk	Very High Risk	Very High Risk
	G4	Severely ↓	15-29	Very High Risk	Very High Risk	Very High Risk
	G5	Kidney failure	<15	Very High Risk	Very High Risk	Very High Risk

1.3 Background

1.3.1 Sodium

The 2012 KDIGO guideline recommends that patients with CKD consume <2 g of sodium per day[18]. There is no recommended lower limit of intake as the relationship with renal outcomes is assumed to be linear. Randomized controlled trials report modest reductions in blood pressure with reduced sodium intake[19-22], also shown in meta-analyses of observational studies[23,24]. At population level, even modest mean reductions in blood pressure (BP) may have significant effects on CKD progression, in addition to effects on clinical and subclinical cardiovascular disease, which may impact an individual's functional status. However, for cardiovascular outcomes, recent evidence suggests a J-shaped relationship between sodium intake and cardiovascular outcomes[25,26], with an increased risk at the low levels recommended by current guidelines[27]. Although reverse causation may partly explain the increase in risk with low sodium intake, it is also known to activate the renin-angiotensin-aldosterone system (RAAS)[28,29], which is associated with adverse cardiovascular and renal effects[30-32], and may offset the benefit of reductions in blood pressure. Therefore, there is conflicting mechanistic evidence as low sodium intake may reduce BP (protective) but activate RAAS (harmful) and there is a need to explore the association between sodium intake and renal outcomes.

1.3.2 Potassium

For potassium intake, KDOQI recommends >4 g per day for patients with GFR>60ml/min/1.73m^2 (stages 1 and 2 CKD) and <2.4 g per day for those with GFR<60ml/min/1.73m^2 (stages 3 and 4 CKD)[33]. In contrast, KDIGO suggests potassium intake <3.0g/day in patients with hyperkalemia[18]. There is a limited evidence base supporting these guidelines, as they are based on the knowledge that those with lower GFR may have difficulties excreting potassium, which may result in hyperkalemia. A cross-sectional study reported higher potassium intake was associated with lower odds of CKD. Three small, short-term clinical trials of higher potassium reported that potassium supplementation was associated with stable or improved serum creatinine [34-36]. However, as higher potassium intake may increase the risks of hyperkalemia and complications such as cardiac arrhythmia, further evidence is needed to inform these guidelines. Importantly, as higher potassium intake is associated with reduced blood pressure and reduced cardiovascular disease, it is possible that excessive potassium restriction in patients without hyperkalemia may be harmful, by denying these patients potential cardiovascular benefits.

1.3.3 Diet Quality

While dietary modifications are recommended for patients with cardiovascular disease and a mainstay of management for advancing CKD and ESRD, few studies have explored the association between diet quality and renal outcomes in the general population. Although it is assumed that observations of the association between diet and cardiovascular disease (CVD) may also hold true for CKD, these assumptions may not be correct. For example, increased potassium intake is recommended to reduce the risk of CVD, but potassium restriction is often recommended to patients with advancing CKD or ESRD (due to the adverse effects of hyperkalemia [37,38]). Moreover, for some specific dietary recommendations (e.g. salt intake), the evidence base for current guidelines is insufficient[39].

1.4 Measurement Issues

Despite multiple guidelines recommending restrictions of dietary sodium intake and/or increases in dietary potassium intake, there is little agreement about the best method to measure adherence to dietary guideline recommendations. Urinary excretion is considered to be the gold standard, as 24-hour urine sodium excretion accounts for 95-98% of dietary intake, but within-person variability in the proportion excreted in urine may be as high as 30%[40]. In addition, 24-hour urine specimens are prone to incomplete collections (which may be verified by administering p-aminobenzoic acid (PABA) to the participant during the 24-hour collection period, as it is almost entirely excreted in 24hours[41]), heavy perspiration and gastrointestinal disturbances, which may result in underestimation and effect the agreement between urinary measures and dietary intake. Dietary measures, including food diary, dietary recall and food frequency questionnaire are easier and quicker to complete but may be prone to recall bias, desirability bias, inaccuracies and omissions (such as added salt at the table or during cooking). As free living people do not consume nutrients in isolation and the presence of many nutrient-nutrient interactions, there is an increasing focus on overall assessments of diet quality, which are associated with reduced risks of chronic disease[42]. Self-reported methods that collect data on food intake are most often used[43,44]. The most economical and practical approach to measurement of nutrition in large studies is the food frequency questionnaire (FFQ)[45], where respondents report their usual frequency of intake of each food from a list of foods (with standardized serving sizes) for a specific period of time. Although the FFQ may not collect as much detail on cooking methods as dietary recalls or food records, the FFQ can give a better overall assessment of usual dietary intake[45]. The approach is well validated in multiple studies, and correlations between FFQ assessments and those from 24-hour dietary recalls are in the range of 0.40-0.74 for most foods and nutrients[46]. Although potentially open to recall bias, the FFQ is usually considered to be the method of choice for measurement of diet in large-scale epidemiologic studies.

1.5 Choice of Outcome Measures

In many areas of medicine, there has been a shift towards the use of surrogate end points (such as changes in laboratory measurements) to reduce the cost and duration of prospective clinical studies. However, in order for a surrogate outcome to be a clinically effective substitute for the patient-important, hard, clinical endpoint, the surrogate must reliably predict the clinical outcome of interest and capture the effect of the exposure or treatment on the outcome[47]. Common problems with surrogate outcomes include that (i) the surrogate outcome is not along the causal pathway of the clinical entity under evaluation, (ii) the intervention does not affect the causal pathway, (iii) the surrogate is insensitive to the effect of the intervention and (iv) the intervention's mechanism of action is independent of the disease process[48]. Even in the ideal setting, the use of a surrogate outcome can yield misleading results. This is particularly challenging in nephrology as surrogate outcomes such as serum creatinine, eGFR and measures of proteinuria are easily available and clinical endpoints (such as death from renal disease and a diagnosis of end-stage kidney disease [including the need for dialysis or renal transplantation]) are relatively rare, as the majority of patients with CKD will die from a cardiovascular event before reaching these endpoints[49], highlighting the importance of considering competing risk methodologies[50].

Preservation of functional independence is a key determinant of 'successful ageing' and in subjective surveys of older adults, functional independence is reported to be more important than absence of disease[51]. Subclinical cardiovascular and cerebrovascular disease, especially covert stroke, has been associated with functional decline, with loss of independence and the ability to perform routine activities of daily living[52]. In addition, functional outcomes are infrequently included in previous clinical studies[53]. While the association between ESRD and functional impairment has been established[54], the relationship between earlier stages of CKD and functional impairment has been understudied in community-dwelling adults. Such information would inform the epidemiology of CKD and the conduct of future clinical research in CKD.

1.6 Overall Objective

Given the burden of CKD, and its covert presentation, population-based interventions targeting modifiable risk factors for CKD are needed. As hypertension, diabetes and cardiovascular disease are major risk factors for CKD, their prevention and treatment are key targets. The modification of diet quality, sodium and potassium intakes may alter the course of kidney disease directly, or through effects on blood pressure, diabetes, metabolic syndrome or cardiovascular disease. In this PhD thesis, I evaluate the associations between diet and renal outcomes, which will expand the existing evidence-base used to inform guidelines that target CKD progression and prevention.

1.7 Specific Objectives

First, a systematic review will be performed to explore the association between sodium intake and measures of renal function (creatinine, GFR, proteinuria, need for dialysis or transplantation) in people with and without CKD.

Second, to build on the existing evidence base, exploration of the associations between dietary sodium and potassium intakes and renal outcomes will be performed using observational data. Two large prospective cohort studies will be used to test the hypothesis that low sodium intake is associated with improved renal outcomes (both surrogate and clinical outcomes) and also to explore the associations with potassium intake. The first cohort study, of patients at high cardiovascular risk, focuses on surrogate outcomes (changes in eGFR, proteinuria and hyperkalemia); the second cohort study, of older adults living in the United States of America, focuses on patient-important clinical outcomes (death due to a renal cause or dialysis).

Third, as outcomes considered important by patients include functional status, an exploration of the association between CKD and functional impairment will be carried out in a cross-sectional study of community-dwelling older adults.

Finally, a protocol for a randomized controlled trial (RCT), based on the results of the previous objectives, will be developed to explore the effect of lowering sodium intake on renal outcomes in patients with CKD, including functional status as an outcome measure. The methodologies developed for this RCT are envisaged to establish an infrastructure for future lifestyle intervention studies in patients with CKD.

Chapter 2: Sodium Intake and Renal Outcomes: A Systematic Review

Smyth A, O'Donnell MJ, Yusuf S, Clase CM, Teo KK, Canavan M, Reddan DN, Mann JFE. Sodium Intake and Renal Outcomes: A Systematic Review. American Journal of Hypertension 2014; 27(10): 1277-84. Reproduced with Permission (Appendix 2)

2.1 Introduction

Given the burden of CKD, and its covert presentation, population-based interventions targeting modifiable risk factors for CKD are needed. There is a complex relationship between sodium intake, blood pressure, cardiovascular disease and kidney disease. CKD shares risk factors with cardiovascular disease (CVD), and CKD may potentiate the effects of cardiovascular risk factors, such as high sodium intake (Figure 2.1). In addition, CVD is the leading cause of death in patients with CKD and patients with CKD are more likely to die from CVD than require RRT for ESRD[55,56]. High sodium intake is associated with increased blood pressure and randomized controlled trials have reported a modest reduction in blood pressure with reduction of sodium intake from moderate to moderate-low levels[19,20]. Based on this evidence linking sodium intake with blood pressure, but without direct evidence linking sodium intake to CKD, guidelines recommend sodium restriction to prevent the onset and progression of CKD.

Most current guidelines, for both the general population and specific populations such as those with CKD or ESRD, recommend a daily sodium intake of <2.3g (<100mmol of sodium, <6g of salt) per day[57-62] and some recommend even lower intake (<1.5g/day)[63] (Figure 2.2). However, these guidelines are mostly extrapolated from studies of populations without CKD, where outcome measures were blood pressure or CV events, or, are based on small, short-term studies that include surrogate measures of renal function[64]. Moreover, recent evidence has suggested that the association between sodium intake and CVD may be J-shaped[26], with an increased risk at the low levels recommended by current guidelines[27]. A recent report from the Institute of Medicine found insufficient evidence to recommend intakes of 1.5-2.3g/day in populations with kidney disease[39].

Given these considerations, a systematic review of cohort studies and clinical trials was undertaken to determine the independent association between sodium intake and renal function (creatinine creatinine-based measures, need for dialysis or transplantation and proteinuria) in populations with and without CKD.

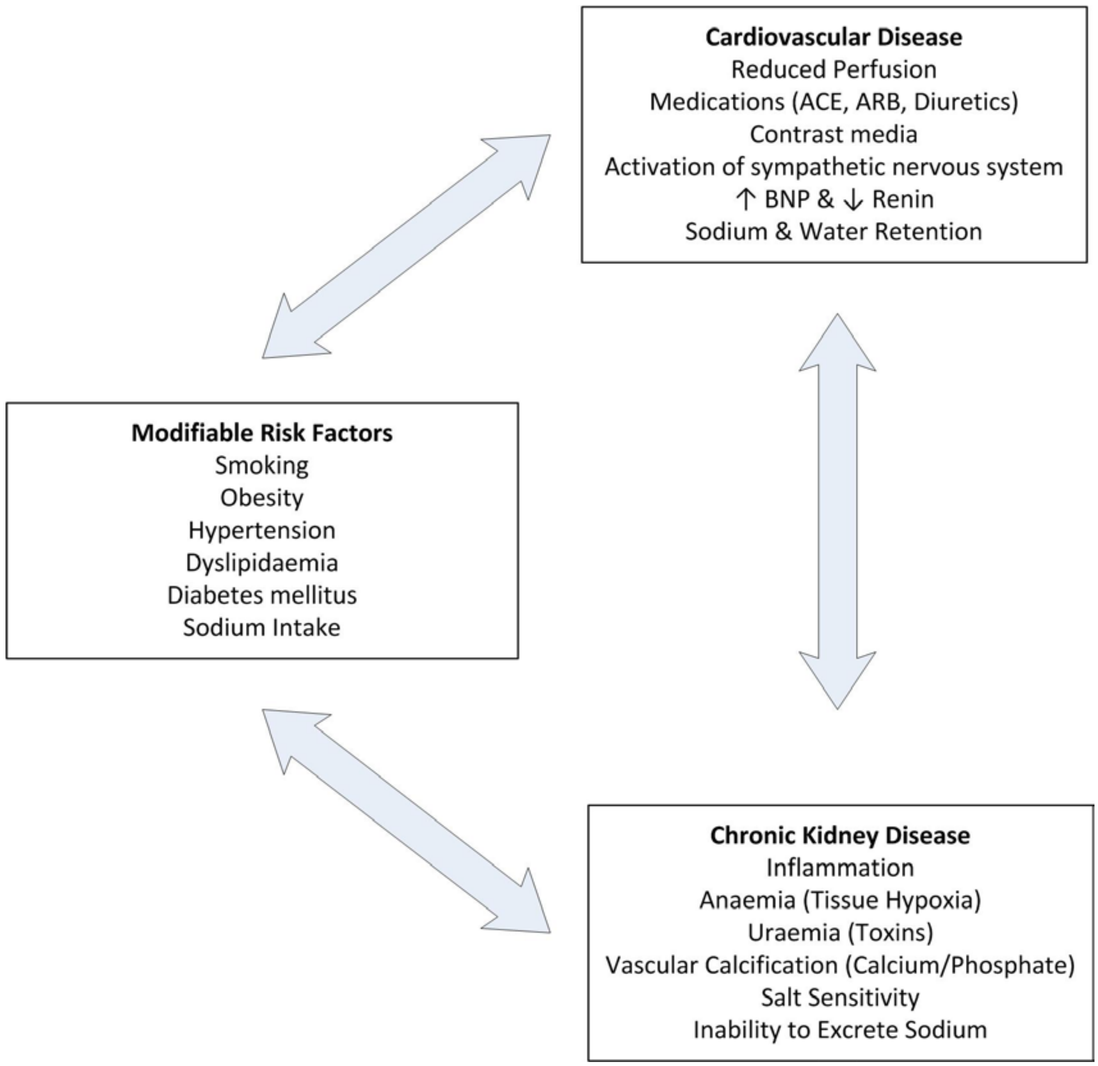

Figure 2.1 Relationship between Cardiovascular Disease and Chronic Kidney Disease

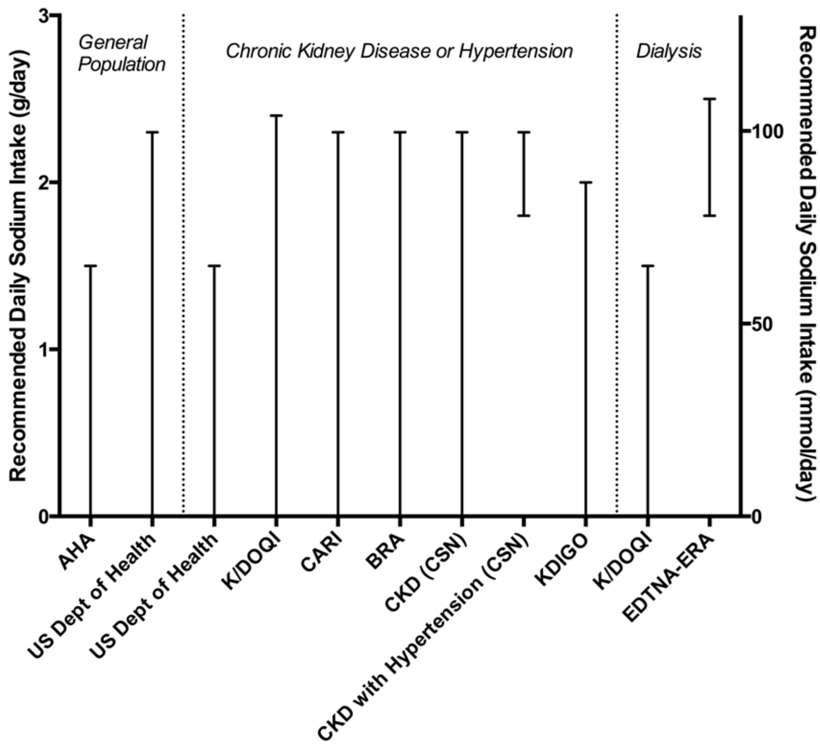

Figure 2.2 Guidelines on Dietary Sodium Intake

Recommend range of intake plotted; guidelines that do not include a recommended lower limit plotted as recommended range (e.g. <2.3g/day plotted as 0-2.3g/day)

X-axis=Target Population (Organization Issuing Guideline)

Legend: AHA=American Heart Association (2012)[63]; US Dept. of Health=United States Department of Health and Human Services (2010)[65]; KDOQI=National Kidney Foundation Disease Outcomes Quality Initiative (2004)[57]; CARI=Caring for Australasians with Renal Impairment (2005)[59]; BRA=British Renal Association (2011)[66]; CSN=Canadian Society of Nephrology (2008)[62]; KDIGO=Kidney Disease Improving Global Outcomes (2012)[61]; EDTNA-ERA=European Dialysis & Transplantation Nurses Association / European Renal Care Association (2002)[58]

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2.2 Methodology

2.2.1 Search Strategy

Searches of electronic databases including the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE (Ovid), EMBASE, CINAHL, PsycINFO, Health Technology Assessment (HTA), PubMed and Abstracts of Reviews of Effects (DARE) from January 1966 to June 2012 were performed by two investigators (AS, MO'D) (Table 2.1).

2.2.2 Eligibility Criteria

Cohort studies (retrospective and prospective) and clinical trials that evaluated the association between sodium ingestion or excretion and renal outcomes were included. The outcome measures included creatinine-based measures of renal function (including eGFR, serum creatinine, creatinine clearance, doubling of creatinine, diagnosis of ESRD and/or the need for dialysis) and proteinuria (including non-specific proteinuria, 24-hour urinary albumin excretion (UAE), 24-hour urinary protein excretion, urine albumin to creatinine ratio (ACR) and urine protein to creatinine ratio (PCR)). Studies that reported quantified measures of sodium intake/excretion (including urinary estimates of sodium, dietary recall and food-frequency questionnaire) and at least one of the aforementioned renal outcomes were eligible. All studies were required to include 50 or more participants (as studies with small sample sizes are likely to be underpowered), and to have a minimum total follow-up of three months (to focus on the medium to long-term effects of sodium intake/excretion on measures of renal function).

Cross-sectional studies were excluded as these studies are the most likely to be affected by confounding by indication, as were studies that included only patients on dialysis or with a renal transplant at the start of the study period, as they are considered a separate population, since the primary outcomes of interest in all subjects were present at baseline. Studies that evaluated a multicomponent dietary or lifestyle intervention were not eligible, unless the independent effect of sodium intake/excretion was reported.

Table 2.1 Full Electronic Search Strategy for Medline (OVID)

Filters applied:

- Human only studies
- Exclude case reports
- Exclude case series
- Exclude editorials
- Exclude abstracts not written in English

No.	Search Term	Results
1	Salt.mp.	267087
2	Sodium.mp.	866236
3	Diet.mp.	724153
4	1 or 2	1034340
5	3 and 4	89236
6	Exp chronic kidney disease	81268
7	Exp glomerular filtration rate	32699
8	Exp creatinine	48325
9	Exp renal failure	128715
10	Exp dialysis	32458
11	Exp albuminuria	12188
12	Exp proteinuria	33152
13	6 or 7 or 8 or 9 or 10 or 11 or 12	239675
14	5 and 13	2113

2.2.3 Data Collection

Two investigators (AS, MOD) independently reviewed full-text articles meeting eligibility criteria and disagreements regarding eligibility were resolved by consensus. Data on study type and design, population characteristics, sodium estimation (method of assessment, actual measurement and categories of intake [low, moderate and high]) and outcome variables were extracted by AS and confirmed by two other reviewers (MOD, CC). Low sodium intake was defined as that recommended by guidelines (<2.3g/day), moderate intake as 2.3-4.6g/day and high intake as twice the upper limit recommended by guidelines (>4.6g/day). A positive association was defined as worsening renal outcomes (i.e. a worsening of renal function or an increase in proteinuria) with increasing sodium intake/excretion and an inverse association was defined conversely. Studies were classified as being of populations with CKD if the population was defined as CKD in the primary paper or if the mean eGFR was <60ml/min/1.73m^2, consistent with the NKF definition of CKD. As renal outcomes differed between studies (e.g. change in creatinine, GFR, creatinine clearance, etc), we report the renal outcomes as defined in the study manuscripts.

2.2.4 Risk of Bias Assessment

Using standardized risk of bias tools, including the Newcastle-Ottawa Scale[67] and the Cochrane Collaboration's Tool for Assessing Risk of Bias[68], two investigators (AS, MC) reviewed all eligible studies.

2.2.5 Statistical Analysis

Due to known heterogeneity in the methods of sodium estimation and the number of different renal outcomes, a decision was made *a priori* not to perform meta-analysis or generate quantitative summary estimates. Instead a qualitative summary in tabular format was planned, with studies divided into those of patients with and without chronic kidney disease. When studies reported multiple adjusted effects, data from the most adjusted model was recorded.

2.3 Results

2.3.1 Search Results

Of 4,337 citations, seven studies (n=8,129) fulfilled the eligibility criteria (Figure 2.3) including one retrospective cohort study (n=57)[69], five prospective cohort studies (n=7,885)[25,70-73] and one clinical trial (n=187)[74]. Four studies (n=1,787) included only participants with CKD[69-71,73]; three studies (n=6,342) included only participants without CKD[25,72,74]. Sodium intake was estimated by 24-hour urinary sodium collection in six studies[25,69-71,73,74] and FFQ in one study[72].

All seven studies reported outcomes for creatinine-based measures of renal function[25,69-74] and six studies reported outcomes for proteinuria [25,69,70,72-74]. Table 2.2 describes the population and method of sodium estimation included in each study and Table 2.3 summarizes the methodological characteristics of included studies.

2.3.2 Risk of Bias

Overall the risk of bias was low in the included observational studies (Table 2.4), but three of the six cohort studies reported only unadjusted estimates for renal outcomes [69,70,73], where the apparent associations are more likely to be affected by confounding. Similarly, the risk of bias was low in the included clinical trial (Table 2.5). Due to differing study designs and outcomes it was not possible to assess for publication bias using a funnel plot.

2.3.3 Studies of Populations with Chronic Kidney Disease

Four studies (n=1,787) reported the association between sodium intake and renal function[69-71,73] and four studies (n=1,787) reported the association between sodium intake and proteinuria[69-71,73] in patients with CKD at baseline (Table 2.6).

(i) Retrospective Cohort Studies: One study (n=57) reported that high sodium intake (>4.6g/day) was associated with a greater decline in creatinine clearance and a greater increase in proteinuria, compared to low intake (<2.3g/day) over a mean follow-up of 3 years. Patients with moderate intake (2.3-4.6g/day) were excluded[69].

(ii) Prospective Cohort Studies: One study (n=500) reported that high sodium intake (>4.6g/day) was associated with the highest risk of ESRD and a greater increase in urinary PCR, compared to moderate sodium intake (2.3-4.6g/day) and low sodium intake (<2.3g/day) over a mean follow-up of 26.2 months. There was no significant difference between moderate and low sodium intake for risk of ESRD[71]. A second study (n=53), that did not include a group with low sodium intake (overall mean intake 2.8±0.7g/day), reported that higher sodium intake was associated with an increased rate of decline in renal function and increased proteinuria over one year follow-up[70]. A third study (n=1,177) reported that the lowest third of sodium intake (mean 3.4g/day, in the moderate range of intake) was associated with the lowest risk of doubling of serum creatinine or ESRD alone, in ARB-treated subjects only, and the greatest reduction in 24-hour ACR over 30 months follow-up[73].

(iii) Clinical Trials: No eligible study was identified.

2.3.4 Studies of Populations without Chronic Kidney Disease

Three studies (n=6,342) reported the association between sodium intake and renal function[25,72,74] and two studies (n=3,535) reported the association between sodium intake and proteinuria[72,74] in patients without CKD at baseline (Table 2.7).

(i) Retrospective Cohort Studies: No eligible study was identified.

(ii) Prospective Cohort Studies: One study (n=3,348) reported that moderate intake (median 2.4g/day, range 2.3-4.9g/day) was associated with an increased risk of decline in GFR, but no change in proteinuria, compared to low intake (median 1.7g/day, range 1.1-1.7g/day) [72]. A second study (n=2,807) reported that low sodium intake (<2.3g) was associated with the highest cumulative incidence of ESRD, and did not report on proteinuria[25].

(iii) Clinical Trials: A cross-over trial (n=187) including patients with mild hypertension, who were randomized to low sodium intake (2.5g/day) or moderate sodium intake (3.8g/day for six weeks each), reported lower serum creatinine (82.3±14.7µmol/L vs. 83.8±15.0µmol/L, p=0.013), but higher UAE (median 10.2mg/24hours vs. 0.9mg/24hours, p<0.001) and ACR (median 0.81mg/mmol vs. 0.66mg/mmol, p<0.001) following randomization to moderate or low intake at six weeks follow-up of each regimen[74].

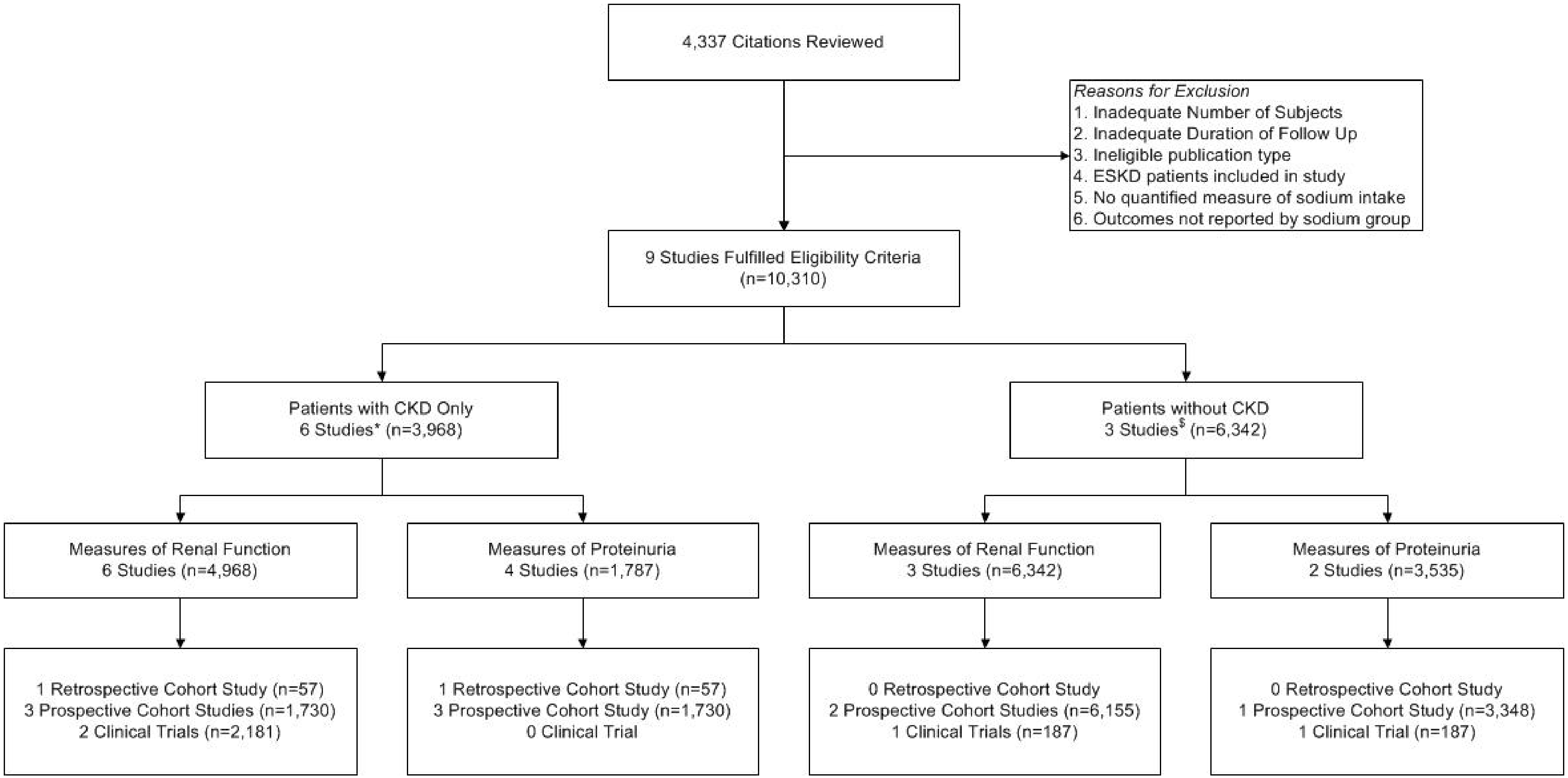


Figure 2.3 Results from Electronic Database Searches

*4 Studies in Patients with Chronic Kidney Disease provided data on measures of renal function and proteinuria; [$]2 Studies in Patients without Chronic Kidney Disease provided data on measures of renal function and proteinuria. Reproduced with permission (Appendix 2)

Table 2.2 Details of included studies

*=Median; µ=mean; NA=not available or reported in primary paper; [$]Defined as eGFR<60ml/min/1.73m^2, estimated using MDRD eGFR from serum creatinine measurements reported in primary paper; [~]All patients had congestive heart failure; ^Mean of five 24hour urine collections; Reproduced with permission (Appendix 2)

	Cianciaruso 1998[69]	**Lin 2010**[72]	**Amaha 2010**[70]	**Thomas 2011**[25]	**Vegter 2012**[71]	**Lambers Heerspink 2012**[73]	**He 2009**[74]
Population	57	3,348	53	2,807	500	1,177	187
Study Design	Cohort	Cohort	Cohort	Cohort	Cohort	Cohort	Trial
Country	Italy	USA	Japan	Finland	Italy	Multiple	UK
Age (years)	µ48-52	µ67	µ64±10	µ38-39	µ52-56	30-70	µ50
Hypertension (%)	22.8%	54.4%	N/A	44.5-53.6%	NA	NA	100%
Proteinuria (%)	100%	6.1%	NA	35%	100%	100%	0%
Reduced GFR (%)	100%	NA	100%	NA	100%	100%	0%
Cardiovascular Disease (%)	NA	6.2%	NA	6.4-9.2%	NA	NA	0%
Diabetes (%)	12.3%	23.7%	0%	100%	0%	100%	0%
ACE/ARB Therapy (%)	NA	15.6%	NA	24-28.9%	100%	42.4%	0%
Diuretic Therapy (%)	NA	NA	NA	8.1-10.0%	NA	61%	0%
Measure of Sodium	Six 24UNa	Five FFQ	Two 24UNa	One 24UNa	Five^ 24UNa	Five^ 24UNa	Six 24UNa
Sodium Intake (g/day)	µ3.7	1.7-2.4*	µ2.8	µ3.5	µ2.8-5.6	µ4.2	µ3.0
Length of Follow-Up (years)	µ3.6	10	1	10*	µ2.2	µ2.5	0.3
Completed Follow-Up (%)	100%	100%	100%	100%	100%	100%	100%

Table 2.3 Detailed methodology of included studies

Reproduced with permission (Appendix 2)

Author, Year	Population
Retrospective Cohort Studies	
Cianciaruso, 1998[69]	Retrospective analyses of patients with CKD (CrCl 10-40ml/min) categorised as (i) Low sodium (<100mEq/day, <2.3g/day) and (ii) High sodium (>200mEq/day, >4.6g/day)
Prospective Cohort Studies	
Lin, 2010[72]	Two sub studies of the Nurses Health Study on renal function and (i) Analgesic use and (ii) Type 2 diabetes
Amaha, 2010[70]	Prospective cohort study of patients with non-diabetic CKD
Thomas, 2011[25]	Subjects with type 1 diabetes but without ESRD
Vegter, 2011[71]	Post-hoc analysis of Ramipril Efficacy in Nephropathy (REIN) trials categorised as (i) Low sodium (<2.9g/day), (ii) Moderate sodium (2.9-5.8g/day) and (iii) High sodium (>5.8g/day)
Lambers Heerspink, 2012[73]	Post-hoc analysis of Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) and Irbesartan Diabetic Nephropathy Trial (IDNT) categorised as (i) Lowest third (µ3.5±1.8g), (ii) Middle third (µ4.1±1.9g) and (iii) Highest third (µ4.8±2.1g)
Clinical Trials	
He, 2009[74]	Randomized, double-blind, crossover trial of two weeks of reduced sodium diet (2g/day) followed by stratified randomization to: (i) Sodium supplementation (2.1g/day) or (ii) Placebo for six weeks

Table 2.4 Risk of Bias in Included Observational Studies

Based on Newcastle-Ottawa Scale for Cohort Studies[67] where ★ is desirable; Reproduced with permission (Appendix 2)

	Cianciaruso, 1998[69]	**Lin, 2010**[72]	**Amaha, 2010**[70]	**Thomas, 2011**[25]	**Vegter, 2012**[71]	**Lambers Heerspink, 2012**[73]
Selection						
Representativeness of exposed cohort	★		★	★	★	★
Selection of Non-Exposed Cohort	★	★	★	★	★	★
Ascertainment of Exposure	★	★	★	★	★	★
Outcome of Interest not Present at Start of Study	★	★	★	★	★	★
Comparability						
Study controls for other factors		★		★	★	
Outcome						
Outcome Assessment	★	★	★	★	★	★
Follow-Up Long Enough	★	★	★	★	★	★
Adequacy of Follow-Up	★	★	★	★	★	★

Table 2.5 Risk of Bias in Included Clinical Trial

Cochrane Collaboration's tool for assessing risk of bias[68] (low, uncertain or high risk); Reproduced with permission (Appendix 2)

	He, 2009[74]
Random sequence generation (selection bias)	Low
Allocation concealment (selection bias)	Low
Blinding of participants and personnel (performance bias)	Low
Blinding of outcome assessment (detection bias)	Low
Incomplete outcome data (attrition bias)	Low
Selective reporting (reporting bias)	Low

Table 2.6 Characteristics and Findings of Studies of Patients with Chronic Kidney Disease

µ=mean; *Positive association defined as lower renal function with higher sodium; [$]Inverse association defined as higher renal function with higher sodium; [~]Positive association defined as higher proteinuria with higher dietary sodium; Reproduced with permission (Appendix 2)

Author, Year	Association	Summary of Findings
Measures of Renal Function		
Retrospective Cohort Studies		
Cianciaruso, 1998[69]	Positive*	Unadjusted analyses reported a greater decline in creatinine clearance in the high sodium group (>4.6g/day, µ5.3g/day) compared to the low sodium group (<2.3g/day, µ1.9g/day).
Effect Size: Mean decline in creatinine clearance 0.51±0.09ml/min in high sodium compared to 0.25±0.07ml/min in the low sodium group (p<0.05).		
Prospective Cohort Studies		
Amaha, 2010[70]	Positive*	Unadjusted analyses reported a faster rate of decline in renal function with increasing sodium (2-5g/day).
Effect Size: Correlation between the rate of renal function decline (i.e. slope of decline in renal function measured as dl/mg/month) and dietary salt intake r=-0.287 (p=0.037).		
Vegter, 2011[71]	Positive*	Multivariable analysis reported a increase in risk of ESRD per 2.3g/day of sodium; unadjusted analyses reported increased risk of ESRD with high sodium vs. low & moderate, but no difference between low and moderate sodium.
Effect Size: HR for progression to ESRD (i) per 2.3g/day HR 1.61 (1.15-2.24); (ii) high sodium (>5.8g/day) versus low (<2.9g/day) HR 3.3 (1.7-6.4); (iii) high sodium versus moderate (2.9-5.8g/day) HR 2.4 (1.4-4.1); (iv) moderate versus low sodium HR 1.4 (0.8-2.4).		
Lambers Heerspink, 2012[73]	Positive*	Unadjusted analyses reported the lowest risk of doubling of creatinine or ESRD in the lowest third of intake (µ3.5g)
Effect Size: HR for doubling of creatinine or ESRD in the lowest third (µ3.5g/day) vs. the highest third (µ4.8g/day) HR 0.75 (0.53-1.05)		
Clinical Trials: No eligible clinical trial was identified.		

Table 2.6 (continued) Characteristics and Findings of Studies of Patients with Chronic Kidney Disease

Author, Year	Association	Summary of Findings
Measures of Proteinuria		
Retrospective Cohort Studies		
Cianciaruso, 1998[69]	Positive~	Unadjusted analyses reported a progression in proteinuria with high sodium (>4.6g/day, µ5.3g/day) but reduction with low sodium (<2.3g/day µ1.9g/day)
Effect Size: Proteinuria increased with high sodium (1.5±0.2g/day to 2.4±0.4g/day (p<0.01)) & decreased with low (2.9±0.3g/day to 1.9±0.3g/day (p<0.005)).		
Prospective Cohort Studies		
Amaha, 2010[70]	Positive~	Unadjusted analyses reported a linear association between urinary protein excretion and sodium intake.
Effect Size: Correlation between urinary protein excretion (g/day) and dietary salt intake (g/day) r=0.528 (p<0.0001).		
Vegter, 2011[71]	Positive~	Unadjusted analyses reported a linear association between increasing urine PCR and increasing sodium excretion.
Effect Size: Urine PCR decreased from baseline by 31% with low sodium (<2.9g/day) (p<0.001), 25% with medium sodium (2.9-5.8g/day) (p<0.001) and 20% with high sodium (>5.8g/day) (p=0.036). There was a significant trend to less proteinuria reduction with increasing intake (p=0.012).		
Lambers Heerspink, 2012[73]	Positive~	Unadjusted analyses reported the greatest decline in 24-hour ACR in the lowest third of sodium intake (µ 3.5g)
Effect Size: 24-hour ACR decline: 44mg/g in the lowest third (µ3.5±1.8g); 16mg/g in the middle third (µ4.1±1.9g); 21mg/g in the highest third (µ 4.8±2.1g).		
Clinical Trials: No eligible clinical trial was identified.		

Table 2.7 Characteristics and Findings of Studies of Patients without Chronic Kidney Disease

µ=mean; *Positive association defined as lower renal function with higher sodium; [$]Inverse association defined as higher renal function with higher sodium' [~]Positive association defined as higher proteinuria with higher sodium; Reproduced with permission (Appendix 2)

Author, Year	Association	Summary of Findings
Measures of Renal Function		
Retrospective Cohort Studies: No eligible study was identified		
Prospective Cohort Studies		
Lin, 2010[72]	Positive*	Multivariable analyses reported a linear association between risk of GFR decline and increasing sodium.
Effect Size: ≥30% decline in GFR over 10 years for high (2.3-4.9g/day) vs. low (1.1-1.7g/day) - OR 1.53 (1.11-2.09).		
Thomas, 2011[25]	Inverse[$]	Multivariable analyses reported a linear association between decreased incidence of ESRD and sodium intake.
Effect Size: Highest sodium associated with lowest ESRD incidence (p<0.001).		
Clinical Trials		
He, 2009[74]	Inverse[$]	Unadjusted analyses reported lower serum creatinine with sodium supplementation (3.8g/d) than placebo (2.5g/d).
Effect Size: Serum creatinine on sodium supplementation was 82.3±14.7µmol/l compared to 83.8±15.0µmol/l while on placebo (p=0.013).		
Measures of Proteinuria		
Retrospective Cohort Studies: No eligible study was identified		
Prospective Cohort Studies		
Lin, 2010[72]	None	Multivariable analyses reported no association between proteinuria and sodium (high vs. low).
Effect Size: OR for onset of microalbuminuria over ten years comparing upper to lower quartile OR 0.94 (0.63-1.41).		

Table 2.7 (continued) Characteristics and Findings of Studies of Patients without Chronic Kidney Disease

Author, Year	Association	Summary of Findings
Clinical Trials		
He, 2009[74]	Positive[~]	Unadjusted analyses reported higher UAE and ACR with sodium supplementation (3.8g/d) than placebo (2.5g/d).
Effect Size: Median 24hr UAE was 10.2mg (IQR 6.8-18.9mg) while on sodium supplementation and 9.1mg (IQR 6.6-14.0mg) while on placebo with an 11% reduction in UAE on switching from sodium supplementation to placebo ($p<0.001$).		

2.4 Discussion

2.4.1 Summary of Findings

In patients with established CKD, consistent associations between high sodium intake (>4.6g/day) and a decline in both creatinine-based measures of renal function (such as GFR) and an increase in proteinuria were observed. However, there was no convincing evidence that low intake (<2.3g/day) was more renoprotective than moderate intake (2.3-4.6g/day), consistent with the findings of the recent report by the Institute of Medicine[39]. In patients without CKD, findings were more inconsistent, with two prospective cohort studies reporting opposing directions of association between sodium intake and renal function[25,72], and one short-term cross-over clinical trial reporting different short-term effects of low sodium intake on creatinine and proteinuria, compared to moderate intake[74].

2.4.2 Evidence to Support Guidelines

Current guidelines recommend that patients with CKD should consume <2.3g/day or <1.5g/day of sodium (Figure 2.2). Hypothetical benefits to low sodium intake include a reduction in the rate of decline in renal function and prevention of CV events, as patients with CKD are at increased CV risk[75]. Evidence included in this review supports the contention that high sodium intake (>4.6g/day) is an important risk factor for decline in renal function in patients with CKD[69-71,73] although the evidence-base is limited as only four studies (n=1,787) that evaluated this association were identified. None of these studies reported more adverse renal outcomes with moderate compared to low intake, meaning that there was no observational evidence to support the <2.3g/day recommended threshold.

No eligible trials in patients with established CKD were identified. A previous systematic review[76], focusing on sodium reduction in patients with diabetes and renal disease, identified a number of clinical trials, but none met the eligibility criteria of this review as all studies reporting renal outcomes were of insufficient sample size (<50 participants)[77-86] and interventions were implemented for short durations in the majority of the trials[77,79-85]. Subsequent to database searches and publication of this review, a double-blind, placebo-controlled, randomized crossover trial of high (median 3.9g/day) versus low (median 1.7g/day) sodium intake in patients with stage 3-4 CKD was published[87]. All patients were counselled to a dietary intake prior to randomization and were randomized to matching placebo or slow-release sodium chloride tablets to raise intake. This trial reported that the low sodium resulted in significant reductions in blood pressure and proteinuria, but intervention was for six weeks only and thus cannot report on long-term renal outcomes. The preliminary results of a Chinese cluster RCT that reduced sodium intake by 0.8g/day was associated with reduced proteinuria over 18 months[88]. Meta-analysis of short studies reported that low sodium intake was associated with increased plasma renin, plasma aldosterone, plasma catecholamines and lipids[89], the effects of which are unknown and may be harmful. Phase 2 of the LowSALT CKD study plans to follow patients for six months, may determine if the counselling to reduce sodium intake was successful in achieving longer term reductions in intake and may provide valuable insights, but is not yet reported[90].

Other trials in patients with CKD were identified but included multi-component interventions, and the independent effect of changes in sodium intake could not be differentiated from other components of the interventions[91-94]. In patients without established CKD, a short-term crossover trial reported that low sodium intake, compared to moderate, was associated with an increase in creatinine but reduction in measures of proteinuria[74]. This finding raises the possibility that sodium intake may have different effects on GFR and proteinuria, which needs to be further determined in longer-term studies. A major finding of this review was the absence of large randomized controlled trials in a population defined by established CKD evaluating the effect of low sodium intake (recommended by current guidelines), compared to moderate intake, on long term renal outcomes. In recognition of the uncertainty, there are ongoing studies in patients with CKD[90,95].

2.4.3 Comparison to Previous Systematic Review

A previous systematic review reported no evidence of a detrimental effect from reduced sodium intake and a link between sodium exposure and kidney tissue injury, and suggested dietary sodium restriction, without a specific guideline on the level of restriction, in patients with CKD[64]. The review presented here, which shows consistent evidence of increased risk with high intake (>4.6g/day) but no evidence of a difference in risk between low (<2.3g/day) and moderate intakes (2.3-4.6g/day), differs from this paper in a number of ways. First, most of the studies (6 of 7) included in this review were published subsequent to the previous review[25,70-74]. Second, the eligibility criteria of this review were stricter, by excluding small studies and confining the review to cohort studies and clinical trials that evaluated the independent association between sodium intake and renal outcomes. Small studies were excluded, as they are more likely to provide spurious results. A crossover clinical trial with a total follow-up, meeting the minimum criteria, was included[74]. Third, only studies with a follow-up of at least three months were included to determine the effect of sodium intake on longer-term change in renal outcomes. Over half of the studies included in the previous review included intervention periods of <1month or were of cross-sectional design. The exclusion of studies with <1year follow-up from this review would mean that no clinical trial was eligible, but the overall results would not have changed significantly.

2.4.4 Sodium and Cardiovascular Disease in Patients with CKD

In general populations, most prospective cohort studies report an association between high sodium intake and hypertension[96,97] and randomized controlled trials reported that low sodium intake reduces blood pressure, compared to moderate or high intake[19,20,98]. A prospective cohort study of patients with CKD (n=217) reported that an increase of 16mmHg in systolic blood pressure was associated with an increased risk of mortality (HR 1.43 (95% CI 1.08-1.90)) and ESRD (HR 3.04 (95% CI 2.13-4.35)), but these estimates were unadjusted and likely affected by confounding as the adjusted estimates were not statistically significant for either mortality (HR 1.13 (95% CI 0.76-1.69)) or ESRD (HR 1.57 (95% CI 0.90-2.75))[99]. Such evidence has been interpreted as representing a compelling case for sodium restriction in patients with CKD, especially in those with hypertension. However, this interpretation assumes that all interventions that lower blood pressure (in a sustained manner) will translate into reduced rate of renal function decline, an assumption that has been challenged by studies published in the last ten years. We know, for example, there are between-class differences in antihypertensive therapies on the rate of disease progression, for a given blood pressure reduction[100]. In particular, medications that inhibit the renin-angiotensin-aldosterone system (RAAS) appear to have renoprotective effects independent of blood pressure lowering[101]. Of note, low sodium intake (<3g/day) has been associated with activation of the RAAS, and increased RAAS activity is associated with a higher incidence of glomerular fibrosis and increased intraglomerular pressure[30]. Therefore, low sodium intake appears to have competing renal effects, i.e. both a reduction in blood pressure, which is expected to be renoprotective, but also an increase in RAAS activity, which is expected to have adverse renal effects.

In addition, salt sensitivity (i.e. some individuals are more sensitive to the hypertensive effects of increased sodium intake) is reported to be more prevalent in patients with CKD due to a reduced ability to excrete sodium, meaning that a given sodium load may additionally increase blood pressure. Patients with CKD may also lose the nocturnal dip in blood pressure[102]. Authors of the LowSALT CKD study drew similar conclusions, although participants in that study had a high use of antihypertensive medications, which may confound their findings[87]. However, salt sensitivity was not defined, measured or reported in any of the studies included in this review. Four studies suggested a trend towards increased blood pressure with increasing sodium intake[25,71,72,74] and two studies adjusted for blood pressure in their analyses[71,72] one of which was performed a population with CKD[71]. In addition, recent genome-wide association studies (GWAS) reported variants in the *UMOD* gene, which encodes uromodulin (Tamm-Horsfall protein) that may modify renal function and hypertension[103-109]. Evidence suggests that this urinary protein, which is secreted by the thick ascending limb of the loop of Henle[110], may regulate the activity of the sodium-potassium-chloride transporter (NKCC) and the renal outer medullary potassium channel (ROMK), which are ion transporters involved in the reabsorption of sodium and chloride[111,112]. However, future studies are needed to investigate this further.

With respect to the prevention of CV events, recent evidence suggests that the association between sodium intake and CV events may be J-shaped, with the lowest CV risk associated with moderate sodium intake[25,26,113,114]. It is unclear if the presence of CKD modifies this association, although a retrospective cohort study of patients with ESRD, not included in our review as it did not meet eligibility criteria, reported an increased all-cause and CV mortality with low sodium intake[27]. The kidney is particularly important in the control of salt and water homeostasis, and patients with CKD may be particularly vulnerable to the effects of extremes of sodium intake, due to a reduced ability to excrete higher sodium loads, which may increase susceptibility to adverse events[115,116]. It is plausible, if not likely, that patients with CKD have different sodium intake requirements to patients without CKD, mandating stand-alone studies in this important population.

2.4.5 Strengths

The main strength of this review is the standardized, guideline-adherent approach to study identification and summarizing data, consistent with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Table 2.8). All items of this internationally accepted guideline document for the reporting of systematic reviews and meta-analyses were considered. Although a structured summary (abstract) is not included in this PhD chapter, it is included in the published manuscript[117]. A decision was made *a priori* to not meta-analyse or generate any summary measures, and instead report studies descriptively, for multiple reasons. First, because a number of measures of renal function are commonly reported (including creatinine, creatinine clearance, estimated glomerular filtration rate, proteinuria and albuminuria) and directional changes are not consistent between measures (e.g. a higher serum creatinine and lower eGFR are in the same direction of association), combining these measures is not appropriate. Second, multiple measures of sodium intake are used, with differing levels of systematic bias. As food frequency questionnaires may underestimate intake and spot urinary measures may overestimate intake, an individual's intake may be classified differently depending on the assessment method used.

2.4.6 Limitations

The main limitation of this review was the low number of studies identified and the small sample sizes included. Only one eligible clinical trial was identified and no trial included a population that was defined by established CKD. Two clinical trials of patients with congestive heart failure, managed with fluid restriction and high-dose diuretics, with high serum creatinine measurements were considered ineligible as sodium intake was assessed by food diary and physician interview (not a quantified assessment[118,119]). In addition, the patients were not considered to be typical of the general population with CKD, despite raised serum creatinine measurements, as all had heart failure and received high-dose diuretics. Both trials reported adverse renal outcomes with low sodium intake (1.8g/day), but these patients may be particularly sensitive to the effects of dietary sodium intake and are not representative of the general population with CKD. Two other similar trials were also identified, but were subsequently excluded after the recent retraction of a meta-analysis of sodium trials in patients with heart failure[120]. Moreover, due to the heterogeneity in study design, population, outcome measure and method of measuring sodium intake, outcome data could not be meta-analyzed. Another challenge in performing this review is the varying methods used to measure renal function. A combination of these factors likely contributes to the inconsistency in the findings.

Multiple methods were used to estimate sodium intake in the included studies. The criterion standard for estimating sodium intake is seven or more 24-hour urine collections conducted over several months[121]. Twenty-four hour urine collections were completed in most studies (6 of 7)[25,69-71,73,74] and only one study performed a single collection[25]. Single measurements of sodium intake are less reliable and susceptible to regression dilution bias due to inter-individual day-to-day variations in intake. However, even with stable day-to-day intake, there are rhythmic changes in sodium excretion and retention, independent of blood pressure, body water and sodium intake, which may affect the reliability of 24hour urine collections, which is especially a problem for individual-level measurement of sodium intake[122]. Measurement bias may influence absolute estimates of sodium intake and affect the strength of association between sodium intake and renal outcomes. Similarly, FFQ, used in one study[72], may be less accurate and may also be prone to recall bias, and therefore also not optimal. Restricting our analysis to studies that used 24-hour urinary measures would not alter our conclusions, but further reduced the already modest number of studies.

Table 2.8 Checklist for Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

Section	#	Checklist Item	Reported In
Title			
Title	1	Identify the report as a systematic review or meta-analysis	Title
Abstract			
Structured summary	2	Provide a structured summary including, as applicable: background, objectives, data sources, study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	See Publication[117]
Introduction			
Rationale	3	Describe rationale in the context of what is already known	Section 2.1
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS)	
Methods			
Protocol & registration	5	Indicate if a review protocol exists, where it can be accessed and, provide registration information/number	Not registered
Eligibility criteria	6	Specify study characteristics (e.g. PICOS, length of follow-up) & report characteristics (e.g. years considered, language, publication status) used as criteria for eligibility	Section 2.2.2
Information sources	7	Describe all information sources (e.g. databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched	Section 2.2.1
Search	8	Present full electronic search strategy for at least one database, including any limits, such that it could be repeated	Table 2.1
Study selection	9	State the process for selecting studies (i.e. screening, eligibility, including in systematic review)	Section 2.2.3
Data collection process	10	Describe method of data extraction from reports and any processes for obtaining and confirming data from investigators	
Data items	11	List and define all variables for which data were sought, any assumptions and simplifications made	
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias (including specification of whether this was done for study or outcome) & how this information is used in data synthesis	Section 2.2.4
Summary measures	13	State the principal summary measures (e.g. risk ratio, difference in means)	None

Table 2.8 (continued) Checklist for Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

Section	**#**	**Checklist Item**	**Reported In**
Synthesis of results	14	Describe the methods of handling data and combining results of studies, including measures of consistency	None
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence	None
Additional analyses	16	Describe methods of additional analyses, if done, indicating which were pre-specified	None
Results			
Study selection	17	Give numbers of studies screened, assessed for eligibility, included, with reasons for exclusions (flow diagram)	Figure 2.3
Study characteristics	18	For each study, present characteristics for which data were extracted and provide the citations	Table 2.2 Table 2.3
Risk of bias within studies	19	Present data on risk of bias of each study an, if available, any outcome level assessment	Table 2.4 Table 2.5
Results of individual studies	20	For all outcomes considered, present for each study: (a) simple summary data for each intervention group, (b) effect estimates and confidence intervals	Table 2.6 Table 2.7
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency	None performed
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies	None performed
Additional analysis	23	Give results of additional analyses, if done	None
Discussion			
Summary of evidence	24	Summarise main findings including the strength of evidence for each outcome; consider their relevance to key groups	Section 2.4.1
Limitations	25	Discuss limitations at study and outcome level (e.g. bias) and at review level (e.g. reporting bias)	Section 2.4.6
Conclusions	26	Provide a general interpretation of the results in context of other evidence & implications for future research	Section 2.4.7
Funding			
Funding	27	Describe sources of funding for the systematic review and other support and role of funders	None

2.4.7 Conclusion

In summary, this review identified a major deficit in the evidence-base to support current guidelines on sodium intake in patients with CKD. While there is reasonable evidence to support an association between high sodium intake and adverse renal outcomes, there is no convincing evidence that low sodium intake is associated with better renal outcomes than moderate intake in patients with and without established CKD. Subsequent to this review, a small (n=20) randomized controlled trial of sodium reduction in hypertensive patients with stage 3-4 CKD reported significant reduction in blood pressure, extracellular fluid volume, albuminuria and proteinuria with sodium restriction, but no long-term renal outcomes were included[87]. Similarly, a cluster randomised controlled trial in China (n=120 villages, n=1,903 participants) reported a reduction in sodium of 0.8g/day which resulted in reduction of albuminuria over 18 month follow-up[88]. Finally, analyses of the MDRD study (n=840) reported no association between 24hour urine sodium and a composite of kidney failure or all-cause mortality[123]. Low sodium diets are difficult for patients, especially renal patients who have other dietary considerations, and have important knock-on implications for other dietary factors. Until further research prospectively compares low versus moderate levels of intake in populations with CKD ideally in RCTs, it would be prudent to recommend moderate levels of salt intake rather than low or very low intakes.

Chapter 3: Sodium and potassium excretion and renal outcomes

Smyth A, Dunkler D, Gao P, Teo KK, Yusuf S, O'Donnell MJ, Mann JFE, Clase CM. Relationship between estimated sodium and potassium excretion and subsequent renal outcomes. Kidney International 2014; 86 (6): 1205-12. Reproduced with Permission (Appendix 3)

3.1 Introduction

Hypertension is a risk factor for chronic kidney disease; prevention and treatment of hypertension are key targets in the prevention and reduction of progression of CKD. Dietary sodium and potassium intake may affect the course of kidney disease directly, or through effects on blood pressure. This study is a post-hoc analysis of the ongoing telmisartan alone and in combination with ramipril global endpoint trial (ONTARGET[124]) and telmistartan randomized assessment study in ACE intolerant subjects with cardiovascular disease (TRANSCEND[125]) studies. We tested the hypothesis that low urinary sodium is associated with improved renal outcomes. As higher potassium has been associated with reductions in blood pressure and cardiovascular disease[126], we examined the association between urinary potassium excretion and renal outcomes.

3.2 Methods

3.2.1 Population

This study includes participants in the ONTARGET[124] (n=25,620) and TRANSCEND[125] (n=5,926) studies with available data on urinary sodium and urinary potassium, excluding those lost to follow up and those with missing or implausible serum creatinine (≤35 umol/L).

These trials enrolled participants aged ≥55years with one of coronary artery disease, peripheral artery disease, cerebrovascular disease or high-risk diabetes mellitus (with evidence of end-organ damage)[127]. Coronary artery disease included previous myocardial infarction, stable angina, unstable angina, multi-vessel percutaneous transluminal coronary angioplasty (PTCA) or multi-vessel coronary artery bypass grafting (CABG). Peripheral artery disease included previous limb bypass surgery or angioplasty, previous limb or foot amputation, intermittent claudication or significant peripheral artery stenosis. Exclusion criteria included inability to discontinue ACE or ARB, known hypersensitivity or intolerance, serum creatinine concentration >265mmol/L, symptomatic congestive heart failure, hemodynamically significant primary valvular or outflow tract obstruction, constrictive pericarditis, complex congenital heart disease, syncopal episodes of unknown etiology, planned PTCA or CABG, uncontrolled hypertension (>160/100mmHg), previous heart transplantation, stroke due to subarachnoid haemorrhage, significant renal artery disease, proteinuria (exclusion criterion for TRANSCEND only), hepatic dysfunction, uncorrected volume or sodium depletion, primary hyperaldosteronism, other major non-cardiac illness expected to reduce life expectancy, simultaneous use of another experimental drug, significant disability, unable or unwilling to provide written informed consent.

Both trials included an active run-in period. For ONTARGET, participants were given ramipril 2.5mg and matching telmisartan 40mg placebo for three days, then ramipril 2.5mg and telmisartan 40mg for seven days followed by ramipril 5mg and telmisartan 40mg for 11-18 days. For TRANSCEND, participants were given telmisartan 80mg placebo for one week followed by telmisartan 80mg daily for 2-3 weeks. For ONTARGET, 29,019 participants were run-in, 3,399 were excluded for reasons of poor compliance (n=1,123), withdrawal from study (n=597), symptoms of hypotension (n=492), hyperkalaemia (n=223), elevated serum creatinine (n=64), death (n=27) and other reasons (n=872). For TRANSCEND, 6,666 participants were run-in, 740 were excluded for reasons of poor compliance (n=311), withdrawal from study (n=135), symptomatic hypotension (n=53), hyperkalaemia (n=26), elevated serum creatinine (n=11), death (n=3) and other reasons (n=201).

After confirming eligibility, 25,620 participants were randomized to one of three arms for ONTARGET (telmisartan alone vs. ramipril alone vs. combination) and 5,926 participants were randomized to one of two arms for TRANSCEND (telmisartan 80mg vs. placebo). For ONTARGET, matching placebo was provided for the other agent (ramipril or telmisartan, as required) and dose titration was required. For the first two weeks participants were given telmisartan 80mg with ramipril 5mg placebo OR ramipril 5mg with telmisartan 80mg placebo OR telmisartan 80 mg and ramipril 5mg. For the remainder of the trial, participants were given telmisartan 80mg with ramipril 10mg placebo OR ramipril 10mg with telmisartan 80mg placebo OR telmisartan 80mg. Follow-up visits were made at six weeks, six months later and then every six months until study closeout. Median follow up was 56 months.

3.2.2 Laboratory Analysis

Serum creatinine and potassium were measured locally before run-in, six weeks after randomization, after two years and at study end and the CKD-EPI formula[12] was used to generate eGFR. Although the accuracy of MDRD and CKD-EPI formulae are reported to be similar[128], the CKD-EPI formula was chosen to estimate GFR in this study as meta-analysis recently reported that CKD-EPI classified fewer individuals as having CKD and more accurately categorized the risk of mortality and ESRD than the MDRD formula[129]. Urine albumin-creatinine ratio (ACR) was measured centrally[130], by a turbidimetric method (Unicel DxC600 Synchron Systems, Beckman Coulter, Bea, CA, USA). The coefficient of variation at 32.2mg/L was 4.4% and at 105.5mg/L was 2.4%. A human serum pool at a concentration of 10.9mg/L gave a coefficient of variation of 9.2% and at 159.8mg/L gave a coefficient of variation of 2.7%. Creatinine in urine was measured centrally by a modified Jaffe method (Unicel DxC600 Synchron Systems). The coefficient of variation at 7.9mg/L was 2.9% and at 23.1mg/L was 2.8%. A human serum pool at a concentration of 10.8mg/L gave a coefficient of variation of 2.6% and at 103 mg/L have a coefficient of variation of 1.8%. Morning fasting urine samples were shipped using STP 250 ambient specimen boxes, and stored at the Hamilton Research Laboratory or a regional laboratory in Beijing, China. Both laboratories measured sodium and potassium identically with indirect potentiometry using the Beckman Coulter Synchron Clinical System. Urinary creatinine was determined with a Roche Hitachi 917 analyzer using an enzymatic colorimetric assay with a sensitivity in urine of 54umol/L, a within-run imprecision of 0.8%, and a between-run imprecision of 2.1% at a urine concentration of 2120umol/L[26].

The Kawasaki formula[131] was used to estimate 24-hour urinary sodium and potassium excretion from a single fasting morning urine sample, taken at the pre-run-in visit. This approach has been used to estimate sodium intake in previous studies of healthy controls[131] and patients taking antihypertensive therapies[132]. Although this formula was developed and validated in Asians, comparison of estimated values with 24-h collections showed Pearson correlations of 0.55 ($p<0.001$) for sodium and 0.43 ($p<0.001$) for potassium[26,133]. These correlation coefficients are higher than those reported in the trial of non-pharmacologic interventions in the elderly (TONE) trial[20], that reported correlations between 24-hour dietary recall and 24-hour urinary measurement of 0.30 for sodium and 0.43 for potassium. In another analysis of ONTARGET and TRANSCEND, urinary measurements were repeated at 2-years in a subset (n=2,625) and showed correlation coefficients between baseline and 2-years of 0.33 for sodium and 0.44 for potassium[26]. A recent analysis of the Prospective Urban Rural Epidemiology (PURE) study compared the Kawasaki[131], Tanaka[134] and INTERSALT[135] formulae for estimating 24hour urinary sodium and potassium excretion and reported the Kawasaki method to be the most valid and least biased method[136]. The Kawasaki formula has also been validated in patients with CKD[137]. Estimates for sodium and potassium excretion were truncated at 10g/day and 5g/day, respectively, as these were hypothesized to be the upper limit of the plausible range of intake.

3.2.3 Outcomes

The primary renal outcome was decline in eGFR of ≥30% or chronic dialysis, chosen as there was a moderate number of dialysis events reported in the ONTARGET and TRANSCEND trials (meaning that it would not be appropriate to analyze this patient-important outcome alone). Although a surrogate outcome, 30% decline in eGFR was chosen based on recent discussions by the NKF and Food and Drug Administration (FDA) about appropriate surrogate renal outcomes for clinical research[138]. Secondary outcomes included: a decline in eGFR of ≥40% or chronic dialysis, rapid progression of renal disease, doubling of serum creatinine or chronic dialysis, progression of proteinuria, and hyperkalemia. Rapid progression of renal disease was defined as a decline of 5%/year in eGFR, with the rate of change defined by regression within participant. Creatinine-based outcomes were defined as change from six weeks after randomization. The six-week post-randomisation eGFR was chosen as the 'baseline' value as it is least likely to be effected by the acute hemodynamic changes associated with the introduction, dose titration and/or withdrawal of ACE or ARB, as occurred during run-in[139]. After the conclusion of the ONTARGET and TRANSCEND trials, a questionnaire was sent to sites that reported dialysis events, and obtained information on duration of dialysis and primary reasons for any acute dialysis events. In three of 162 cases there was missing information on whether dialysis was acute or chronic[130]. Chronic dialysis was defined as dialysis occurring for two months or longer. Progression of proteinuria was defined as progression from normal (ACR<30mg/g) to micro- (ACR≥30 to <300mg/g) or macro-albuminuria (UACR≥300mg/g) or from micro- to macro-albuminuria, based on change from ACR measured at run in. Hyperkalemia was defined as a serum potassium of >5.5mmol/L on laboratory measurement either at two years or study end, and included as a secondary outcome as elevations in serum potassium were a potential safety concern with increased dietary potassium intake.

For laboratory-based outcomes, participants with a 'baseline' value (i.e. six-week laboratory measurement of eGFR or run-in ACR) and either the outcome event, death, or laboratory measurement at the end of follow up were included. For the outcome progression of proteinuria participants with macroalbuminuria at baseline (defined as ACR >300mg/g) were excluded, as these participants could not have the outcome of interest, as they were in the highest category of proteinuria at baseline. For missing values for laboratory outcomes at study end, to maximize power, the two-year value was used. This approach is conservative as it is likely that change in eGFR and ACR are underestimated, meaning that outcome events may have been missed, with these participants classified as not having had the event of interest, potentially biasing our results towards the null.

3.2.4 Statistical Analysis

Continuous data are presented as mean with standard deviation (SD) or median with interquartile range, and categorical data as frequencies and percentages. Baseline differences were compared using χ^2 test and analysis of variance. All p values are two-sided and $p \leq 0.05$ was considered statistically significant, without adjustment for multiple comparisons.

Analyses included multinomial logit regression modelling with the multivariable fractional polynomial algorithm[140] with a significance level of 0.05 for the continuous variables sodium or potassium excretion, age, body mass index (BMI), six-week eGFR and run-in ACR, to allow non-linear effects, and the categorical variables sex, ethnicity, diabetes, smoking, diuretic use (loop, thiazide and thiazide-like diuretics but information on potassium sparing diuretics was not collected) and RAAS blockade to identify odds ratios (OR) and 95% confidence intervals (CI) for the effects of sodium and potassium excretion on each renal outcomes with the competing risk of death. Multinomial logit regression was chosen over proportional sub-distribution hazard models for competing risk data, according to Fine and Gray[141], because the exact dates of change in eGFR, proteinuria or hyperkalemia are not known, and because rapid progression is calculated over the whole study and does not occur at a particular time. Sodium and potassium excretion were also analysed in thirds of excretion, as much of the existing literature reports sodium and potassium as low, moderate and high. Rather than apply cut-points demarking low, moderate and high intake *a priori,* thirds of estimated excretion are reported. Potential confounders or explanatory variables were pre-specified (including age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade) and forced into every model (i.e. no forward selection or backwards elimination approaches to regression analyses were adopted).

A priori it was specified that renal outcomes may be affected or be a consequence of other outcome adverse events, and not be reflective of or predicted by baseline sodium and potassium. To account for these effects, the definition of the competing event was expanded (i.e. by adding other adverse events to the baseline competing risk of death, participants who experienced renal outcomes but also had one of these adverse outcomes were counted in the competing risk group). First, hospitalisation for any reason was included with death as the competing risk event for each outcome. Second, myocardial infarction, stroke or hospitalisation for heart failure was included with death as the competing risk. Finally, only myocardial infarction or hospitalisation for heart failure was included with death as the competing risk, as stroke would be the least likely cardiovascular event to affect renal outcomes.

To assess whether the effects of sodium and potassium differed according to patients' baseline characteristics, two-way interaction terms were added one at a time to the base model, based on the hypothesis that associations may differ by levels of diabetes, RAAS blockade, age, baseline eGFR and baseline ACR. The false discovery rate[140,142] of 5% was used to adjust for multiple testing. The interaction terms were used as the formal test for subgroup effects, but for presentation the effect within subgroups defined categorically was also analysed.

3.3 Results

Of the 31,546 participants in the ONTARGET and TRANSCEND studies, 28,879 (91.5%) were included, with no clinically meaningful differences between included and excluded participants (Table 3.1). The mean age was 66.5 (7.2) years, mean eGFR was 68.4 (17.6) ml/min/1.73m^2, mean sodium excretion was 4.76 (1.55) g/day and the mean potassium excretion was 2.18 (0.54) g/day. Patients with higher estimated sodium excretion had higher eGFR, ACR and systolic blood pressure (Table 3.2). Patients with higher estimated potassium excretion also had higher eGFR and ACR (Table 3.3).

Table 3.1 Comparison of baseline characteristics of included and excluded participants

Microalbuminuria = ACR>30mg/g; macroalbuminuria = ACR>300 mg/g; obesity = BMI≥30 kg/m^2; renin-angiotensin system (RAAS) blockade = assignment to telmisartan, ramipril, or both; diuretics included loop, thiazide or thiazide-like diuretics. §Mean on-study systolic blood pressure = mean of all systolic blood pressures recorded during the study (including after outcomes); reproduced with permission (Appendix 3)

	Overall (n=28,879)	Creatinine & Potassium Outcomes			Proteinuria Outcome		
		Included (n=27,077)	**Excluded (n=1,802)**	**p-value**	**Included (n=25,891)**	**Excluded (n=2,988)**	**p-value**
Sodium, mean (SD), g/d	4.76 (1.55)	4.77 (1.55)	4.59 (1.58)	<0.0001	4.75 (1.53)	4.78 (1.70)	0.3401
Potassium, mean (SD), g/d	2.18 (0.54)	2.18 (0.54)	2.21 (0.57)	0.0128	2.18 (0.54)	2.25 (0.56)	<0.0001
Age, mean (SD), years	66.5 (7.2)	66.5 (7.2)	67.1 (7.4)	0.0002	66.5 (7.2)	66.9 (7.3)	0.0010
Creatinine, mean (SD), mg/dL	1.06 (0.28)	1.06 (0.27)	1.05 (0.30)	0.1052	1.05 (0.26)	1.13 (0.36)	<0.0001
eGFR*, mean (SD), mL/min/1.73m^2	68.4 (17.6)	68.4 (17.5)	68.7 (18.6)	0.5156	68.8 (17.3)	64.0 (19.9)	<0.0001
eGFR categories, n(%)							
≥90 mL/min/1.73m^2	3,584 (12.7%)	3,434 (12.7%)	150 (13.4%)	0.4914	3,277 (12.9%)	307 (11.2%)	0.0156
60-90 mL/min/1.73m^2	15,481 (54.9%)	14,863 (54.9%)	618 (55.1%)	0.8754	14,224 (55.9%)	1,257 (46.0%)	<0.0001
45-60 mL/min/1.73m^2	6,380 (22.6%)	6,148 (22.7%)	231 (20.6%)	0.0992	5,723 (22.5%)	657 (24.1%)	0.0584
30-45 mL/min/1.73m^2	2,363 (8.4%)	2,266 (8.4%)	97 (8.7%)	0.7364	1972 (7.7%)	391 (14.3%)	<0.0001
<30 mL/min/1.73m^2	390 (1.4%)	365 (1.3%)	25 (2.2%)	0.0132	272 (1.1%)	118 (4.3%)	<0.0001
ACR, mean (SD), mg/dL	5.33 (24.21)	5.22 (22.79)	7.06 (39.88)	0.0017	1.95 (4.3)	35.0 (67.8)	<0.0001
ACR categories, n(%)							
Normoalbuminuria	24,197 (83.9%)	22,736 (84.1%)	1,461 (81.3%)	0.0023	22,548 (87.1%)	1,649 (55.0%)	<0.0001
Microalbuminuria	3,622 (12.6%)	3,357 (12.4%)	265 (14.8%)	0.0037	3,343 (12.9%)	279 (9.5%)	<0.0001
Macroalbuminuria	1,020 (3.5%)	950 (3.5%)	70 (3.9%)	0.3928	0 (0.0%)	1,020 (34.6%)	<0.0001
Diabetes, n(%)	10,717 (37.1%)	10,010 (37.0%)	707 (39.5%)	0.0350	9,147 (35.3%)	1,570 (52.7%)	<0.0001
Body mass index, mean (SD)	28.1 (4.6)	28.1 (4.5)	28.4 (4.9)	0.0013	28.1 (4.5)	28.6 (5.0)	<0.0001
Obesity, n(%)	8,537 (29.6%)	7,947 (29.4%)	590 (32.8%)	0.0024	7,543 (29.2%)	993 (33.4%)	<0.0001
Women, n(%)	8,503 (29.4%)	7,835 (28.9%)	668 (37.1%)	<0.0001	7,444 (29.8%)	1,059 (33.4%)	<0.0001
RAAS blockade(study drug), n(%)	26,160 (90.6%)	24,549 (90.7%)	1,611 (89.4%)	0.0755	23,472 (90.7%)	2,688 (90.0%)	0.2166

Table 3.1 (continued) Comparison of baseline characteristics of included and excluded participants

	Overall (n=28,879)	Creatinine & Potassium Outcomes			Proteinuria Outcome		
		Included (n=27,077)	Excluded (n=1,802)	p-value	Included (n=25,891)	Excluded (n=2,988)	p-value
Open label ACEi /AIIA, n(%)	18,715 (64.8%)	17,577 (64.9%)	1,138 (63.2%)	0.1256	16,637 (64.3%)	2,078 (69.6%)	<0.0001
Diuretics, n(%)	8,298 (28.7%)	7,720 (28.5%)	578 (32.1%)	0.0012	7,224 (27.9%)	1,074 (35.9%)	<0.0001
Systolic Blood Pressure							
Baseline, mean (SD), mmHg^	134.50 (18.80)	134.39 (18.71)	136.55 (20.30)	<0.0001	134.00 (18.59)	139.12 (19.99)	<0.0001
Mean on-study[§] >140 mmHg, n(%)	9,890 (34.3%)	9,186 (33.9%)	704 (39.2%)	<0.0001	8,555 (33.0%)	1,335 (44.8%)	<0.0001

Table 3.2 Baseline patient characteristics by estimated sodium excretion

Microalbuminuria = ACR>30mg/g; macroalbuminuria = ACR>300 mg/g; obesity = BMI≥30 kg/m^2; renin-angiotensin system (RAAS) blockade = assignment to telmisartan, ramipril, or both; diuretics included loop, thiazide or thiazide-like diuretics. §Mean on-study systolic blood pressure = mean of all systolic blood pressures recorded during the study (including after outcomes); reproduced with permission (Appendix 3)

	Estimated Sodium Excretion						
	Overall (n=28,879)	**<2g/day (n=818)**	**2-3.99g/day (n=8,353)**	**4-5.99g/day (n=14,155)**	**6-8g/day (n=4,706)**	**>8g/day (n=847)**	**p-value**
Sodium, mean (SD), g/d	4.76 (1.55)	1.55 (0.35)	3.24 (0.53)	4.93 (0.56)	6.71 (0.53)	8.97 (0.70)	<0.0001
Age, mean (SD), years	66.5 (7.2)	67.6 (7.6)	67.0 (7.4)	66.5 (7.2)	65.8 (7.0)	65.4 (6.8)	<0.0001
Creatinine, mean (SD), mg/dL	1.06 (0.28)	1.08 (0.31)	1.07 (0.28)	1.06 (0.27)	1.06 (0.27)	1.06 (0.29)	0.2225
eGFR*, mean (SD), mL/min/1.73m^2	68.4(17.6)	64.6 (18.3)	67.2 (17.8)	68.7 (17.4)	69.8 (17.5)	69.7 (17.9)	<0.0001
eGFR categories, n(%)							
≥90 mL/min/1.73m^2	3,584 (12.7%)	80 (10.0%)	947 (11.6%)	1,757 (12.7%)	676 (14.6%)	124 (15.1%)	<0.0001
60-90 mL/min/1.73m^2	15,481 (54.9%)	388 (48.7%)	4,367 (53.7%)	7,706 (55.7%)	2,562 (55.5%)	458 (55.7%)	0.0002
45-60 mL/min/1.73m^2	6,380 (22.6%)	215 (27.0%)	1,898 (23.3%)	3,106 (22.5%)	998 (21.6%)	163 (19.8%)	0.0014
30-45 mL/min/1.73m^2	2,363 (8.4%)	89 (11.2%)	794 (9.8%)	1,074 (7.8%)	342 (7.4%)	64 (7.8%)	<0.0001
<30 mL/min/1.73m^2	390 (1.4%)	25 (3.1%)	127 (1.6%)	183 (1.3%)	41 (0.9%)	14 (1.7%)	<0.0001
ACR, mean (SD), mg/g	5.33 (24.21)	3.90 (10.83)	3.85 (14.48)	4.72 (20.16)	7.81 (30.06)	17.73 (76.07)	<0.0001
ACR categories, n(%)							
Normoalbuminuria	24,197 (83.9%)	654 (80.5%)	7,123 (85.5%)	12,106 (85.6%)	3,746 (79.6%)	568 (67.3%)	<0.0001
Microalbuminuria	3,622 (12.6%)	139 (17.1%)	974 (11.7%)	1,587 (11.2%)	723 (15.4%)	199 (23.6%)	<0.0001
Macroalbuminuria	1,020 (3.5%)	19 (2.3%)	234 (2.8%)	454 (3.2%)	236 (5.0%)	77 (9.1%)	<0.0001
Diabetes, n (%)	10,717 (37.1%)	320 (39.1%)	2,691 (32.2%)	5,128 (36.2%)	2,141 (45.5%)	437 (51.6%)	<0.0001
Body mass index, mean (SD)	28.1 (4.6)	27.3 (4.6)	27.5 (4.5)	28.1 (4.4)	29.1 (4.7)	30.2 (5.1)	<0.0001
Obesity, n(%)	8,537 (29.6%)	216 (26.4%)	2,080 (24.9%)	4,045 (28.6%)	1,812 (38.7%)	384 (45.7%)	<0.0001
Women, n(%)	8,503 (29.4%)	438 (53.5%)	3,172 (38.0%)	3,763 (26.6%)	952 (20.2%)	178 (21.0%)	<0.0001
RAAS blockade(study drug), n(%)	26,160 (90.6%)	712 (87.0%)	7,575 (90.7%)	12,837 (90.7%)	4,288 (91.1%)	748 (88.3%)	0.0009

Table 3.2 (continued) Baseline patient characteristics by estimated sodium excretion

	Estimated Sodium Excretion						
	Overall (n=28,879)	**<2g/day (n=818)**	**2-3.99g/day (n=8,353)**	**4-5.99g/day (n=14,155)**	**6-8g/day (n=4,706)**	**>8g/day (n=847)**	**p-value**
Open label ACEi /AIIA, n(%)	18,715 (64.8%)	550 (67.2%)	5,456 (65.3%)	9,603 (64.0%)	3,062 (65.1%)	584 (68.9%)	0.0103
Diuretics, n(%)	8,298 (28.7%)	335 (41.0%)	2,568 (30.7%)	3,662 (25.9%)	1,366 (29.0%)	367 (43.3%)	<0.0001
Systolic Blood Pressure							
Week 6, mean (SD), mmHg	134.50 (18.80)	134.74 (19.46)	134.09 (19.06)	134.31 (18.66)	135.19 (18.58)	137.60 (18.64)	<0.0001
Mean on-study[§] >140 mmHg, n(%)	9,890 (34.3%)	287 (35.1%)	2,794 (33.5%)	4,731 (33.4%)	1,722 (36.6%)	356 (42.0%)	<0.0001

Table 3.3 Baseline patient characteristics by estimated potassium excretion

Microalbuminuria = ACR>30mg/g; macroalbuminuria = ACR>300 mg/g; obesity = BMI≥30 kg/m^2; renin-angiotensin system (RAAS) blockade = assignment to telmisartan, ramipril, or both; diuretics included loop, thiazide or thiazide-like diuretics; §Mean on-study systolic blood pressure = mean of all systolic blood pressures recorded during the study (including after outcomes); reproduced with permission (Appendix 3)

	Estimated Potassium Excretion						
	Overall (n=28,879)	**<1.5g/day (n=2,194)**	**1.5-1.99g/day (n=9,710)**	**2.0-2.49g/day (n=9,876)**	**2.5-3.0g/day (n=4,850)**	**>3.0g/day (n=2,249)**	**p-value**
Potassium, mean (SD), g/d	2.18 (0.54)	1.36 (0.12)	1.77 (0.14)	2.24 (0.14)	2.72 (0.14)	3.38 (0.37)	<0.0001
Age, mean (SD), years	66.5 (7.2)	66.9 (7.4)	66.9 (7.3)	66.9 (7.2)	65.8 (7.0)	64.5 (6.6)	<0.0001
Creatinine, mean (SD), mg/dL	1.06 (0.28)	1.07 (0.31)	1.07 (0.28)	1.07 (0.27)	1.05 (0.26)	1.03 (0.25)	<0.0001
eGFR*, mean (SD), mL/min/1.73m^2	68.4 (17.6)	65.2 (18.2)	66.6 (17.7)	68.2 (17.3)	71.0 (17.1)	73.9 (16.9)	<0.0001
eGFR categories, n(%)							
≥90 mL/min/1.73m^2	3,584 (12.7%)	221 (10.3%)	1,043 (11.0%)	1,175 (12.2%)	718 (15.2%)	427 (19.4%)	<0.0001
60-90 mL/min/1.73m^2	15,481 (54.9%)	1,062 (49.4%)	5,006 (52.9%)	5,353 (55.4%)	2,762 (58.5%)	1,298 (59.1%)	<0.0001
45-60 mL/min/1.73m^2	6,380 (22.6%)	576 (26.8%)	2,341 (24.7%)	2,218 (23.0%)	897 (19.0%)	348 (15.8%)	<0.0001
30-45 mL/min/1.73m^2	2,363 (8.4%)	237 (11.0%)	921 (9.7%)	803 (8.3%)	292 (6.2%)	110 (5.0%)	<0.0001
<30 mL/min/1.73m^2	390 (1.4%)	55 (2.6%)	155 (1.6%)	114 (1.2%)	52 (1.1%)	14 (0.6%)	<0.0001
ACR, mean (SD), mg/g	5.33 (24.21)	2.52 (10.53)	4.15 (18.86)	5.53 (21.02)	6.62 (26.55)	9.49 (48.87)	<0.0001
ACR categories, n(%)							
Normoalbuminuria	24,197 (83.9%)	1,969 (89.8%)	8,408 (86.7%)	8,143 (82.6%)	3,928 (81.0%)	1,749 (77.9%)	<0.0001
Microalbuminuria	3,622 (12.6%)	185 (8.4%)	1,004 (10.4%)	1,336 (13.5%)	715 (14.8%)	382 (17.0%)	<0.0001
Macroalbuminuria	1,020 (3.5%)	36 (1.6%)	281 (2.9%)	385 (3.9%)	204 (4.2%)	114 (5.1%)	<0.0001
Diabetes, n(%)	10,717 (37.1%)	696 (31.7%)	3,560 (36.7%)	3,672 (37.2%)	1,860 (38.4%)	929 (41.3%)	<0.0001
Body mass index, mean (SD)	28.1 (4.6)	26.2 (4.4)	27.5 (4.4)	28.3 (4.4)	29.0 (4.6)	29.9 (4.8)	<0.0001
Obesity, n(%)	8,537 (29.6%)	380 (17.3%)	2,411 (24.9%)	2,965 (30.1%)	1,802 (37.3%)	979 (43.8%)	<0.0001
Women, n(%)	8,503 (29.4%)	1,156 (52.7%)	3,645 (37.5%)	2,518 (25.5%)	888 (18.3%)	296 (13.2%)	<0.0001
RAAS blockade(study drug), n(%)	26,160 (90.6%)	1,909 (87.0%)	8,701 (89.6%)	9,013 (91.3%)	4,460 (92.0%)	2,077 (92.4%)	<0.0001
Open label ACEi /AIIA, n(%)	18,715 (64.8%)	1,422 (64.8%)	6,293 (64.8%)	6,354 (64.3%)	3,164 (65.3%)	1,482 (65.9%)	0.6304

Table 3.3 (continued) Baseline patient characteristics by estimated potassium excretion

	Estimated Potassium Excretion						
	Overall (n=28,879)	**<1.5g/day (n=2,194)**	**1.5-1.99g/day (n=9,710)**	**2.0-2.49g/day (n=9,876)**	**2.5-3.0g/day (n=4,850)**	**>3.0g/day (n=2,249)**	**p-value**
Diuretics, n(%)	8,298 (28.7%)	690 (31.4%)	2,944 (30.3%)	2,768 (28.0%)	1,236 (25.5%)	660 (29.3%)	<0.0001
Systolic Blood Pressure							
Baseline, mean (SD), mmHg	134.50 (18.80)	133.93 (19.22)	134.77 (18.74)	134.55 (18.66)	134.27 (19.04)	134.15 (18.70)	0.2438
Mean on-study[§] >140 mmHg, n(%)	9,890 (34.3%)	722 (32.9%)	3,442 (35.5%)	3,383 (34.3%)	1,610 (33.2%)	733 (32.6%)	0.0109

During follow up, mean decline in eGFR was 1.3 (SD 15.6) mL/min/1.73m^2 and median increase in ACR was 0.6 mg/g (IQR -2.1 to 6.4). Outcome data were available for 27,077 participants (93.8%) for creatinine-based outcomes and hyperkalemia, and for 25,891 (89.7%) for progression of proteinuria (Table 3.1). The primary renal outcome occurred in 2,052 (7.6%).

3.3.1 Sodium

There was no association between estimated 24-hour urinary sodium excretion and the primary outcome (Figure 3.1), progression of proteinuria (Figure 3.2) or hyperkalemia (Figure 3.3) on adjusted models. Similarly, there was no association with eGFR decline of ≥40% or chronic dialysis (Figure 3.4), doubling of creatinine or chronic dialysis (Figure 3.5), but the association with rapid progression of renal disease was U shaped (p=0.0334) (Figure 3.6); participants with the lowest and highest sodium excretion were at increased odds of rapid progression of renal disease.

Compared with the third of patients with the lowest sodium excretion (median 3.3 g/day), there was no association between moderate (median 4.7 g/day) or high (median 6.1 g/day) sodium excretion and the primary outcome (odds ratio (OR) and confidence interval (CI) 0.98 (0.92-1.04) and 0.99 (0.89-1.09), respectively) (Table 3.4). There was no association between sodium excretion and secondary outcomes (eGFR decline of ≥40% or chronic dialysis, rapid progression of renal disease, doubling of creatinine or chronic dialysis, progression of proteinuria, and hyperkalemia).

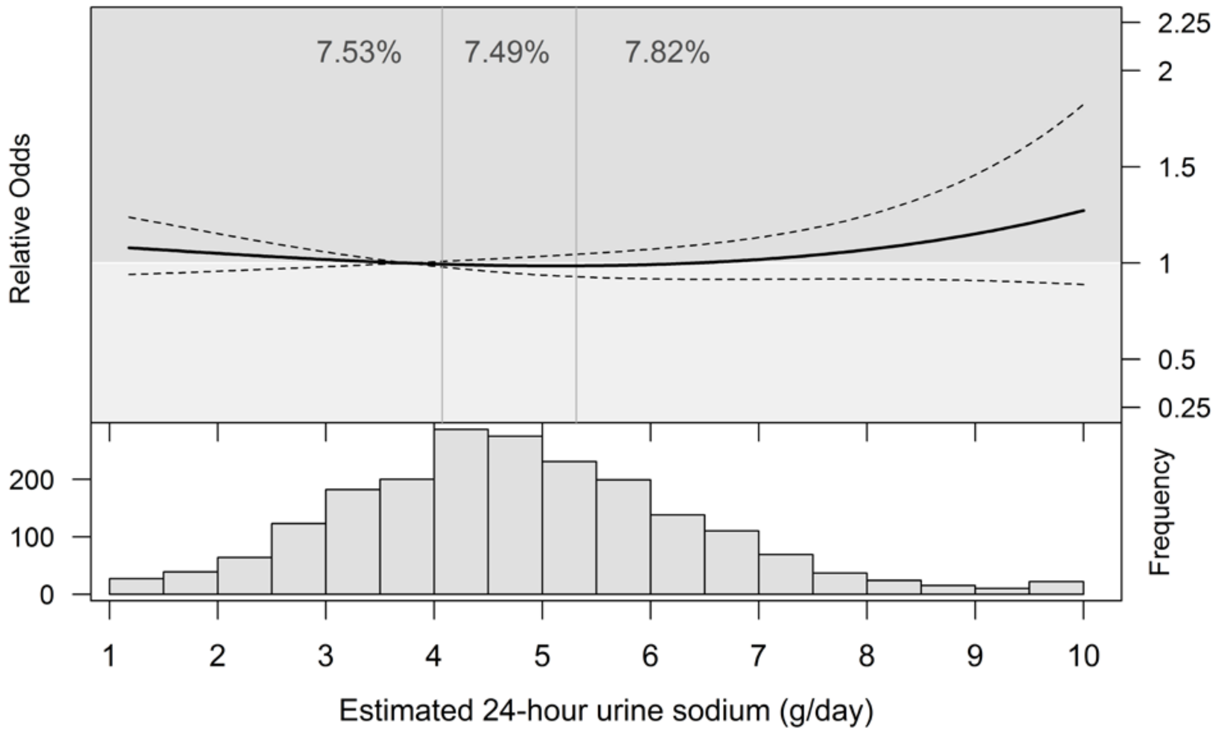


Figure 3.1 24-hour urinary sodium excretion and primary outcome

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of primary outcome (defined as eGFR decline >=30% or chronic dialysis), observed in 7.6% (n=2,052) (p=0.3659) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of sodium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

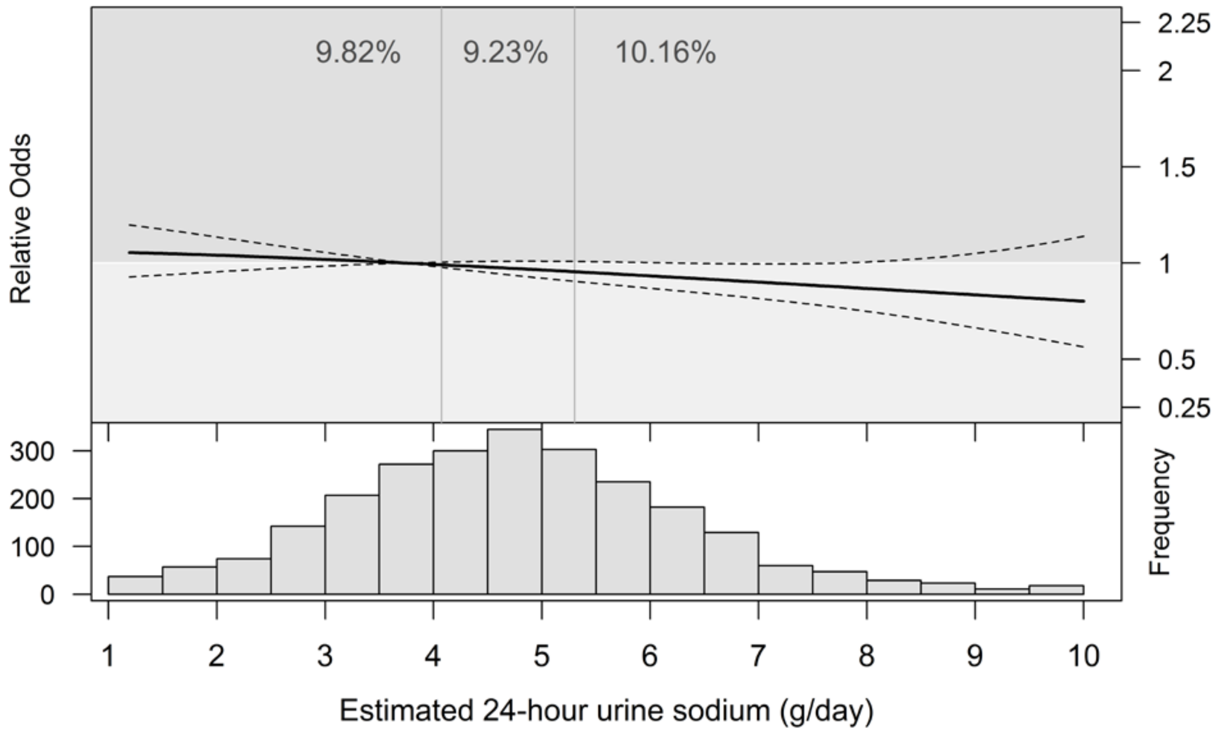


Figure 3.2 24-hour urinary sodium excretion and proteinuria

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of proteinuria (defined as progression in proteinuria), observed in 9.5% (n=2,471) (p=0.1203) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of sodium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

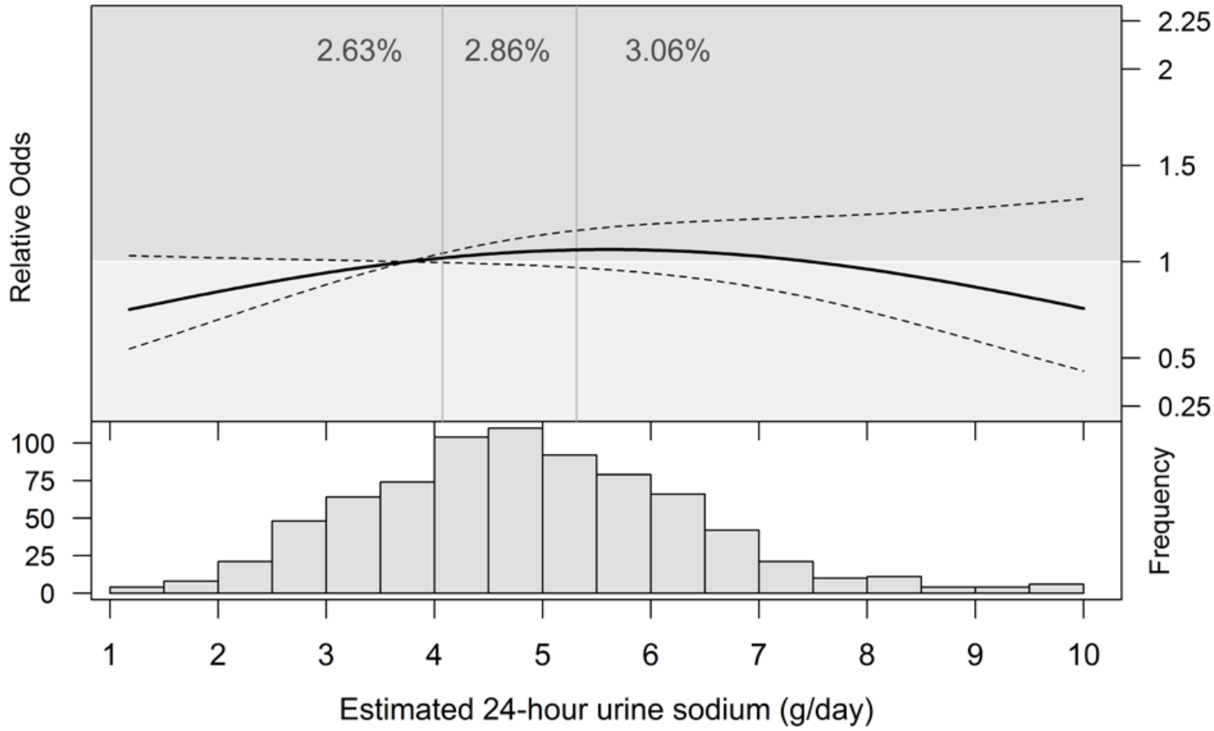


Figure 3.3 24-hour urinary sodium excretion and hyperkalemia

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of hyperkalemia (defined as serum potassium measurement of >5.5mmol/L at two years or study end), observed in 2.7% (n=768) (p=0.209) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of sodium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

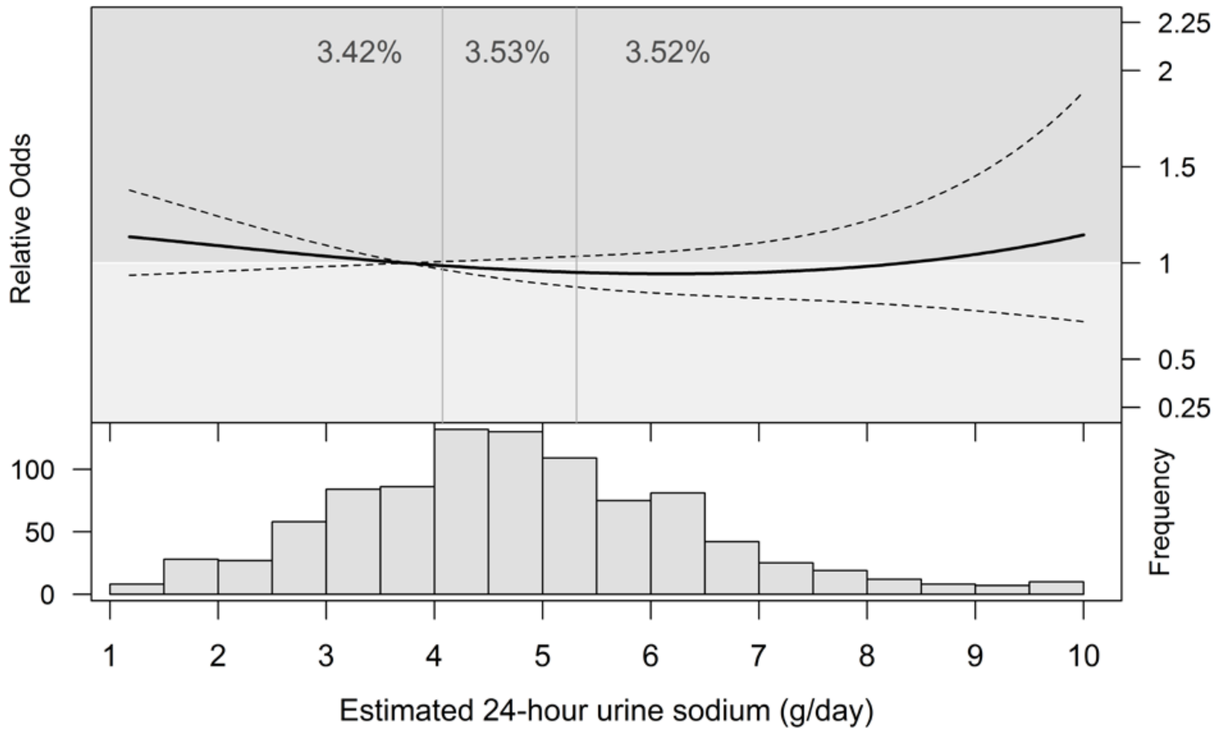


Figure 3.4 24-hour urinary sodium excretion and ≥40% or chronic dialysis

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of eGFR decline ≥40% or chronic dialysis, observed in 3.5% (n=941) (p=0.4305) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of sodium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

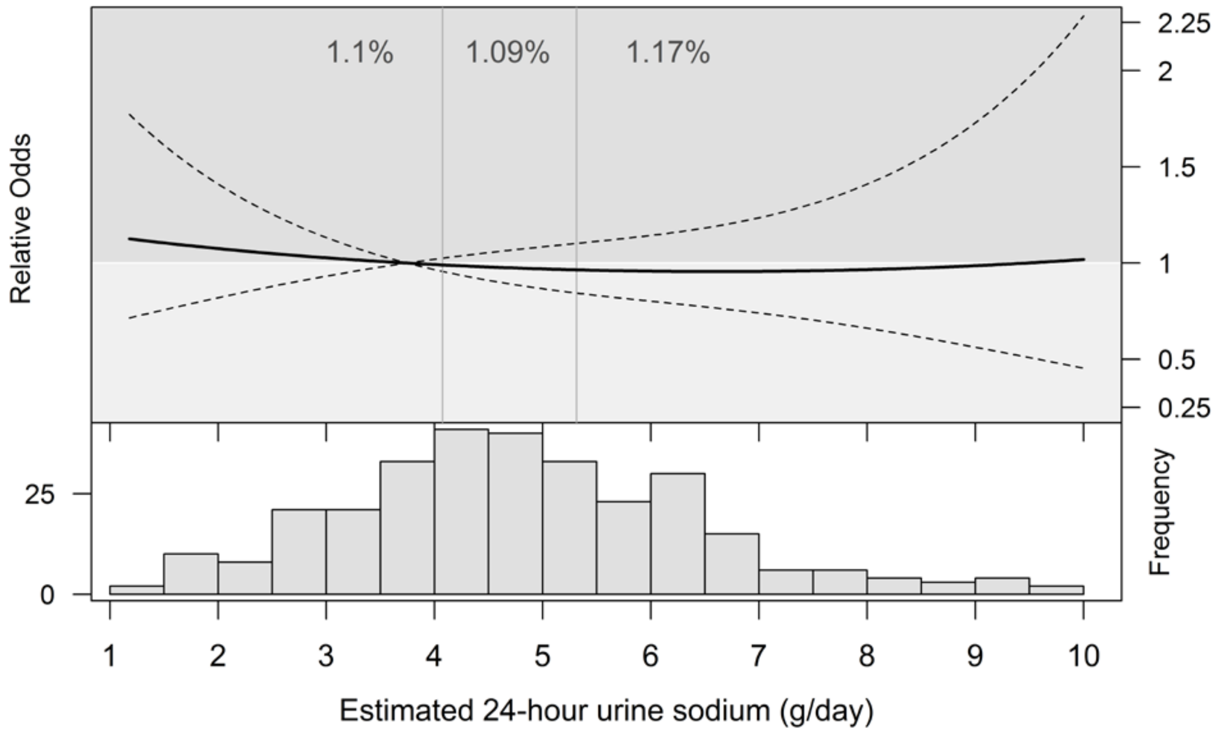


Figure 3.5 24-hour urinary sodium excretion and doubling of creatinine or chronic dialysis

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of doubling of creatinine or chronic dialysis, observed in 1.1% (n=302) (p=0.858) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of sodium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

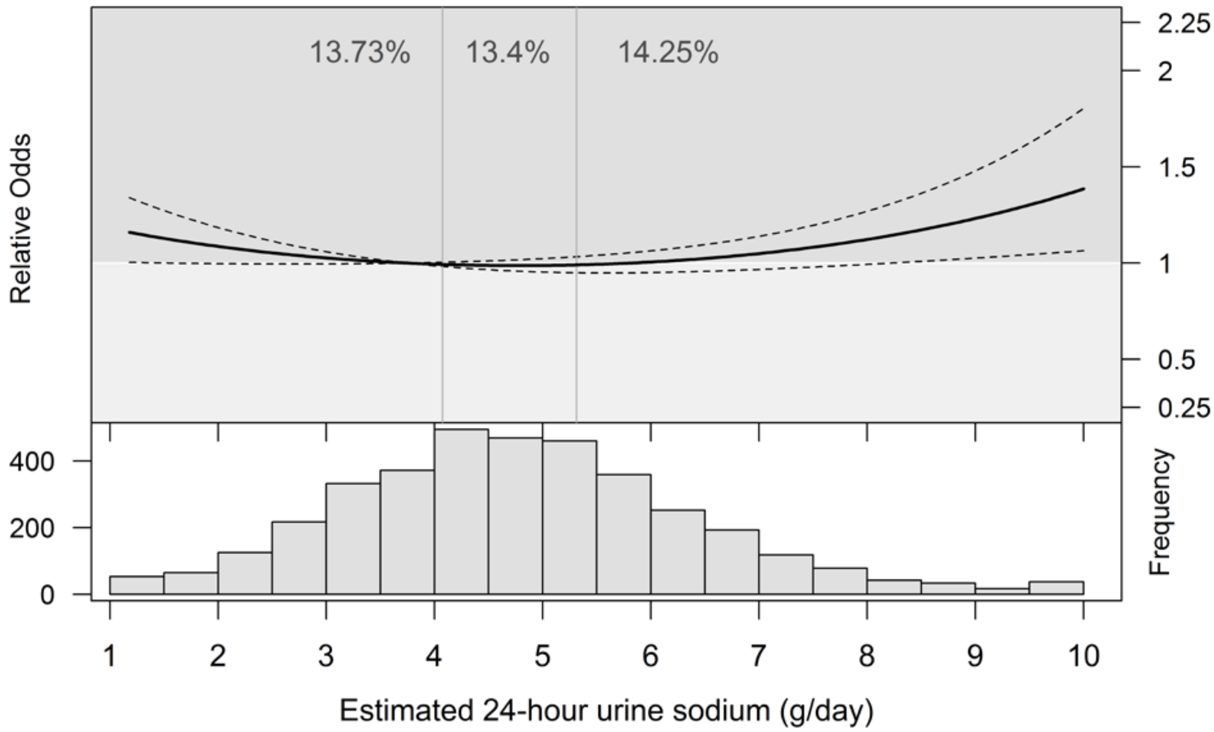


Figure 3.6 24-hour urinary sodium excretion and rapid progression of renal disease

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of rapid progression of renal disease (defined as decline of 5% per year in eGFR), observed in 13.7% (n=3,717) (p=0.0334) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of sodium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

Table 3.4 Sodium Excretion and Renal Outcomes

eGFR decline defined as a change from 6 week creatinine/GFR measurement; Rapid Progression of Renal Disease defined as decline of 5% per year in eGFR from 6 week measurement; Progression of proteinuria defined as change from normal to microalbuminuria (ACR>30mg/g) or macroalbuminuria (ACR>300mg/g) or from microalbuminuria to macroalbuminuria; Hyperkalemia defined as a serum potassium >5.5mmol/L. *Adjusted for known risk factors including age, sex, ethnicity, six week eGFR, run-in ACR, diabetes, BMI, smoking, use of on-study RAAS blockade, diuretic use and potassium excretion; reproduced with permission (Appendix 3).

Outcome	No. of Outcomes (%)	Reference	Unadjusted		Adjusted*	
		Low (Median 3.3g)	Moderate (Median 4.7g)	High (Median 6.2g)	Moderate (Median 4.7g)	High (Median 6.2g)
eGFR decline of ≥30% or Chronic Dialysis	2,052 (7.6%)	1.0	0.96 (0.91-1.01)	1.00 (0.92-1.09)	0.98 (0.92-1.04)	0.99 (0.89-1.09)
Progression of Proteinuria	2,471 (9.5%)	1.0	0.95 (0.91-0.99)	0.98 (0.91-1.05)	0.96 (0.91-1.02)	0.92 (0.84-1.01)
Hyperkalemia	768 (2.7%)	1.0	1.08 (0.99-1.19)	1.14 (0.99-1.32)	1.08 (0.98-1.19)	1.09 (0.93-1.28)
eGFR decline of ≥40% or Chronic Dialysis	941 (3.5%)	1.0	0.94 (0.87-1.02)	0.98 (0.87-1.11)	0.95 (0.87-1.03)	0.93 (0.80-1.07)
Doubling of Creatinine or Chronic Dialysis	302 (1.1%)	1.0	0.96 (0.84-1.09)	1.00 (0.81-1.24)	0.96 (0.84-1.11)	0.94 (0.75-1.19)
Rapid Progression of Renal Disease	3,717 (13.7%)	1.0	0.96 (0.92-1.00)	1.00 (0.94-1.07)	0.97 (0.93-1.02)	1.00 (0.93-1.07)

3.3.2 Potassium

There were almost-linear, strong, statistically significant associations between estimated 24-hour urinary potassium excretion and the primary outcome (Figure 3.7) and progression of proteinuria (Figure 3.8) on adjusted models; participants with higher potassium excretion had significantly lower odds of the outcomes. The association with hyperkalemia was non-significant, but there was a tendency towards increased odds of hyperkalemia with higher potassium excretion (Figure 3.9). Similarly, there were significant associations with eGFR decline of ≥40% or chronic dialysis (Figure 3.10), doubling of creatinine or chronic dialysis (Figure 3.11) or rapid progression of renal disease (Figure 3.12).

Compared with the lowest potassium excretion (median 1.7g), both moderate (median 2.1g/day), and high (median 2.7g/day) potassium excretion were associated with a dose-dependent reduction in the odds of the primary outcome (OR 0.88 (95% CI 0.84-0.92) and 0.74 (95% CI 0.67-0.82), respectively) (Table 3.5). The magnitude of the reduction in the odds was similar for eGFR decline of ≥40% or chronic dialysis, rapid progression of renal disease, doubling of creatinine or chronic dialysis, and progression of proteinuria. Both moderate and high potassium excretion were associated with a tendency towards an increase in the odds of hyperkalemia (OR 1.07 (95% CI 1.00-1.14) and 1.16 (95% CI 0.99-1.36), respectively).

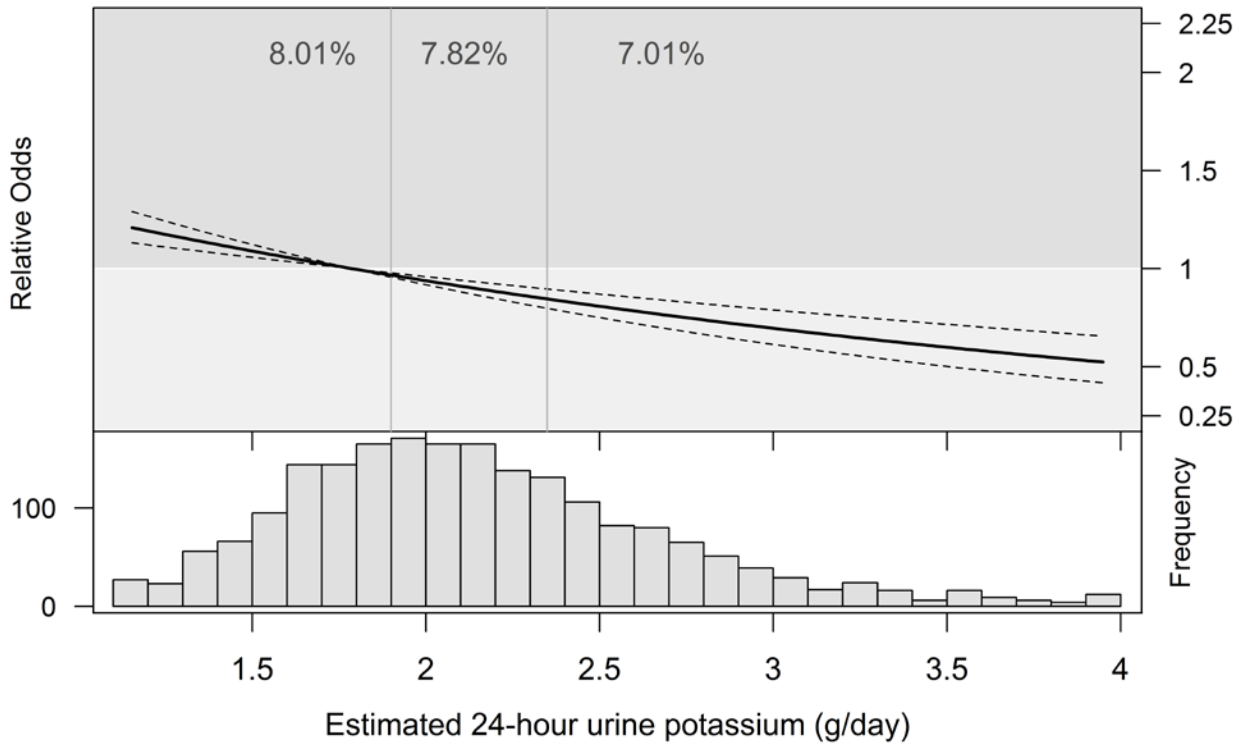


Figure 3.7. 24-hour urinary potassium excretion and primary outcome

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of primary outcome (defined as eGFR decline >=30% or chronic dialysis), observed in 7.6% (n=2,052) (p<0.0001) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of potassium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

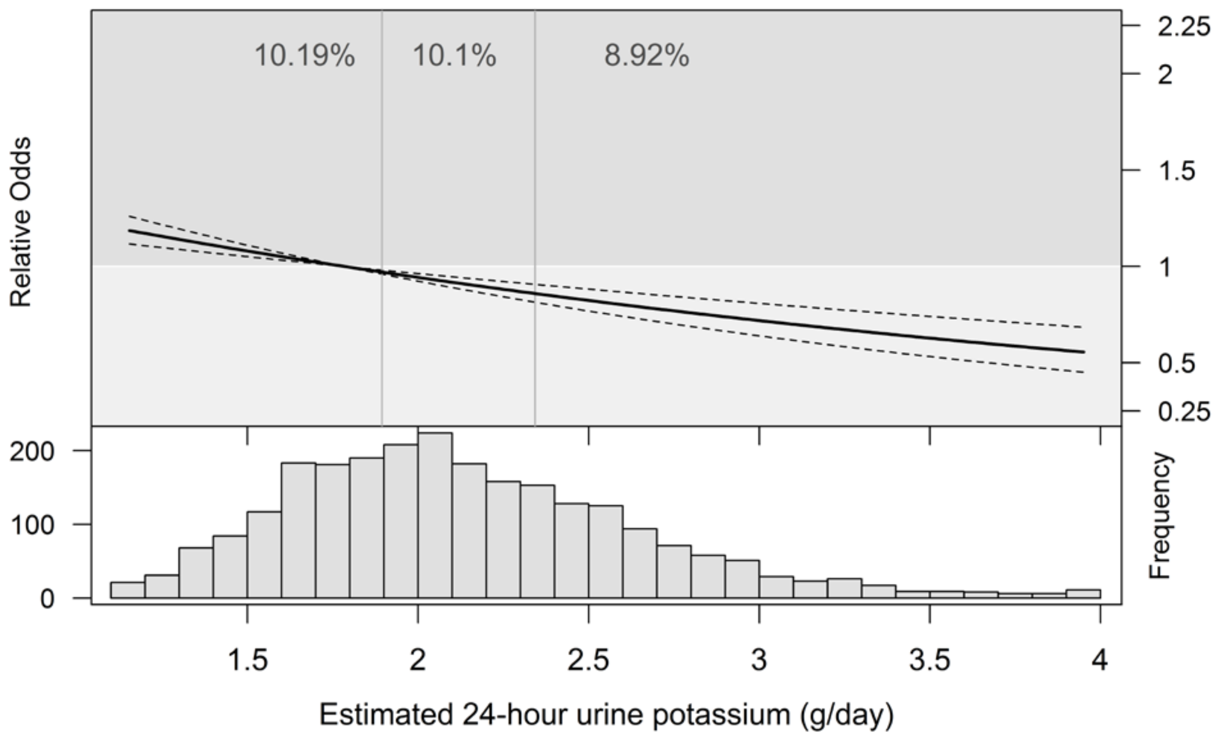


Figure 3.8 24-hour urinary potassium excretion and proteinuria

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of proteinuria (defined as progression in proteinuria), observed in 9.5% (n=2,471) (p<0.0001) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of potassium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

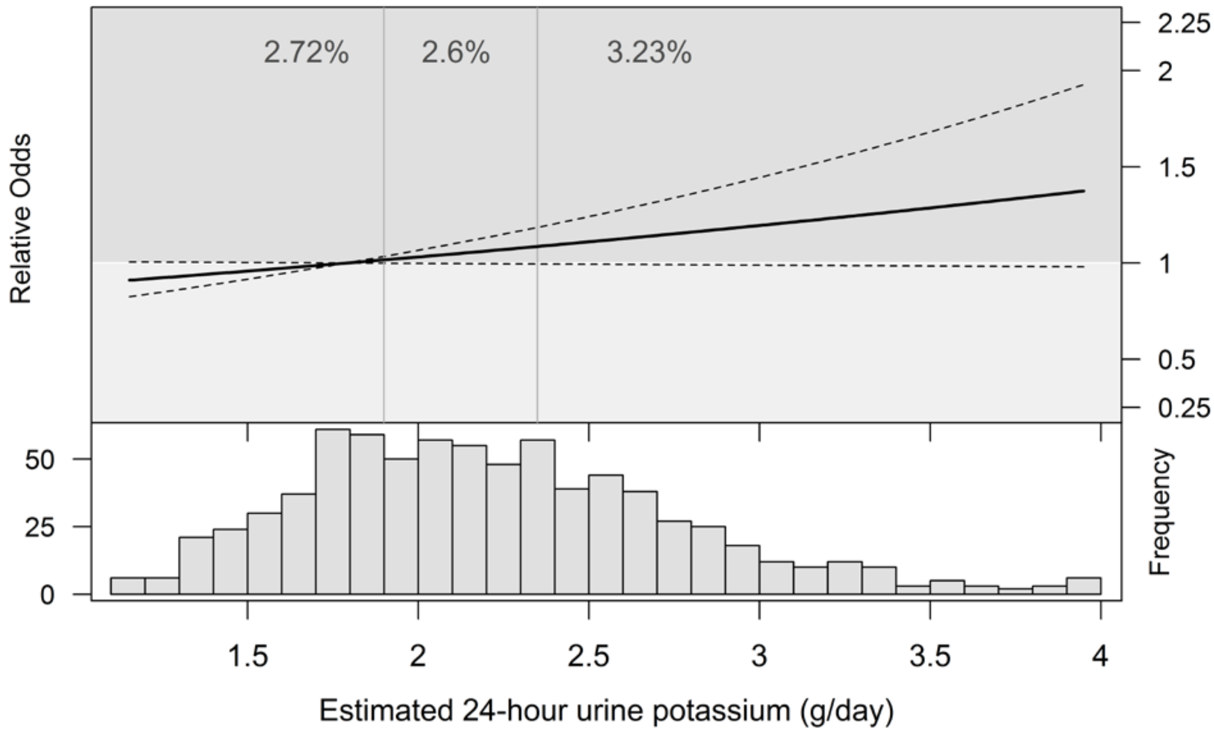


Figure 3.9 24-hour urinary potassium excretion and hyperkalemia

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of hyperkalemia (defined as serum potassium measurement of >5.5mmol/L at two years or study end), observed in 2.7% (n=768) (p=0.0649) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of potassium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

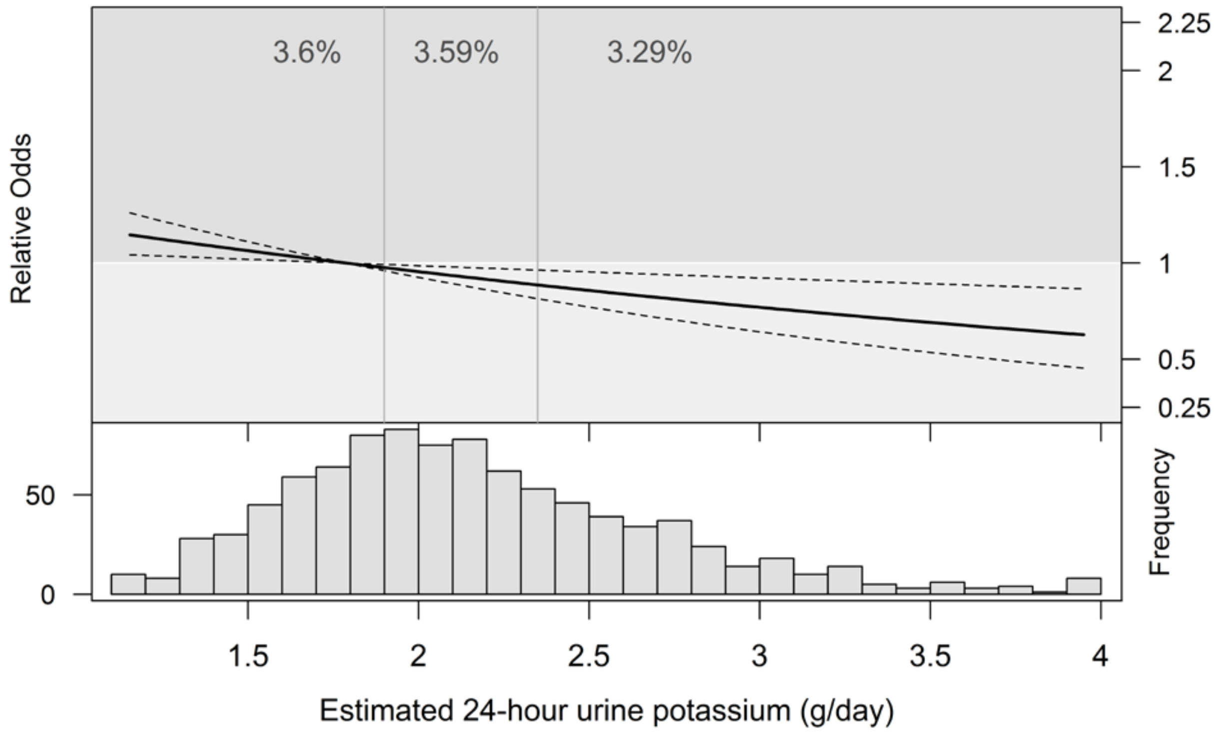


Figure 3.10 24-hour urinary potassium excretion and ≥40% or chronic dialysis

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of eGFR decline ≥40% or chronic dialysis, observed in 3.5% (n=941) (p=0.0046) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of potassium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

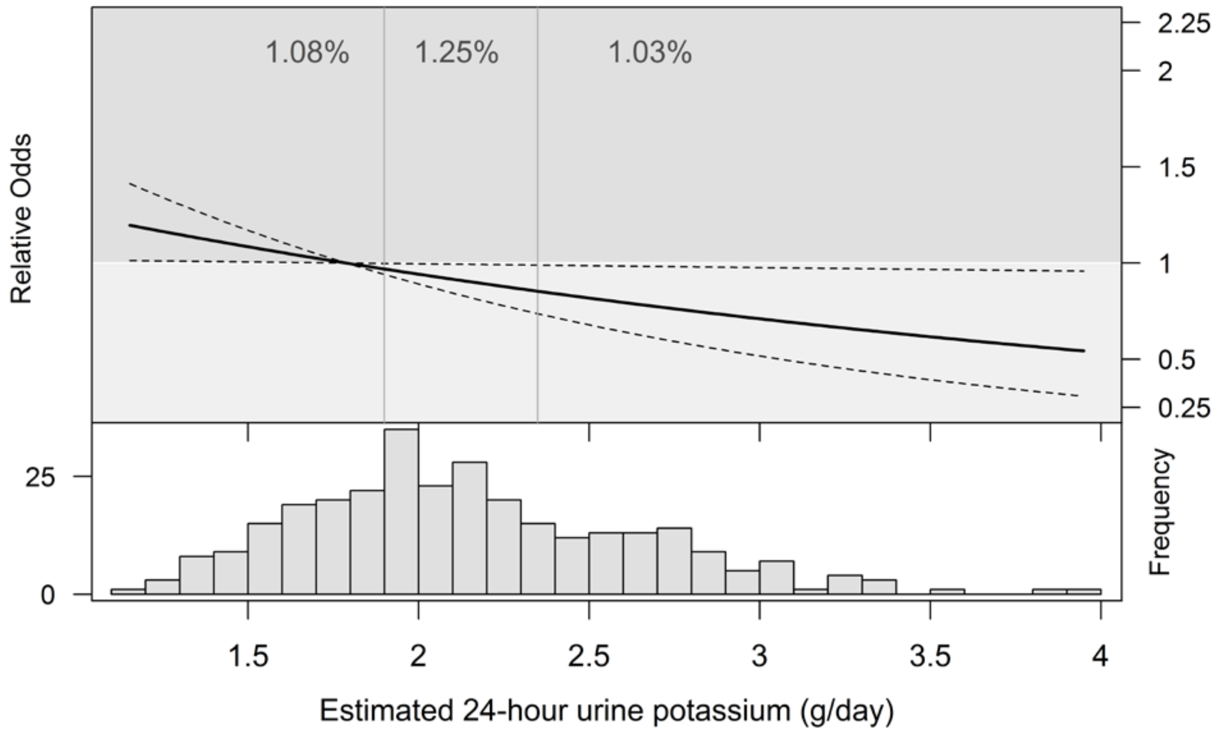


Figure 3.11 24-hour urinary potassium excretion and doubling of creatinine or chronic dialysis

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of doubling of creatinine or chronic dialysis, observed in 1.1% (n=302) (p=0.035) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of potassium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

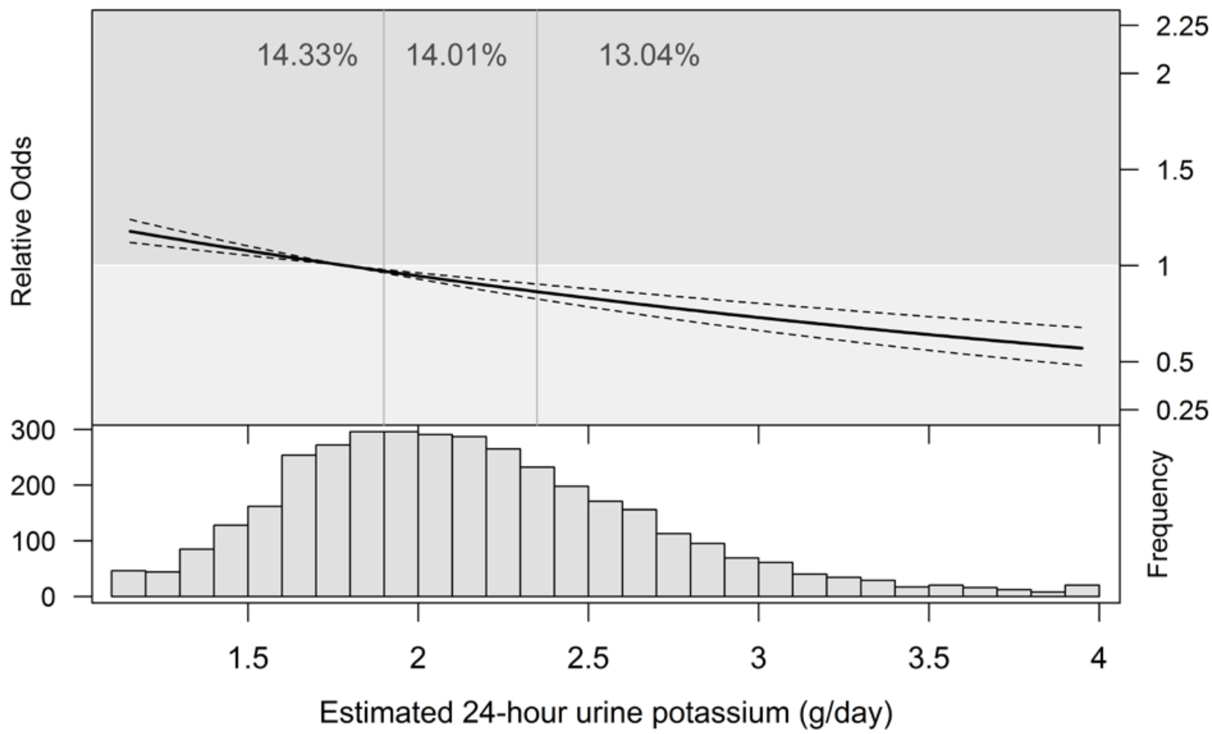


Figure 3.12 24-hour urinary potassium excretion and rapid progression of renal disease

Plot shows relative odds (solid line) with 95% confidence interval (dotted line) of rapid progression of renal disease, defined as decline of 5% per year in eGFR, observed in 13.7% (n=3,717) (p<0.001) and is based on the model for sodium and potassium excretion (adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use and use of on-study RAAS blockade). Histogram presents the distribution of potassium excretion in participants with the outcome. The vertical lines show tertiles and the reported percentages within each tertile indicate the percentage of participants experiencing the outcome. Reproduced with permission (Appendix 3).

Table 3.5 Potassium Excretion and Renal Outcomes

eGFR decline defined as a change from 6 week creatinine/GFR measurement; Rapid Progression of Renal Disease defined as decline of 5% per year in eGFR from 6 week measurement; Progression of proteinuria defined as change from normal to microalbuminuria (ACR>30mg/g) or macroalbuminuria (ACR>300mg/g) or from microalbuminuria to macroalbuminuria; Hyperkalemia defined as a serum potassium >5.5mmol/L. *Adjusted for known risk factors including age, sex, ethnicity, six week eGFR, run-in ACR, diabetes, BMI, smoking, use of on-study RAAS blockade, diuretic use and sodium excretion; reproduced with permission (Appendix 3).

Outcome	**No. of Outcomes (%)**	**Reference**	**Unadjusted**		**Adjusted***	
		Low (Median 1.7g)	**Moderate (Median 1.7g)**	**High (Median 2.7g)**	**Moderate (Median 2.1g)**	**High (Median 2.7g)**
eGFR decline of ≥30% or Chronic Dialysis	2,052 (7.6%)	1.0	0.94 (0.91-0.98)	0.87 (0.80-0.95)	0.88 (0.84-0.92)	0.74 (0.67-0.82)
Progression of Proteinuria	2,471 (9.5%)	1.0	0.93 (0.90-0.96)	0.84 (0.78-0.91)	0.89 (0.85-0.93)	0.76 (0.69-0.84)
Hyperkalemia	768 (2.7%)	1.0	1.06 (1.00-1.12)	1.14 (1.00-1.30)	1.07 (1.00-1.14)	1.16 (0.99-1.36)
eGFR decline of ≥40% or Chronic Dialysis	941 (3.3%)	1.0	0.97 (0.91-1.02)	0.92 (0.81-1.04)	0.91 (0.85-0.97)	0.81 (0.69-0.94)
Doubling of Creatinine or Chronic Dialysis	302 (1.1%)	1.0	0.95 (0.87-1.05)	0.89 (0.72-1.11)	0.88 (0.79-0.99)	0.75 (0.58-0.98)
Rapid Progression of Renal Disease	3,717 (13.7%)	1.0	0.96 (0.93-0.98)	0.90 (0.84-0.96)	0.89 (0.86-0.92)	0.77 (0.71-0.84)

3.3.3 Sensitivity analyses

Sodium

The inclusion of hospitalisation with death as the competing risk event (Sensitivity Analysis 1) had no significant effect on the association with the primary outcome (Table 3.6). There was a reduction in the odds ratio for progression of proteinuria with high sodium intake, which became statistically significant (OR 0.86 (95% CI 0.76-0.97)) and a reduction in the odds of hyperkalemia. The inclusion of myocardial infarction, stroke or hospitalisation for heart failure with death as the competing risk event (Sensitivity Analysis 2) had no significant effect on the primary outcome, progression of proteinuria or hyperkalemia. The inclusion of myocardial infarction and hospitalisation for heart failure with death as the competing risk event (Sensitivity Analysis 3) also had no effect on the associations.

Potassium

The inclusion of hospitalisation with death as the competing risk event (Sensitivity Analysis 1) reduced the odds ratio for the primary outcome and hyperkalemia and increased the odds ratio for progression of proteinuria, but changes were not statistically significant (Table 3.6). The inclusion of myocardial infarction, stroke or hospitalisation for heart failure with death as the competing risk event (Sensitivity Analysis 2) had no significant effect on the primary outcome, progression of proteinuria or hyperkalemia. The inclusion of myocardial infarction and hospitalisation for heart failure with death as the competing risk event (Sensitivity Analysis 3) also had no effect on the associations.

Table 3.5 Sensitivity analyses of effect of sodium & potassium excretion on select renal outcomes

Main Model: Competing Risk = All-Cause Mortality; Sensitivity Analysis 1: Competing Risk = Any Hospitalisation or All-Cause Mortality; Sensitivity Analysis 2: Competing Risk = Incident MI, Incident Stroke, Incident Hospitalisation for Heart Failure or All-Cause Mortality; Sensitivity Analysis 3: Competing Risk = Incident MI, Incident Hospitalisation for Heart Failure or All-Cause Mortality. *Model for sodium and potassium adjusted age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, use of on-study RAAS blockade and diuretic use; reproduced with permission (Appendix 3).

Analysis	No. of Outcomes (%)	Sodium Excretion*			Potassium Excretion*		
		Low (Ref) (Median 3.3g)	Moderate (Median 4.7g)	High (Median 6.1g)	Low (Ref) (Median 1.7g)	Moderate (Median 2.1g)	High (Median 2.7g)
Outcome: Primary Outcome (eGFR decline of ≥30% or Chronic Dialysis)							
Main Model	2,052 (7.6%)	1.0	0.98 (0.92-1.04)	0.99 (0.89-1.09)	1.0	0.88 (0.84-0.92)	0.74 (0.67-0.82)
Sensitivity Analysis 1	710 (2.7%)	1.0	1.01 (0.94-1.08)	1.01 (0.87-1.18)	1.0	0.85 (0.79-0.92)	0.69 (0.58-0.82)
Sensitivity Analysis 2	1,463 (5.7%)	1.0	0.97 (0.90-1.03)	0.97 (0.86-1.08)	1.0	0.87 (0.82-0.91)	0.72 (0.64-0.81)
Sensitivity Analysis 3	1,525 (5.9%)	1.0	0.97 (0.91-1.04)	0.99 (0.89-1.10)	1.0	0.86 (0.82-0.91)	0.71 (0.63-0.80)
Outcome: Progression of Proteinuria							
Main Model	2,471 (9.5%)	1.0	0.96 (0.91-1.02)	0.92 (0.84-1.01)	1.0	0.89 (0.85-0.93)	0.76 (0.69-0.84)
Sensitivity Analysis 1	1,097 (4.2%)	1.0	0.93 (0.88-0.98)	0.86 (0.76-0.97)	1.0	0.91 (0.85-0.96)	0.80 (0.69-0.92)
Sensitivity Analysis 2	2,114 (8.2%)	1.0	0.99 (0.93-1.04)	0.95 (0.86-1.06)	1.0	0.89 (0.85-0.93)	0.76 (0.69-0.84)
Sensitivity Analysis 3	2,213 (8.6%)	1.0	0.98 (0.92-1.04)	0.94 (0.86-1.04)	1.0	0.89 (0.85-0.93)	0.76 (0.69-0.84)
Outcome: Hyperkalemia							
Main Model	768 (2.7%)	1.0	1.08 (0.98-1.19)	1.09 (0.93-1.28)	1.0	1.07 (1.00-1.14)	1.16 (0.99-1.36)
Sensitivity Analysis 1	301 (1.2%)	1.0	1.02 (0.90-1.17)	0.94 (0.73-1.20)	1.0	1.04 (0.93-1.15)	1.08 (0.85-1.38)
Sensitivity Analysis 2	609 (2.4%)	1.0	1.06 (0.95-1.18)	1.08 (0.90-1.29)	1.0	1.05 (0.97-1.13)	1.11 (0.93-1.32)
Sensitivity Analysis 3	629 (2.4%)	1.0	1.07 (0.96-1.18)	1.07 (0.90-1.26)	1.0	1.05 (0.97-1.13)	1.12 (0.94-1.33)

3.3.4 Subgroup analyses

Sodium

There were no statistically significant subgroup effects comparing high sodium intake to low sodium intake for the primary outcome (Figure 3.13). For progression of proteinuria, there was a reduction in the odds in patients without diabetes (p=0.04), but no other significant subgroup effects (Figure 3.14). There were no statistically significant subgroup effects identified for hyperkalemia (Figure 3.15).

Potassium

There was a loss of statistical significance of reduced odds for the primary outcome in patients with eGFR ≤45 mL/min/1.73m^2 (p=0.01) and macroalbuminuria (p=0.02) (Figure 3.16). There were no statistically significant subgroup effects for progression of proteinuria (Figure 3.17). For hyperkalemia, there was an increase in the odds of in patients without diabetes (p=0.03) and normal baseline ACR (p=0.02) (Figure 3.18).

	Odds ratio	Lower limit	Upper limit
Diabetes (p=0.50)			
Yes	0.994	0.865	1.143
No	0.978	0.853	1.122
RAAS Blockade (p=0.28)			
Yes	0.987	0.890	1.095
No	0.940	0.657	1.344
Age (p=0.28)			
>69 years	0.963	0.821	1.130
62-69 years	0.922	0.787	1.080
<62 years	1.095	0.904	1.326
eGFR at 6 weeks (p=0.09)			
>60ml/min/1.73m^2	1.003	0.891	1.130
45-60ml/min/1.73m^2	0.823	0.667	1.015
<45ml/min/1.73m^2	1.273	0.948	1.709
Baseline UACR (p=0.54)			
Normal	1.038	0.931	1.158
Microalbuminuria	0.891	0.736	1.079
Macroalbuminuria	1.225	0.856	1.753
Overall Effect	0.985	0.891	1.089

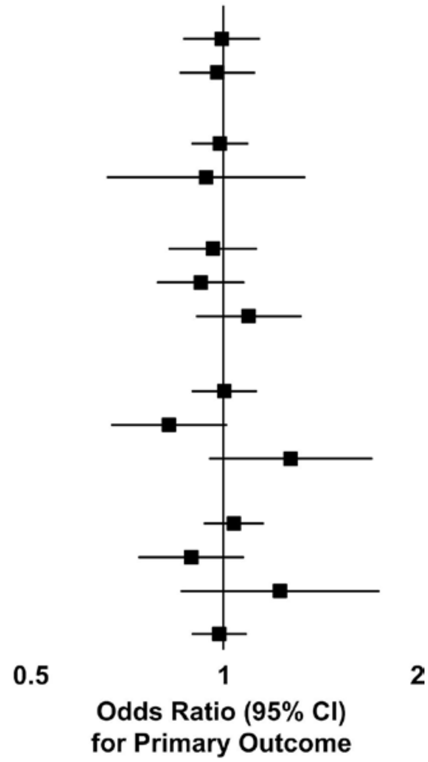


Figure 3.13 Subgroup analyses for sodium and primary outcome

Association between sodium excretion (high (median 6.2g/day) vs. low (median 3.3g/day) and primary outcome (eGFR decline ≥30% or chronic dialysis). All models adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use, use of on-study RAAS blockade, and potassium excretion. Reproduced with permission (Appendix 3).

	Odds ratio	Lower limit	Upper limit
Diabetes (p=0.04)			
Yes	0.978	0.861	1.110
No	0.848	0.747	0.963
RAAS Blockade (p=0.97)			
Yes	0.917	0.832	1.010
No	0.928	0.719	1.198
Age (p=0.22)			
>69 years	0.852	0.740	0.980
62-69 years	0.930	0.798	1.084
<62 years	1.012	0.853	1.200
eGFR at 6 weeks (p=0.54)			
>60ml/min/1.73m^2	0.906	0.807	1.017
45-60ml/min/1.73m^2	1.001	0.844	1.187
<45ml/min/1.73m^2	0.790	0.621	1.005
Baseline UACR (p=0.07)			
Normal	0.980	0.894	1.075
Microalbuminuria	1.018	0.824	1.257
Macroalbuminuria	-	-	-
Overall Effect	0.918	0.838	1.006

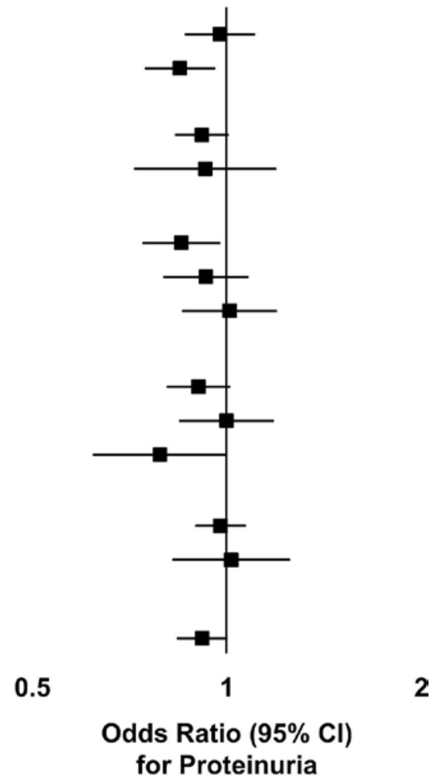


Figure 3.14 Subgroup analyses for sodium and proteinuria

Association between sodium excretion (high (median 6.2g/day) vs. low (median 3.3g/day) and progression of proteinuria. No estimate provided for patients with macroalbuminuria at baseline as these patients were excluded from the outcome of progression of proteinuria. All models adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use, use of on-study RAAS blockade, and potassium excretion. Reproduced with permission (Appendix 3).

	Odds ratio	Lower limit	Upper limit
Diabetes (p=0.87)			
Yes	1.053	0.857	1.293
No	1.132	0.904	1.417
RAAS Blockade (p=0.51)			
Yes	1.111	0.947	1.304
No	0.760	0.389	1.485
Age (p=0.30)			
>69 years	1.160	0.891	1.511
62-69 years	1.113	0.865	1.432
<62 years	0.938	0.723	1.217
eGFR at 6 weeks (p=0.42)			
>60ml/min/1.73m^2	1.073	0.872	1.320
45-60ml/min/1.73m^2	1.285	0.969	1.705
<45ml/min/1.73m^2	0.875	0.624	1.227
Baseline UACR (p=0.99)			
Normal	1.090	0.914	1.300
Microalbuminuria	1.161	0.822	1.639
Macroalbuminuria	1.022	0.562	1.858
Overall Effect	1.092	0.934	1.277

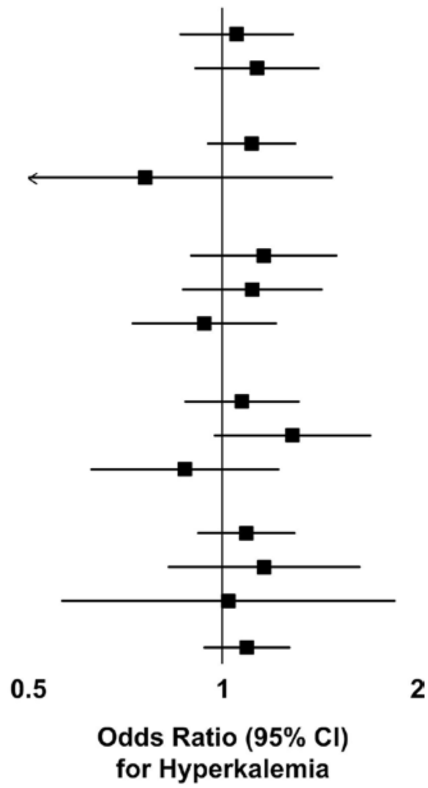


Figure 3.15 Subgroup analyses for sodium and hyperkalemia

Association between sodium excretion (high (median 6.2g/day) vs. low (median 3.3g/day) and hyperkalemia. All models adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use, use of on-study RAAS blockade, and potassium excretion. Reproduced with permission (Appendix 3).

	Odds ratio	Lower limit	Upper limit
Diabetes (p=0.87)			
Yes	0.748	0.650	0.860
No	0.736	0.642	0.844
RAAS Blockade (p=0.93)			
Yes	0.740	0.665	0.824
No	0.729	0.516	1.030
Age (p=0.51)			
>69 years	0.748	0.632	0.886
62-69 years	0.780	0.668	0.911
<62 years	0.686	0.578	0.814
eGFR at 6 weeks (p=0.01)			
>60ml/min/1.73m^2	0.728	0.648	0.817
45-60ml/min/1.73m^2	0.783	0.624	0.983
<45ml/min/1.73m^2	1.143	0.853	1.532
Baseline UACR (p=0.02)			
Normal	0.820	0.731	0.919
Microalbuminuria	0.584	0.467	0.730
Macroalbuminuria	0.769	0.552	1.071
Overall Effect	0.741	0.667	0.823

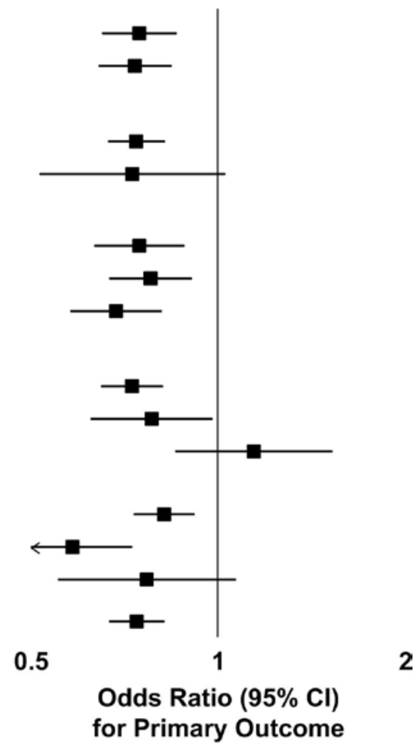


Figure 3.16 Subgroup analyses for potassium and primary outcome

Association between potassium excretion (high (median 2.7g/day) vs. low (median 1.7g/day) and primary outcome (eGFR decline ≥30% or chronic dialysis). All models adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use, use of on-study RAAS blockade, and sodium excretion. Reproduced with permission (Appendix 3).

	Odds ratio	Lower limit	Upper limit
Diabetes (p=0.46)			
Yes	0.788	0.693	0.896
No	0.739	0.650	0.840
RAAS Blockade (p=0.43)			
Yes	0.753	0.680	0.833
No	0.837	0.650	1.077
Age (p=0.54)			
>69 years	0.745	0.638	0.869
62-69 years	0.727	0.624	0.848
<62 years	0.812	0.698	0.945
eGFR at 6 weeks (p=0.48)			
>60ml/min/1.73m^2	0.771	0.689	0.863
45-60ml/min/1.73m^2	0.777	0.647	0.933
<45ml/min/1.73m^2	0.651	0.497	0.853
Baseline UACR (p=0.07)			
Normal	0.951	0.862	1.050
Microalbuminuria	0.767	0.612	0.962
Macroalbuminuria	-	-	-
Overall Effect	0.761	0.691	0.839

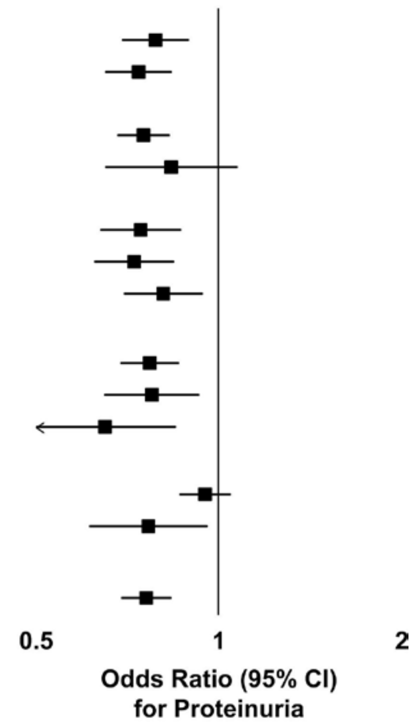


Figure 3.17 Subgroup analyses for potassium and proteinuria

Association between potassium excretion (high (median 2.7g/day) vs. low (median 1.7g/day) and progression of proteinuria. No estimate provided for patients with macroalbuminuria at baseline as these patients were excluded from the outcome of progression of proteinuria. All models adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use, use of on-study RAAS blockade, and sodium excretion. Reproduced with permission (Appendix 3).

	Odds ratio	Lower limit	Upper limit
Diabetes (p=0.03)			
Yes	1.000	0.811	1.234
No	1.351	1.100	1.660
RAAS Blockade (p=0.47)			
Yes	1.170	0.996	1.374
No	0.921	0.486	1.745
Age (p=0.75)			
>69 years	1.061	0.812	1.386
62-69 years	1.201	0.942	1.531
<62 years	1.181	0.931	1.498
eGFR at 6 weeks (p=0.36)			
>60ml/min/1.73m^2	1.074	0.890	1.296
45-60ml/min/1.73m^2	1.356	1.020	1.802
<45ml/min/1.73m^2	1.099	0.763	1.584
Baseline UACR (p=0.02)			
Normal	1.305	1.103	1.544
Microalbuminuria	0.804	0.560	1.154
Macroalbuminuria	0.805	0.460	1.409
Overall Effect	1.159	0.990	1.356

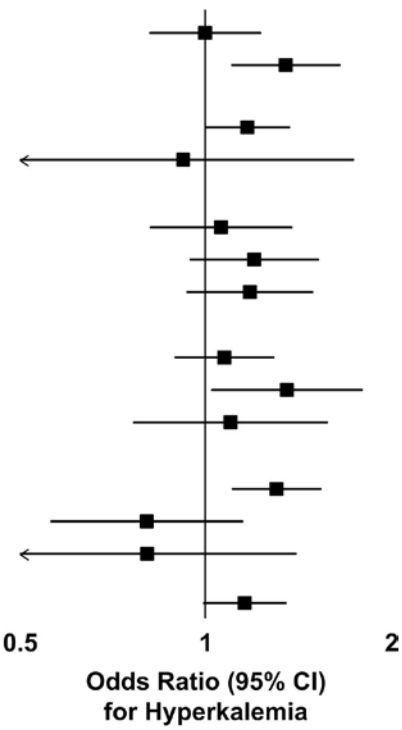


Figure 3.18 Subgroup analyses for potassium and hyperkalemia

Association between potassium excretion (high (median 2.7g/day) vs. low (median 1.7g/day) and hyperkalemia. All models adjusted for age, sex, ethnicity, six-week eGFR, run-in ACR, diabetes, BMI, smoking, diuretic use, use of on-study RAAS blockade, and sodium excretion. Reproduced with permission (Appendix 3).

3.4 Discussion

3.4.1 Summary of Findings

No support was found for the hypothesis that lower sodium excretion is associated with reduced renal risk. Sodium excretion varied widely and there were a substantial number of outcomes that included changes of eGFR, chronic dialysis and proteinuria. A new observation is the robust protective effect of higher potassium excretion.

3.4.2 Sodium

There was a non-significant tendency towards a J-shaped association between sodium excretion and the primary outcome (eGFR decline of ≥30% or chronic dialysis), and a significant U-shaped association with rapid progression of renal disease (p=0.0334). These results are consistent with previous studies that report increased risk with high intake (>4.6 g/day)[69-71]. A possible increase in risk below 4 g/day is consistent with the systematic review[117] presented in Chapter 2, and with findings in patients with type 1 diabetes[25], and does not support current recommendations of low intakes (<2.3g/day, with no lower limit suggested)[18].

A US-based cross-sectional study (n=13,917) observed an increased prevalence of low GFR in those with sodium intake <2.1g compared with higher sodium intakes[143], and the same effect was observed cross-sectionally in the baseline data, perhaps due to confounding by indication as those with higher burdens of disease may have lower sodium intakes due to lower overall dietary intake. However, unlike that study, which reported that higher sodium intake was associated with lower odds of CKD, a protective association between higher sodium and any outcome was not observed here. This may be due to differences in population characteristics (high-cardiovascular risk patients vs. general population) and study design (sodium may have differential effects on GFR in acute [detected cross-sectionally] and chronic [detected on follow-up] settings).

No association between sodium excretion and progression of proteinuria was observed. The acute antiproteinuric action of RAAS inhibition is greater in those with a negative sodium balance[144] and there are trial data showing the risk of kidney outcomes decreases with lower sodium intake in patients with proteinuric nephropathies on RAAS blockade[71]. The effect of sodium excretion on proteinuria may have been missed as most participants were on RAAS blockade at enrolment and the effect of RAAS blockade on proteinuria is acute.

3.4.3 Potassium

The association of potassium excretion with reduction in odds of renal outcomes was strong, dose dependent, and consistent across outcomes. The odds ratio for the primary outcome varied substantially with potassium excretion from approximately 1.2 to 0.5, with narrow confidence intervals. No previous large longitudinal study has examined the association between potassium intake and prospectively recorded renal outcomes. A cross-sectional study (n=13,917) reported a protective association between high potassium intake (>3.3g/day) and CKD[143], confirmed here (Table 3.4). Three small short term controlled trials reported that the addition of approximately 60mmol/day of potassium chloride compared to placebo, resulted in either no change[34,35] or a decrease in serum creatinine[36]. However, the subgroup analysis presented here showed a loss of statistical significance in those with eGFR <45ml/min/1.73m^2 (p=0.01) or macroalbuminuria (p=0.02) (Figure 3.16), suggesting that the protective association may not extend to those with more advanced CKD.

ONTARGET previously reported that a diet rich in fruits and vegetables was associated with better renal outcomes[145]. Obviously such a diet will lead to higher potassium excretion. Thus, the potassium effect observed here might entirely be due to healthy diet, mediated by dietary antioxidants that have anti-inflammatory activity, which may decrease kidney disease progression[146]. However, a high potassium intake may also increase plasma potassium, which is reported to reduce renal vascular resistance and increase glomerular filtration rate[147]. Another possibility is that higher potassium excretion itself is renoprotective, mediated by up regulation of renal kinins, such as kallikrein. In a protein-overload rat model of kidney injury, high potassium intake increased renal kallikrein expression, decreased blood pressure, pro-fibrotic tissue growth factor beta, and tubulointerstitial fibrosis; the last three of which effects were blunted by kinin antagonism[148]. Finally, confounding by indication may explain findings, namely patients with more severe renal disease or worse prognosis may have limited potassium intake due to limited overall intake or have been advised to restrict potassium intake due to hyperkalemia. Several facts speak against this last interpretation. The number of participants with clearly elevated serum creatinine, which prompts physicians to advise reduced potassium intake, was low in these trials; furthermore participants with elevated serum potassium were excluded during run in (Section 3.2.1).

A high potassium intake has the obvious concern of promoting hyperkalemia and cardiac arrhythmia. There was a trend towards increased odds of hyperkalemia with higher potassium excretion that, although not statistically significant, may be clinically important. Subgroup analyses showed that association with hyperkalemia was similar at all levels of GFR (p=0.36), but those without diabetes (p=0.03) and with normal ACR appeared to be at increased risk (p=0.02) (Figure 3.18). However these findings have borderline statistical significance, have questionable biological plausibility and are limited by multiple testing. Previous analyses of ONTARGET reported that the risk of hyperkalemia increases with lower GFR and higher ACR[149]. A prospective cohort study of patients with CKD (n=840) associated both low and high serum potassium with risk of death and end-stage renal disease[150].

This study suggests that a diet associated with a high potassium intake may benefit the kidney, and a previous analysis exhibited cardiovascular benefits[26]. It may therefore be important not to restrict potassium intake prematurely in people with CKD. However, our findings are limited by the low proportion of participants with eGFR<30ml/min/1.73m^2, corresponding to stages four and five of chronic kidney disease, where metabolic complications including hyperkalemia are more common. The effect of potassium intake on kidney outcomes in other large longitudinal trials, including studies of people with lower GFR than studied here, will be of great interest, in particular as our subgroup analysis suggested a loss of the protective association with higher potassium excretion in those with eGFR <45ml/min/1.73m^2. Interventional studies of a healthy, high-potassium diet or potassium supplementation to prevent kidney outcomes would not be straightforward, but the absolute risk of severe hyperkalemia (>6.5 mmol/L) in our participants on RAAS blockade was low (0.02 per 100 patient-years in the monotherapy groups in ONTARGET).

3.4.4 Strengths

This study has a number of strengths. First, the large sample size with prospective follow-up, for a mean of 56 months, corresponds to over 120,000 person-years of follow up. Second, the cohort is composed of participants from 40 countries, with representation from multiple ethnicities and geographical regions. Third, as this cohort is based on the participants of two large, simple clinical trials, there was a high completion of follow-up and a high completeness of data. Finally, the use of a central laboratory for urinary electrolytes, used to estimate 24-hour sodium and potassium excretion, standardized this assessment despite the international nature of the cohort.

3.4.5 Limitations

This study has a number of limitations. First, estimating electrolyte excretion from a single morning urine specimen is not the gold standard. Seven or more urine collections or dietary recalls may be necessary to accurately estimate usual intake[121], although a recent study suggest that there are rhythmic changes in sodium excretion and retention that are independent of blood pressure, body water and dietary intake[122]. The method used to estimate sodium and potassium intake is a simple, objective method, previously validated for use in patients with[137] and without CKD[131]. Although lacking in precision for individuals, the method provides a measure of intake feasible for large-scale studies[121] and previous studies reported overall consistency with 24-hour urine collections[26,136]. While a strong effect for sodium excretion on renal outcomes was not identified, potassium excretion, assessed with the same methodology, yielded robust results.

Second, this cohort includes only participants at high cardiovascular, but relatively low renal risk, who were recruited into a randomized controlled trial. Ideally our outcome measure would be patient important, such as dialysis or transplantation, but there were a moderate number of these events during follow-up. As a result, the outcomes are predominantly derived from changes of eGFR, and proteinuria, which are surrogate outcomes. Third, measurement of serum creatinine and potassium were performed locally. However, as there was no inter-laboratory variation for study participants and the outcomes are based on proportional changes from the 6-week value, it is unlikely that the results are affected by the testing method. Fourth, no interactions between sodium excretion and other variables were observed; however it is possible that even in studies as large as this, such analyses may be statistically underpowered. Fifth, as this study is observational in nature, conclusions about association may be drawn but not causation. Finally, it is not possible to generalize these findings to patients with more severe CKD, for example, to patients with very low GFR or nephrotic-range proteinuria.

3.4.6 Conclusions

This observational analysis can generate hypotheses but is not the optimal study design to inform treatment or guidelines. However, the data do not support recommendations to reduce sodium intake substantially below 3 g/day in order to protect the kidney – recommendations largely based on observational data. Even more important, the data speak against a general recommendation of a low potassium diet in patients with CKD and certainly not in those with serum potassium values well within the normal range. However, there were a limited number of participants with low eGFR or macroalbuminuria. Randomized trials of healthy diets, or sodium or potassium intake, in patients with CKD (in particular with lower eGFR), powered for kidney outcomes, are needed. The feasibility of a diet trial was recently exemplified for cardiovascular outcomes[151] and the strong circumstantial evidence for potassium-rich diet from this and other studies[145] supports this approach.

Chapter 4: Diet and Major Renal Outcomes: The NIH-AARP Diet and Health Study

Smyth A, Canavan M, Griffin M, O'Donnell MJ. Diet and Major Renal Outcomes: The NIH-AARP Diet and Health Study. Presented at American Society of Nephrology Kidney Week 2014 Annual Meeting, Philadelphia, PA, USA, November 13th 2014

4.1 Introduction

While dietary modifications are a mainstay of management for patients with advancing CKD and ESRD, studies examining the association of diet factors with renal outcomes in the general population are limited. Reasons to account for deficits in our knowledge include a lack of large prospective cohort studies that measure clinical renal outcomes, the under-recognition of CKD[152], the slow nature of CKD progression and the competing risk of other events related to diet quality, namely cardiovascular death. Moreover, for some specific dietary recommendations (e.g. salt intake), the evidence to support current guidelines is insufficient and conflicting[39].

In general, it is assumed that observations of the association between diet and CVD may also hold true for CKD. However, these assumptions may not be correct, as the association between dietary factors and renal outcomes may differ to associations with CVD. For example, increased potassium intake is recommended to reduce the risk of CVD, but potassium restriction is often recommended to patients with advanced CKD, as potassium excretion is reduced and high serum potassium may have adverse effects[37,38]. A further complication is that higher dietary quality is associated with higher potassium intake[153], meaning that a dietary recommendation for low potassium intake may have adverse effects on overall diet quality. Alternatively, adopting a high quality diet, rich in potassium, may have differing associations with CVD and CKD.

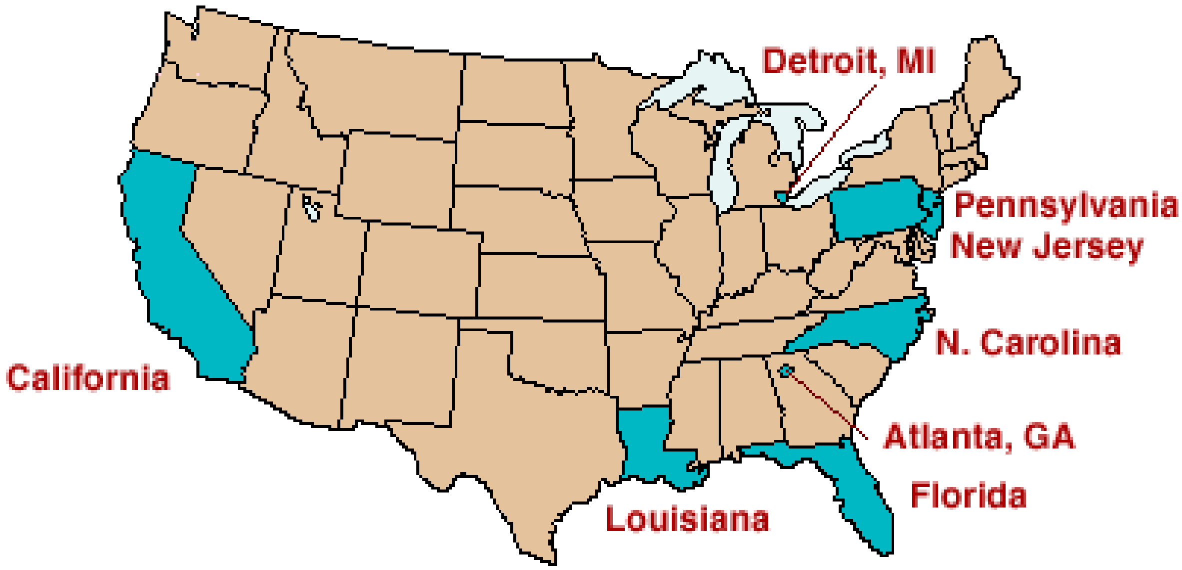
A key limitation of prior studies in this area is that almost all have relied on surrogate markers (including eGFR[145,154] and proteinuria[145,154,155]) rather than the most clinically relevant outcomes such as death from renal disease or the need for dialysis. The objective of this prospective cohort study was to report the associations between diet quality, sodium intake and potassium intake and clinical renal endpoints (including the incidence of death due to a renal cause and of dialysis).

4.2 Methodology

4.2.1 Study Population

In these analyses, data from the U.S. National Institutes of Health American Association of Retired Persons (NIH-AARP) Diet and Health Study were used[156]. This prospective cohort study, primarily designed to investigate the association between diet and cancer, was established in 1995-6 when 18% of AARP members (567,169 of 3.5 million people invited), living in six states (California, Florida, Louisiana, New Jersey, North Caroline and Pennsylvania) and two metropolitan areas (Atlanta, Georgia and Detroit, Michigan) (Figure 4.1), completed questionnaires eliciting information on demographics, anthropometrics, dietary intake and health behaviours. The original design of this study was to include a two-stage sampling frame: (i) collection of baseline diet intake for a large pool of respondents and (ii) selection of a cohort for follow-up based on respondent characteristics, to specifically include over-sampling of those with extremes of dietary intake[156]. However, this was changed as the actual response rate to the baseline questionnaire was low (18%) and the range of intake in respondents was wider than anticipated. Records were excluded for duplicate questionnaires (n=179), persons who moved from the study area (n=321), records not completed by the intended respondent (n=16,021), participants that required dialysis at baseline (n=1,278) and those with extreme high and low total energy intakes (identified as >2-times the inter-quartile ranges of sex-specific log transformed calories per day, using Box-Cox methodology) (n=4,998). The cohort used in these analyses comprised 544,635 people (321,640 men and 222,995 women), representing 96.2% of the entire cohort. The Special Studies Institutional Review Board of the U.S. National Cancer Institute approved the NIH-AARP Diet and Health Study.

Figure 4.1 Areas included in the NIH-AARP Study



Areas included highlighted

4.2.2 Dietary assessment

At baseline, study participants completed a validated[159] 124-item FFQ to assess dietary intake over the preceding year. The FFQ sought information on the frequency of intake of 124 food items and on portion sizes for 116 food items. The validation study was carried out to explore the extent of measurement error, which can be substantial when using FFQ to assess diet intake[160] and may reduce power to detect diet-disease associations[161,162]. A stratified sampling strategy, based on FFQ-derived estimates of intake of fat, fruit and vegetables, was used to identify a cohort of 1,953 individuals who completed two, non-consecutive, unannounced 24-hour diet recalls, randomly assigned by day of the week that were separated by a median of 21 days, to capture usual intake. The 24-hour diet recalls involved recalling all foods and beverages consumed on the day before the interview, using a food probe list of over 100 food categories. The calibration cohort was representative of the overall study cohort (with respect to demographics and environmental exposures (such as smoking, BMI, physical activity)). Correlations between the FFQ and diet recalls for 26 nutrients ranged between 0.36 and 0.76, similar to other large prospective cohort studies, showing the validity of the FFQ.

The food and nutrient variables were merged with the MyPyramid Equivalents Database (version 1.0) to calculate pyramid equivalents for dietary constituents[163]. Pyramid equivalents were calculated for grains, vegetables, fruit, dairy, meat and beans, oils, solid fat, added sugars, and alcohol and nutrient estimates were created for saturated fat (SFA), polyunsaturated fat (PUFA), monounsaturated fat (MUFA), trans fat, cereal fibre, sodium and potassium. Using the pyramid equivalents and other variables, component and index scores were calculated for the Alternate Healthy Eating Index (AHEI)[164], Healthy Eating Index (HEI) 2010[165], Mediterranean Diet Score (MDS)[166,167] and Recommended Food Score (RFS)[168] (Table 4.1). Information from the FFQ was used to estimate sodium and potassium intake. Sodium intake was truncated at 0.5-10g/day and potassium intake at 7.5g/day.

The AHEI was designed to reflect the Healthy Eating Pyramid[169] and scores nine components to generate a total of 87.5 points. Eight components are worth ten points (vegetables, fruit, nuts and soy, white:red meat ratio, trans fat, PUFA:SFA ratio, cereal fibre and alcohol) and multivitamin use is scored as 7.5 for regular intake or 2.5 for intake less than every other day.

The HEI 2010 was designed to reflect the Dietary Guidelines for Americans 2010[170] and scores 12 components, adjusted for energy, to generate a total of 100 points. Six components are worth five points (total fruit, whole fruit, total vegetables, greens and beans, total protein foods and seafood and plant proteins), five components are worth ten points (whole grains, dairy, fatty acids, refined grains and sodium) and one component is worth twenty points (empty calories).

The MDS was designed to reflect the key components of a Mediterranean diet pattern and assesses nine components to generate a total of nine points. Sex-specific medians, based on the energy-adjusted distribution of intake within a given population, are used to generate scores. One point is given for intake at or greater than the median for whole grains, vegetables, fruit, fish, legumes and nuts. One point is given for intake at or less than the median for red and processed meat and the MUFA:SFA ratio.

The RFS is based on the intake of 45 specified foods, defined by dietary guidelines, to generate a total score of 45 points. One point is given for at least weekly consumption of each food listed including whole-grain read/cereal items, vegetable items, fruit items, milk items and lean-meat items. The score used in these analyses is larger than the RFS designed by Kant, because the FFQ used by the NIH-AARP included 124 items compared to 62-items in the FFQ used by Kant[168].

Table 4.1 Components and Optimal Scores for Each Component of the HEI 2010, AHEI, MDS and RFS

PUFA=polyunsaturated fatty acids; MUFA = monounsaturated fatty acids; SFA = saturated fatty acids

*Each component of the Mediterranean Diet Score is adjusted for energy intake; Median values calculated separately for males and females

	Alternate Healthy Eating Index[164]	**Healthy Eating Index 2010[165]**	**Mediterranean Diet Score*[166,167]**	**Recommended Food Score[168]**
Fruit				
Total	≥2 cups/day	≥0.8 cups/1,000 kcal	≥Median	15 Items
Whole		≥0.4 cups/1,000 kcal		
Vegetables				
Total	≥2.5 cups/day	≥1.1 cups/1,000 kcal	≥Median (excl. potatoes)	18 Items
Greens and Beans		≥0.2 cups/1,000kcal		
Grains				
Whole		≥1.5 ounces/1,000 kcal	≥Median	5 items
Refined		≤1.8 ounces/1,000 kcal		
Dairy		≥1.3 cups/1,000 kcal		2 Items
Proteins				
Total		≥2.5 ounces/1,000 kcal		5 items
Seafood & Plant		≥0.8 ounces/1,000 kcal		
White:Red Meat	Ratio=4			
Red/Processed Meat			<Median	
Fish			>Median	
Nuts & Soy	1 ounce/day		>Median	
Fats & Fatty Acids	PUFA:SFA ≥1	(PUFA+MUFA)/SFA > 2.5	MUFA:SFA <Median	
Sodium		≤1.1g /1,000 kcal		
Fibre	Cereal Fibre 15g/day			
Empty Calories		≤19% of energy		
Alcohol	1.5-2.5 drinks/day (males) 0.5-1.5 drinks/day (females)		5-25g/day	
Multivitamins	Regular use			

4.2.3 Follow-Up and Outcome Measures

The primary outcome measure for these analyses was a composite of renal death (censored at 31/12/2011) and dialysis (self-reported and censored 2004, when the follow-up questionnaire was completed by 329,728 (60.5%)). Vital status was ascertained by annual linkage to the Social Security Administration Death Master File (SSA DMF) and follow-up searches of the National Death Index (NDI) were used to determine the contributing cause of death. Death due to a renal cause was defined as death where renal disease was listed as the primary or contributing cause of death, based on International Classification of Diseases (ICD) coding[171] (Table 4.2). Due to the long duration of study follow-up, both ICD-9 and ICD-10 were used. Participants were invited to complete a follow-up questionnaire in 2004, which included the question "Have you ever been told by a doctor that you had any of the following conditions?" followed by a list including "Renal disease requiring dialysis", in two time periods (1995-1999 and 2000-2004)[172]. Secondary outcomes included death due to a renal cause and dialysis.

Table 4.2 International Classification of Diseases to Identify Renal Deaths

Code	Label
ICD-9	
585	Chronic kidney disease
586	Renal failure, unspecified
ICD-10	
N180	Chronic kidney failure, unspecified
N183	Chronic kidney disease, stage 3
N185	Chronic kidney disease, stage 5
N188	Chronic kidney disease, unspecified
N189	Chronic kidney disease, unspecified
N19	Unspecified kidney failure

4.2.4 Statistical Analysis

Baseline descriptive statistics of the cohort categorized by quintile of each of the diet variables (AHEI, HEI, MDS, RFS, sodium and potassium intakes) were compared using the chi-square test or analysis of variance, as appropriate. Pearson's correlation coefficients were calculated to compare the four diet indexes and to compare sodium and potassium intakes. Competing risks regression, according to the method of Fine and Gray[141], was used to calculate the sub-hazard ratio (SHR) and 95% confidence intervals (CI) for the primary, composite outcome. First, restricted cubic splines and a Wald-type test to test for non-linear relationships (i.e. non-linearity beyond the first knot) with the dietary variables were used[173]. Next, analyses by quintile of diet variables with the lowest quintile as the reference category were completed. In order to account for differing lengths of follow-up for components of the composite outcome, the date of dialysis events was imputed as the midpoint of the reported time periods (i.e. July 1st 1997 for the period 1995-1999 and July 1st 2002 for the period 2000-2004). For the secondary outcomes, competing risks regression was used to calculate the sHR for death due to a renal cause, but [141]as the exact date of dialysis events was not available, multinomial logistic regression was used to calculate the relative risk ratio (RRR) for dialysis. Death from a non-renal cause was considered to be the competing event in all models.

To further explore the effect of potential confounders, two models are reported: (i) unadjusted and (ii) fully adjusted (age, sex, BMI, smoking, education, ethnicity, activity levels, diabetes and CVD). Age and BMI were included as continuous variables; categorical variables included sex, smoking (never (reference), former or current smoking), education (<12 years (reference), high school, post-high school, some college or college and postgraduate), ethnicity (non-Hispanic white (reference), non-Hispanic Black, Hispanic or Asian), physical activity (never (reference), rarely, 1-3 times per month, 1-2 times per week, 3-4 times per week or >4 times per week), diabetes and CVD (self-reported). For nutrient-specific models (sodium and potassium) we also adjusted for the other nutrient (i.e. potassium in the sodium model and sodium in the potassium model). Tests for interactions, defined *a priori,* between different diet variables and between the diet variables and categories of age (dichotomized at 63 years (median)), sex, BMI (dichotomized at 30kg/m^2) and diabetes were completed using logistic models. As sensitivity analyses, first, participants who reported dialysis or those who died from a renal cause during the follow-up period (1995-9) were excluded, to address the potential for reverse causation (i.e. excluded early events). Next, analyses were completed to determine if the addition of hypertension as a covariate modified associations. Finally, as diabetes and CVD may be on the causal pathway, analyses exploring these variables from the model were performed to determine if associations were materially modified.

4.3 Results

Of 544,635 participants included in the analyses, all were followed up for mortality to 31/12/2011 (mean follow-up 14.3 years) and 329,728 (60.5%) for self-reported dialysis up to 2004. Overall mean age was 62.2 (5.4) years and 59.1% (n=321,640) were male.

4.3.1 Cross-Sectional Analyses

The four scores of diet quality were correlated, with similar correlations in men (Figure 4.2) and women (Figure 4.3). The correlations were strongest between AHEI, HEI and MDS, all of which were weakly correlated with RFS. Sodium and potassium intakes were similarly correlated in men and women (Figure 4.4).

Compared to the lowest quintile, those in the highest quintiles of AHEI, HEI and MDS were more likely to be older, more active, non-smokers, who have completed a higher level of formal education and have a history of heart disease. Those in the highest quintile of RFS were more likely to be older, more active, non-smokers and have a history of heart disease. Those in the highest quintiles of sodium and potassium intake were more likely to be younger, more active, male, and have a history of diabetes, hypertension, high cholesterol and heart disease (Table 4.3)

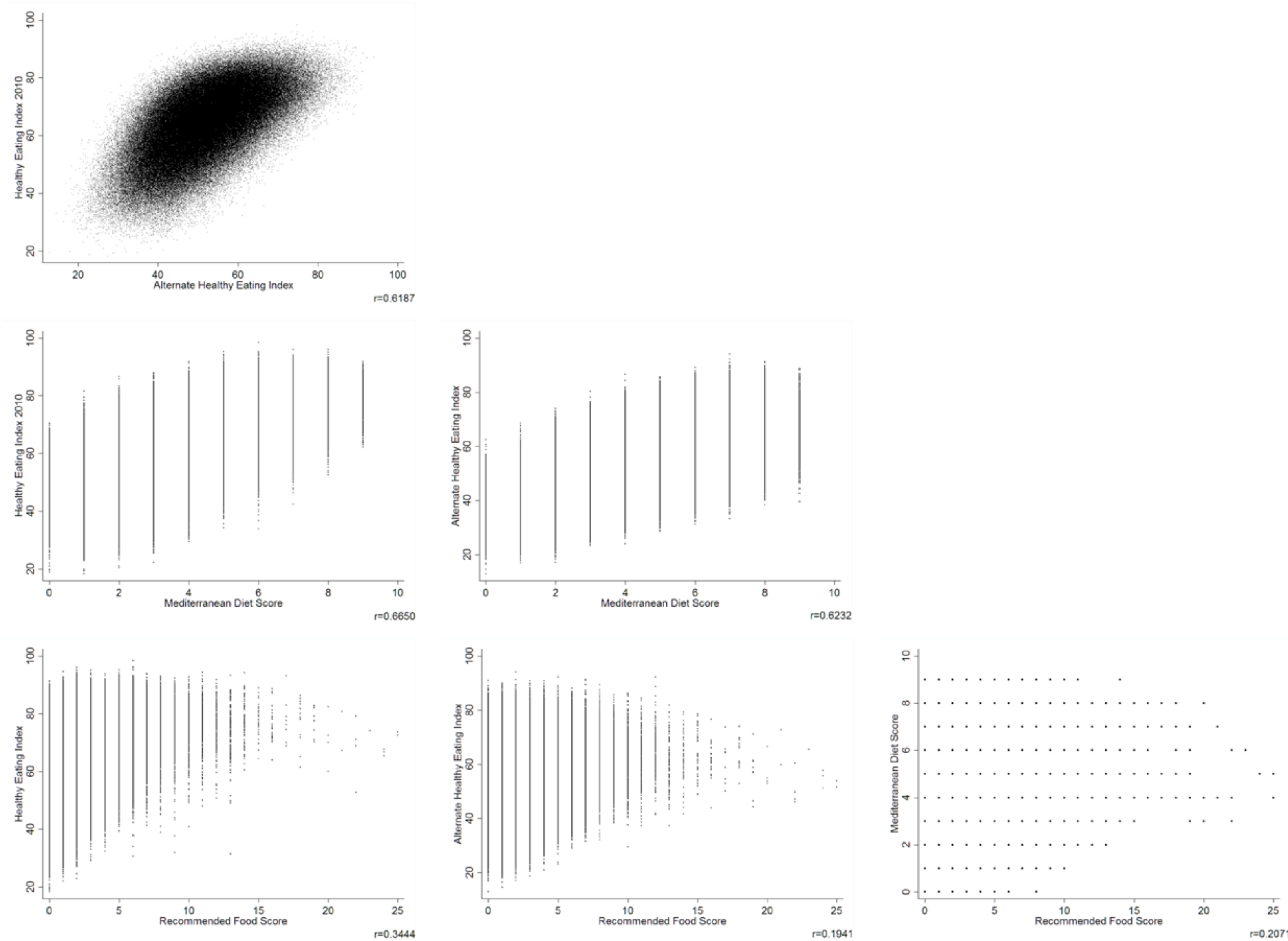


Figure 4.2 Scatterplot Matrix Comparing Diet Indexes in Men

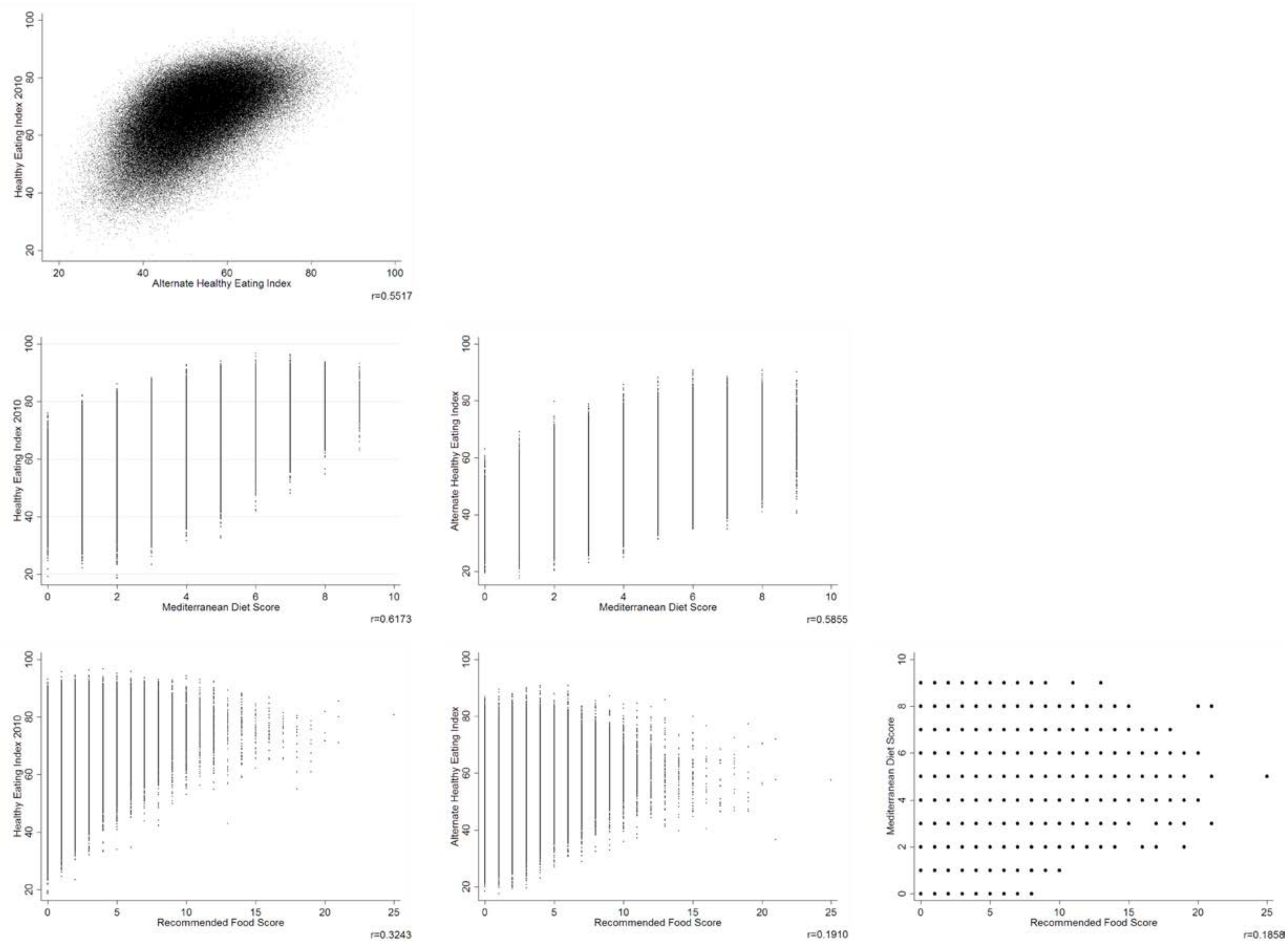


Figure 4.3 Scatterplot Matrix Comparing Diet Indexes in Women

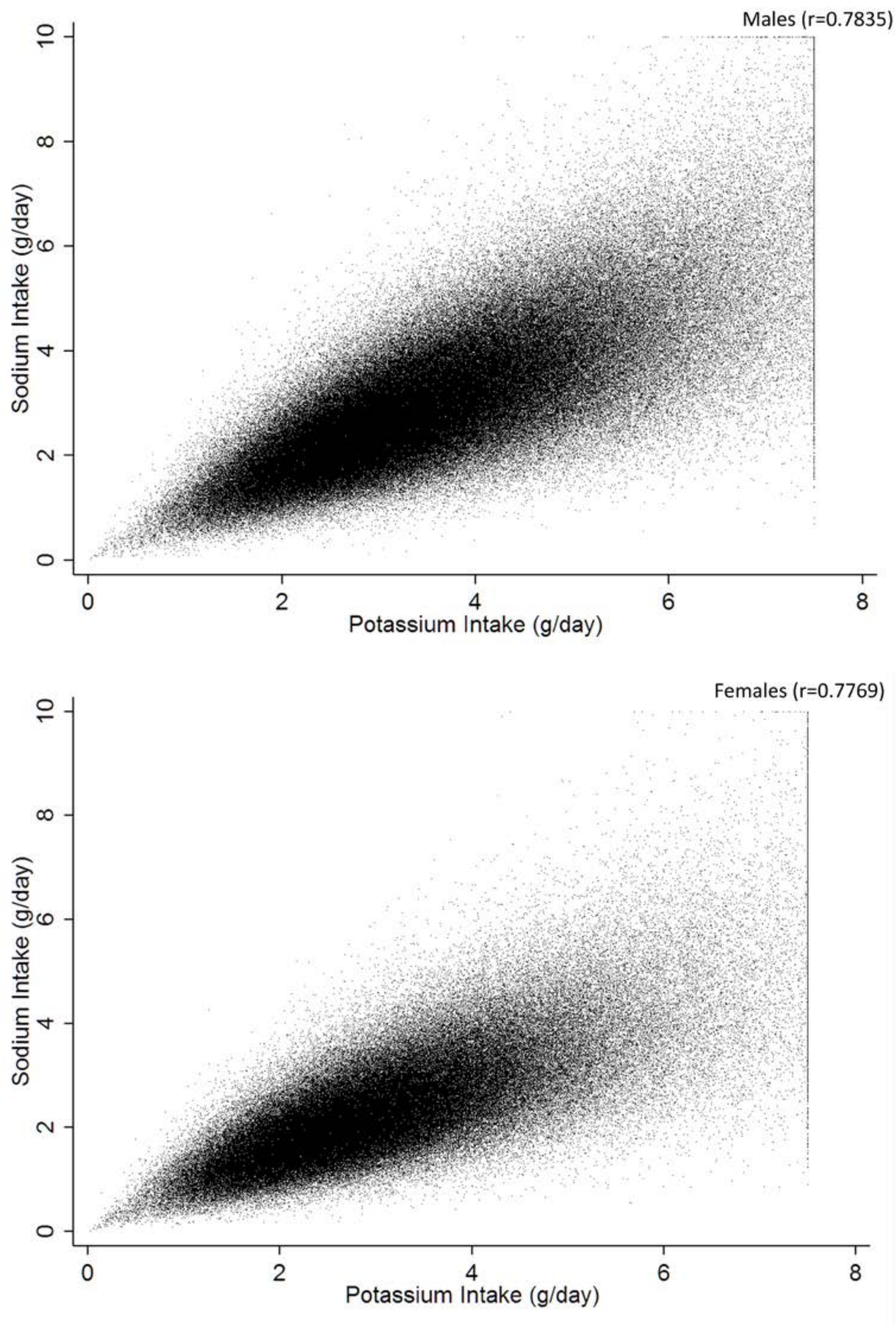


Figure 4.4 Correlation between sodium and potassium intakes

Table 4.3 Cross-sectional Analyses

		Alternative Health Eating Index[164]			Healthy Eating Index 2010[165]		
Characteristic	**Overall**	**Q1**	**Q5**	**p**	**Q1**	**Q5**	**p**
Total (n)	544,635	108,927	108,927	-	108,927	108,927	-
Mean Diet Index (SD)	-	39.1 (4.0)	66.3 (4.7)	<0.001	49.7 (6.0)	79.6 (3.3)	<0.001
Age, (SD) years	62.2 (5.4)	61.9 (5.5)	62.4 (5.3)	<0.001	61.5 (5.5)	62.9 (5.2)	<0.001
Male, % (n)	59.1% (321,640)	62.7% (68,240)	59.1% (64,395)	<0.001	68.1% (74,203)	50.9% (55,462)	<0.001
White, % (n)	92.7% (497,922/537,477)	93.0% (99,770)	93.0% (100,046)	<0.001	93.1% (99,975)	92.3% (99,326)	<0.001
Education, % (n)							
<=11 years	6.2% (32,499/528,656)	8.7% (9,195)	3.8% (4,015)	<0.001	9.3% (9,848)	4.2% (4,419)	<0.001
12yrs/High school	20.1% (106,253/528,656)	26.2% (27,575)	13.7% (14,499)		25.9% (27,253)	16.1% (17,020)	
Post High school	10.1% (53,402/528,656)	11.4% (12,036)	8.2% (8,745)		11.5% (12,088)	9.0% (9,557)	
Some college	23.9% (126,492/528,656)	23.2% (24,437)	23.6% (25,021)		23.8% (25,062)	23.5% (24,915)	
College/Postgraduate	39.7% (210,073/528,656)	30.5% (32,195)	50.8% (53,902)		29.5% (31,136)	47.3% (50,096)	
Smoking Status, % (n)							
Never	36.2% (189,803/524,026)	34.4% (36,044)	36.1% (37,811)	<0.001	28.0% (29,306)	42.1% (44,302)	<0.001
Former	51.5% (269,844/524,026)	46.5% (48,675)	57.0% (59.756)		47.9% (50,028)	52.2% (54,950)	
Current	12.3% (64,599/524,206)	19.1% (20,015)	6.9% (7,273)		24.1% (25,181)	5.7% (5,941)	
BMI, mean (SD) kg/m^2	27.1 (5.1)	27.5 (5.3)	26.4 (4.9)	<0.001	27.4 (5.3)	26.3 (4.8)	<0.001
Physical Activity >20min in past year, % (n)							
Never	4.6% (24,748/538,735)	7.5% (7,999)	2.4% (2,647)	<0.001	8.2% (8,789)	2.4% (2,610)	<0.001
Rarely	13.8% (74,573/538,735)	19.6% (21,095)	8.3% (8.949)		20.8% (22,310)	8.4% (9,032)	
1-3 times/ month	13.6% (73,427/538,735)	16.5% (17,749)	9.8% (10,609)		16.8% (18,101)	9.7% (10,489)	
1-2 times/week	21.7% (116,718/538,735)	22.1% (23,753)	19.4% (20,993)		21.5% (23,145)	19.6% (21,180)	
3-4 times/week	27.0% (145,306/538,735)	20.7% (22,219)	32.5% (35,086)		19.2% (20,668)	33.6% (36,272)	
>=5 times/week	19.3% (103,963/538,735)	13.6% (14,614)	27.5% (29,737)		13.4% (14,471)	26.3% (28,414)	
Baseline Comorbidities, % (n)							
Diabetes	9.2% (49,824/544,635)	8.6% (9,403)	8.6% (9,357)	<0.001	7.0% (7,650)	9.6% (10,479)	<0.001
Hypertension	43.5% (127,711/293,693)	45.5% (25,708)	40.2% (24,593)	<0.001	43.5% (24,147)	42.1% (26,084)	<0.001
Heart disease	14.0% (76,368/544,635)	13.4% (14,635)	14.4% (15,697)	<0.001	12.2% (13,309)	15.6% (16,958)	<0.001
Stroke	2.1% (11,591/544,635)	2.5% (2,747)	1.7% (1,827)	<0.001	2.4% (2,626)	1.9% (2,066)	<0.001

Table 4.3 (continued). Cross-sectional Analyses

		Mediterranean Diet Score[166,167]			Recommended Food Score[168]		
Characteristic	**Overall**	**Q1**	**Q5**	**p**	**Q1**	**Q5**	**p**
Total (n)	544,635	194,846	49,636	-	151,939	64,655	-
Mean Diet Index (SD)	-	2.3 (0.8)	7.3 (0.5)	<0.001	0 (0)	5.3 (1.8)	<0.001
Age, (SD) years	62.2 (5.4)	61.8 (5.4)	62.6 (5.2)	<0.001	61.5 (5.4)	63.2 (5.2)	<0.001
Male, % (n)	59.1% (321,640)	58.9% (114,817)	64.5% (32,013)	<0.001	59.9% (91,006)	57.3% (37,055)	<0.001
White, % (n)	92.7% (497,922/537,477)	93.5% (179,494)	92.8% (45,588)	<0.001	91.5% (136,898)	90.8% (57,858)	<0.001
Education, % (n)							
<=11 years	6.2% (32,499/528,656)	8.0% (15,066)	3.3% (1,598)		6.6% (9,719)	6.5% (4,053)	
12yrs/High school	20.1% (106,253/528,656)	24.6% (46,311)	12.8% (6,201)		21.0% (30,920)	19.0% (11,868)	
Post High school	10.1% (53,402/528,656)	11.0% (20,815)	8.1% (3,937)	<0.001	10.4% (15,336)	9.5% (5,900)	<0.001
Some college	23.9% (126,492/528,656)	24.2% (45,665)	22.6% (10,950)		25.9% (38,086)	21.9% (13,679)	
College/Postgraduate	39.7% (210,073/528,656)	32.2% (60,773)	53.3% (25,854)		36.1% (53,121)	43.1% (26,922)	
Smoking Status, % (n)							
Never	36.2% (189,803/524,026)	32.6% (61,045)	40.6% (19,468)		32.2% (47,097)	42.4% (26,281)	
Former	51.5% (269,844/524,026)	48.6% (91,008)	55.0% (26,374)	<0.001	50.4% (73,633)	51.5% (31,907)	<0.001
Current	12.3% (64,599/524,206)	18.8% (35,180)	4.5% (2,141)		17.4% (25,354)	6.0% (3,740)	
BMI, mean (SD) kg/m^2	27.1 (5.1)	27.5 (5.3)	26.1 (4.3)	<0.001	27.4 (5.2)	26.3 (5.1)	<0.001
Physical Activity >20min in past year, % (n)							
Never	4.6% (24,748/538,735)	6.8% (13,039)	1.6% (809)		5.3% (7,951)	4.0% (2,556)	
Rarely	13.8% (74,573/538,735)	18.4% (35,351)	6.8% (3,329)		16.6% (24,887)	9.7% (6,202)	
1-3 times/ month	13.6% (73,427/538,735)	16.2% (31,060)	8.9% (4,403)	<0.001	16.1% (24,117)	8.9% (5,704)	<0.001
1-2 times/week	21.7% (116,718/538,735)	22.0% (42,381)	19.7% (9,710)		22.9% (34,302)	17.3% (11,031)	
3-4 times/week	27.0% (145,306/538,735)	21.9% (42,116)	35.1% (17,292)		24.6% (36,919)	29.2% (18,655)	
>=5 times/week	19.3% (103,963/538,735)	14.8% (28,425)	27.9% (13,763)		14.6% (21,974)	30.8% (19,658)	
Baseline Comorbidities, % (n)							
Diabetes	9.2% (49,824/544,635)	9.0% (17,568)	7.3% (3,638)	<0.001	8.0% (12,114)	10.9% (7,073)	<0.001
Hypertension	43.5% (127,711/293,693)	44.3% (45,152)	40.5% (11,513)	<0.001	42.7% (33,697)	43.6% (15,514)	<0.001
Heart disease	14.0% (76,368/544,635)	12.4% (24,159)	16.9% (8,367)	<0.001	12.5% (19,010)	15.6% (10,052)	<0.001
Stroke	2.1% (11,591/544,635)	2.4% (4,605)	1.7% (831)	<0.001	2.1% (3,187)	2.3% (1,454)	0.006

Table 4.3 (continued). Cross-sectional Analyses

		Sodium Intake			Potassium Intake		
Characteristic	**Overall**	**Q1**	**Q5**	**p**	**Q1**	**Q5**	**p**
Total (n)	544,635	108,927	108,927	-	108,928	108,925	-
Mean Intake (SD) mg/day	-	1354.7 (293.7)	4690.0 (1116.1)	<0.001	1801.3 (364.0)	5519.5 (1221.3)	<0.001
Age, (SD) years	62.2 (5.4)	62.4 (5.4)	61.7 (5.4)	<0.001	62.2 (5.4)	61.9 (5.4)	<0.001
Male, % (n)	59.1% (321,640)	39.9% (43,414)	76.9% (83,770)	<0.001	45.9% (49,949)	70.3% (76,547)	<0.001
White, % (n)	92.7% (497,922/537,477)	89.8% (95,793)	92.4% (99,447)	<0.001	89.6% (95,672)	92.0% (98,856)	<0.001
Education, % (n)							
<=11 years	6.2% (32,499/528,656)	6.5% (6,802)	7.8% (8,177)		6.6% (6,930)	7.2% (7,574)	
12yrs/High school	20.1% (106,253/528,656)	22.6% (23,678)	20.0% (21,079)		23.2% (24,331)	19.1% (20,088)	
Post High school	10.1% (53,402/528,656)	10.0% (10,504)	10.4% (10,985)	<0.001	10.0% (10,551)	10.2% (10,724)	<0.001
Some college	23.9% (126,492/528,656)	25.2% (26,480)	23.0% (24,303)		25.1% (26,323)	23.2% (24,431)	
College/Postgraduate	39.7% (210,073/528,656)	35.7% (37,461)	38.8% (40,992)		35.2% (36,954)	40.3% (42,452)	
Smoking Status, % (n)							
Never	36.2% (189,803/524,026)	39.3% (40,987)	32.7% (34,178)		39.5% (41,277)	34.5% (36,052)	<0.001
Former	51.5% (269,844/524,026)	47.6% (49,645)	53.5% (55,904)	<0.001	47.6% (49,673)	52.4% (54,729)	
Current	12.3% (64,599/524,206)	13.1% (13,620)	13.8% (14,365)		12.9% (13,461)	13.0% (13,610)	
BMI, mean(SD) kg/m^2	27.1 (5.1)	26.5 (5.1)	28.0 (5.4)	<0.001	27.0 (5.3)	27.5 (5.3)	<0.001
Physical Activity >20min in past year, % (n)							
Never	4.6% (24,748/538,735)	5.9% (6,298)	4.6% (4,975)		6.6% (7,100)	4.0% (4,289)	
Rarely	13.8% (74,573/538,735)	16.5% (17,606)	12.8% (13,800)		18.4% (19,681)	11.1% (11,989)	
1-3 times/ month	13.6% (73,427/538,735)	15.2% (16,278)	12.2% (13,104)	<0.001	16.5% (17,701)	11.0% (11,797)	<0.001
1-2 times/week	21.7% (116,718/538,735)	21.0% (22,420)	21.4% (23,069)		21.5% (22,974)	20.2% (21,718)	
3-4 times/week	27.0% (145,306/538,735)	25.2% (26,919)	26.6% (28,645)		23.3% (24,908)	28.2% (27,608)	
>=5 times/week	19.3% (103,963/538,735)	16.4% (17,517)	22.5% (24,262)		13.8% (14,762)	25.6% (27,608)	
Baseline Comorbidities, % (n)							
Diabetes	9.2% (49,824/544,635)	7.2% (7,833)	12.7% (13,792)	<0.001	8.6% (9,337)	10.7% (11,635)	<0.001
Hypertension	43.5% (127,711/293,693)	42.2% (24,611)	46.6% (26,248)	<0.001	43.8% (25,116)	44.8% (25,708)	<0.001
Heart disease	14.0% (76,368/544,635)	12.8% (13,910)	15.3% (16,642)	<0.001	13.4% (14,544)	15.0% (16,318)	<0.001
Stroke	2.1% (11,591/544,635)	2.3% (2,456)	2.4% (2,556)	<0.001	2.4% (2,568)	2.2% (2,434)	<0.001

4.3.2 Diet Quality and Outcomes

The composite outcome (death due to a renal cause or dialysis) occurred in n=4,848 participants (71 per 100,000 person-years). Restricted cubic splines showed linear associations between the composite outcome and the AHEI (test for non-linearity p=0.5422, Figure 4.5), HEI (test for non-linearity p=0.6103, Figure 4.6), and MDS (test for non-linearity p=0.3659, Figure 4.7) and the composite outcome, with higher scores associated with reduced hazards. A linear association was also seen between the composite outcome and the RFS, but the association was not statistically significant (test for non-linearity p-=0.0962, Figure 4.8). Compared to the lowest quality diet (quintile 1), the highest quality diet (quintile 5) was associated with a statistically significant reduction in the hazard of the composite for AHEI (SHR 0.71 (95% CI 0.65-0.79)), HEI (SHR 0.82 (95% CI 0.74-0.91)), and MDS (SHR 0.84 (95% CI 0.74-0.95)) (Table 4.4). There was no significant association between RFS and the composite.

Death due to a renal cause occurred in 4,042 participants (52.7 per 100,000 person-years) over a mean follow-up of 14.3 (SD 3.6) years. Compared to the lowest quality diet, there was a significant reduction in the hazard of renal death in the highest quintile of AHEI (SHR 0.67 (95% CI 0.60-0.75)), HEI (SHR 0.78 (95% CI 0.70-0.87)), and MDS (SHR 0.82 (95% CI 0.72-0.94)) (Table 4.4). There was no association between RFS and renal death.

Dialysis was reported by 0.2% (n=946) up to 2004, when the follow-up questionnaire was completed. Compared to the lowest quality diet (quintile 1), estimates were consistent with those for the composite outcome, but not statistically significant, with hazards for dialysis in the highest quintile of AHEI (RRR 0.85 (95% CI 0.67-1.06)), HEI (RRR 0.91 (95% CI 0.71-1.08)), and MDS (RRR 0.82 (95% CI 0.83-1.07)) (Table 4.4). There was no association between RFS and self-reported dialysis.

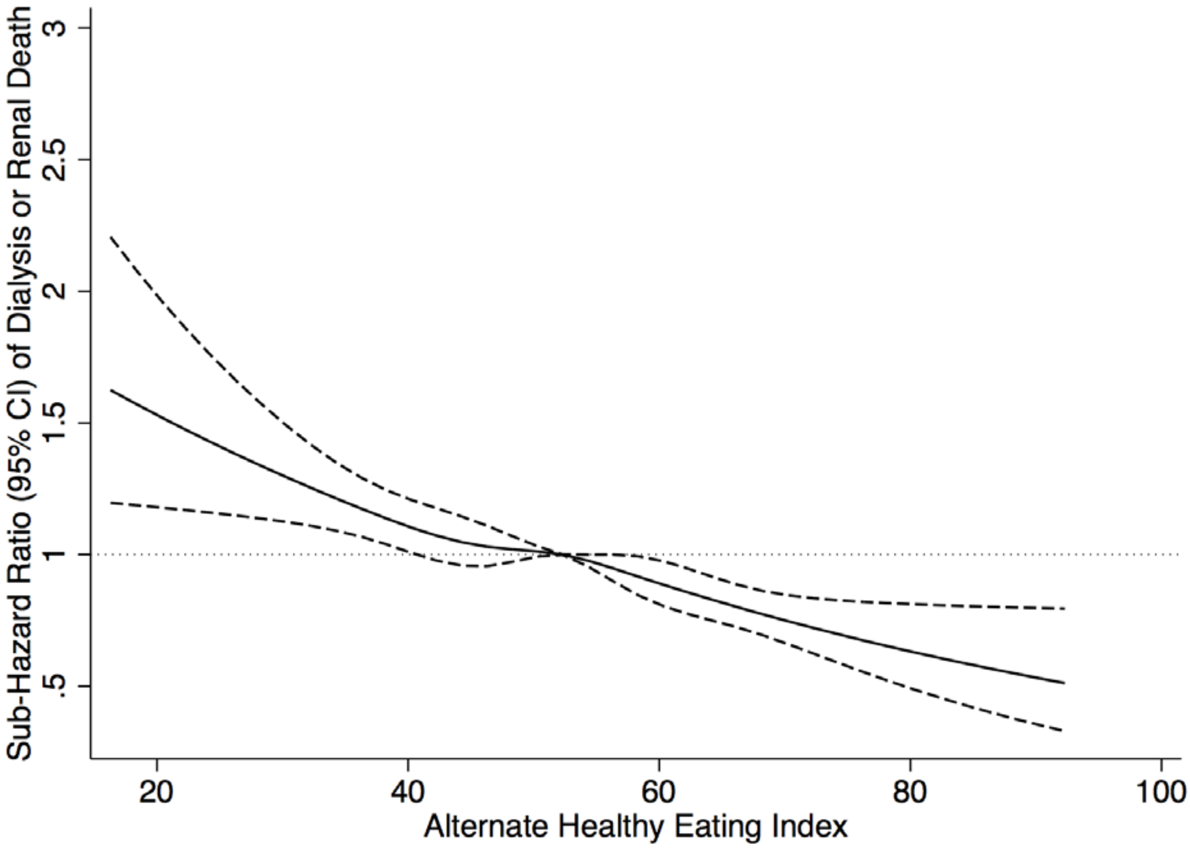


Figure 4.5 Alternate Healthy Eating Index and Composite Outcome

Reference point = median Alternate Healthy Eating Index (52.2); adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke. Wald test for non-linearity, p=0.5422, indicating the relationship is linear.

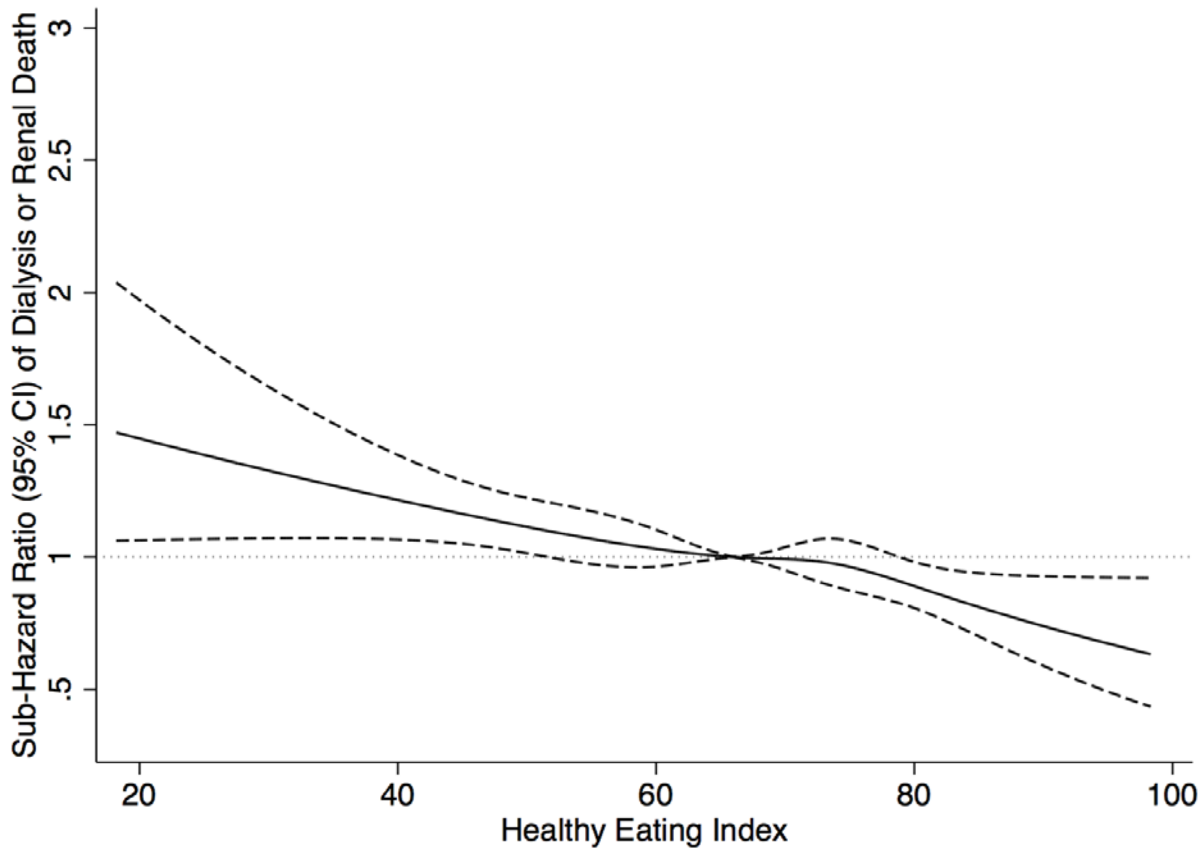


Figure 4.6 Healthy Eating Index and Composite Outcome

Reference point = median Healthy Eating Index (67.2); adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke. Wald test for non-linearity, p=0.6103, indicating the relationship is linear.

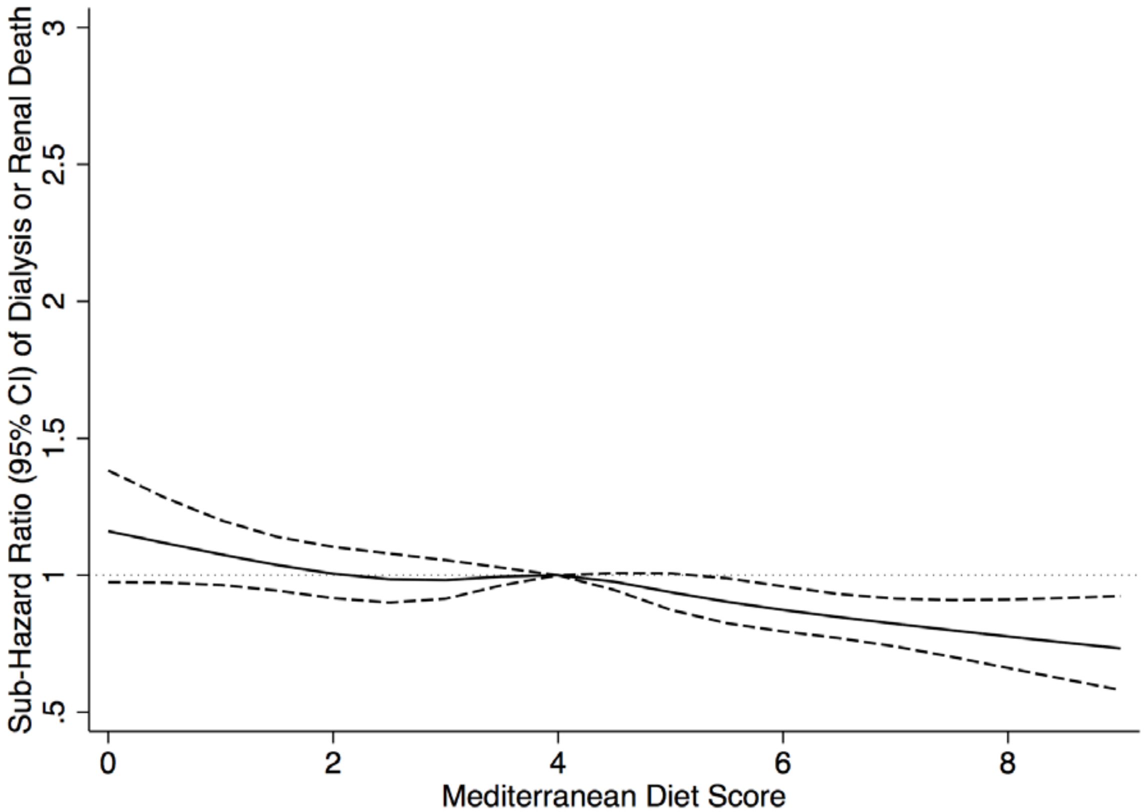


Figure 4.7 Mediterranean Diet Score and Composite Outcome

Reference point = median Mediterranean Diet Score (4); adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke. Wald test for non-linearity, p=0.3659, indicating the relationship is linear.

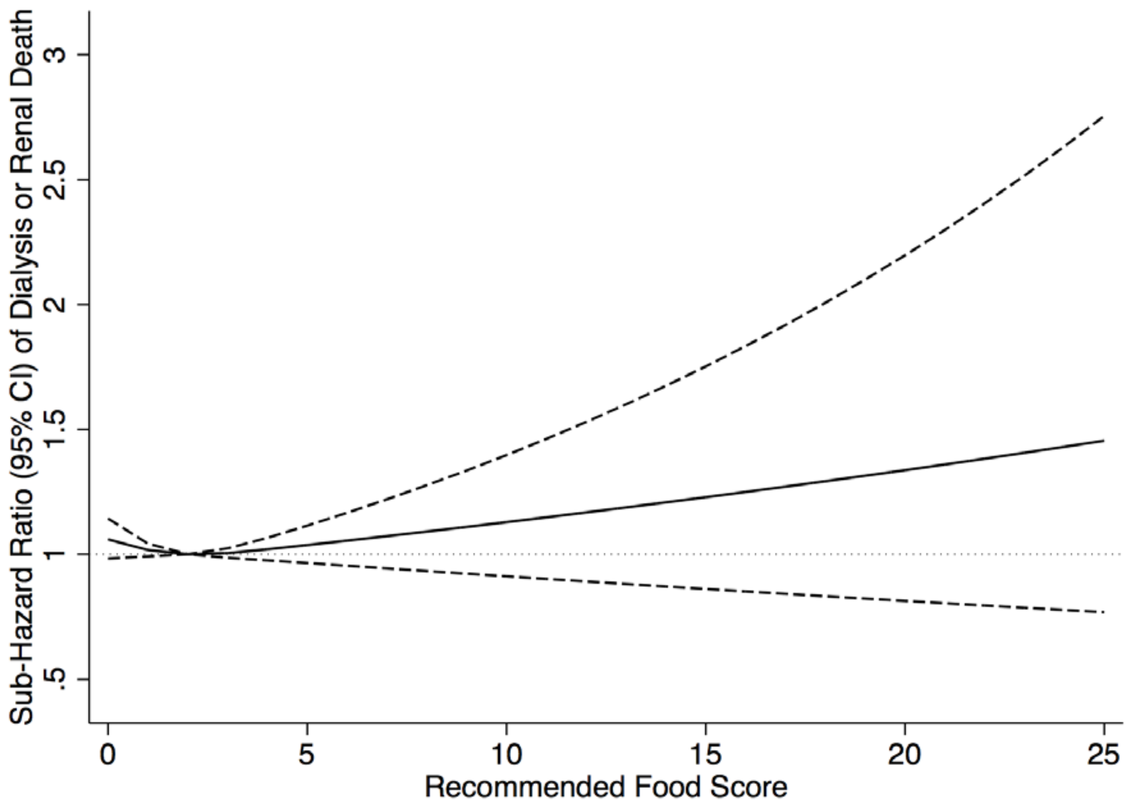


Figure 4.8 Recommended Food Score and Composite Outcome

Reference point = median Recommended Food Score (2); adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke. Wald test for non-linearity, p=0.0962, indicating the relationship is linear.

Table 4.4 Dietary indexes and composite outcome (death due to a renal cause and dialysis)

*Adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke

	Range	Composite Outcome (SHR, 95% CI)		Death due to a Renal Cause (SHR, 95% CI)		Self-Reported Dialysis (RRR, 95% CI)	
		Unadjusted	**Adjusted***	**Unadjusted**	**Adjusted***	**Unadjusted**	**Adjusted***
Healthy Eating index 2010[165]							
Q1	18.2-57.0	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)
Q2	57.0-64.4	0.99 (0.91-1.08)	0.99 (0.90-1.09)	0.97 (0.89-1.07)	0.98 (0.88-1.08)	1.03 (0.84-1.27)	0.98 (0.79-1.22)
Q3	64.4-69.9	0.91 (0.83-0.99)	0.91 (0.83-1.00)	0.87 (0.79-0.96)	0.89 (0.80-0.98)	1.01 (0.82-1.24)	0.99 (0.79-1.23)
Q4	69.9-73.3	0.92 (0.84-1.00)	0.93 (0.85-1.02)	0.85 (0.77-0.94)	0.87 (0.78-0.97)	1.05 (0.85-1.28)	1.05 (0.85-1.31)
Q5	>73.3	0.77 (0.70-0.84)	0.82 (0.74-0.91)	0.71 (0.65-0.79)	0.78 (0.70-0.87)	0.89 (0.73-1.10)	0.91 (0.72-1.14)
p for trend		<0.0001	0.0013	<0.001	<0.001	0.365	0.646
Alternate Healthy Eating Index[164]							
Q1	12.8-44.1	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)
Q2	44.1-49.7	0.91 (0.84-0.99)	0.91 (0.83-1.00)	0.89 (0.81-0.97)	0.89 (0.81-0.98)	0.98 (0.80-1.20)	0.93 (0.75-1.16)
Q3	49.7-54.7	0.86 (0.70-0.94)	0.87 (0.79-0.95)	0.81 (0.74-0.88)	0.82 (0.74-0.91)	1.11 (0.91-1.35)	1.07 (0.87-1.32)
Q4	54.7-60.7	0.78 (0.71-0.85)	0.81 (0.74-0.89)	0.72 (0.66-0.79)	0.77 (0.69-0.85)	0.95 (0.77-1.16)	0.98 (0.79-1.21)
Q5	>60.7	0.62 (0.57-0.68)	0.71 (0.65-0.79)	0.57 (0.51-0.63)	0.67 (0.60-0.75)	0.81 (0.65-0.99)	0.85 (0.67-1.06)
p for trend		<0.0001	<0.001	<0.001	<0.001	0.049	0.289
Mediterranean Diet Score[166,167]							
Q1	0-3	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)
Q2	4	1.01 (0.94-1.09)	1.01 (0.93-1.09)	1.00 (0.92-1.08)	1.01 (0.93-1.10)	0.99 (0.83-1.17)	0.96 (0.80-1.16)
Q3	5	0.86 (0.80-0.93)	0.92 (0.84-1.00)	0.83 (0.76-0.90)	0.89 (0.81-0.98)	0.92 (0.77-1.10)	0.94 (0.77-1.14)
Q4	6	0.80 (0.73-0.87)	0.87 (0.79-0.96)	0.75 (0.68-0.84)	0.82 (0.74-0.92)	0.88 (0.72-1.07)	0.95 (0.76-1.18)
Q5	>6	0.73 (0.65-0.81)	0.84 (0.74-0.95)	0.69 (0.61-0.79)	0.82 (0.72-0.94)	0.74 (0.58-0.95)	0.82 (0.63-1.07)
p for trend		<0.0001	0.0023	<0.001	<0.001	0.012	0.186
Recommended Food Score[168]							
Q1	0 (0)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)
Q2	1 (0)	0.99 (0.92-1.07)	0.97 (0.90-1.05)	1.00 (0.92-1.08)	0.97 (0.89-1.06)	0.89 (0.75-1.05)	0.89 (0.74-1.06)
Q3	2 (0)	0.97 (0.89-1.06)	0.94 (0.86-1.03)	0.98 (0.89-1.07)	0.94 (0.85-1.04)	0.86 (0.71-1.04)	0.84 (0.69-1.04)
Q4	3 (0)	1.05 (0.95-1.16)	0.97 (0.87-1.08)	1.07 (0.96-1.19)	0.97 (0.86-1.10)	0.91 (0.72-1.14)	0.89 (0.69-1.14)
Q5	5.3 (1.8)	1.05 (0.95-1.15)	0.97 (0.87-1.08)	1.02 (0.92-1.14)	0.95 (0.85-1.07)	1.16 (0.95-1.43)	1.08 (0.86-1.35)
p for trend		0.4890	0.7361	0.6847	0.7898	0.366	0.855

4.3.3 Sodium and Potassium Intakes and Outcomes

Restricted cubic splines showed non-linear associations between sodium and potassium intakes and the composite outcome (tests for non-linearity p=0.0027 (Figure 4.9) and p=0.0225 (Figure 4.10), respectively). The highest quintile of sodium intake was associated with an increased hazard (sHR 1.17 (95% CI 1.02-1.33)) but the highest quintile of potassium intake was associated with a reduced hazard (sHR 0.83 (95% CI 0.73-0.95)).

For death due to a renal cause, there was a trend towards an increase in the hazard with the highest quintile of sodium intake (sHR 1.15 (95% CI 0.99-1.33)) and a significant reduction in the hazard with the highest quintile of potassium intake (sHR 0.78 (95% CI 0.67-0.90)).

For dialysis, a similar trend towards increased hazard was seen in the highest quintile of sodium intake (RRR 1.16 (95% CI 0.86-1.56)) and potassium intake (RRR 1.10 (95% CI 0.83-1.48)).

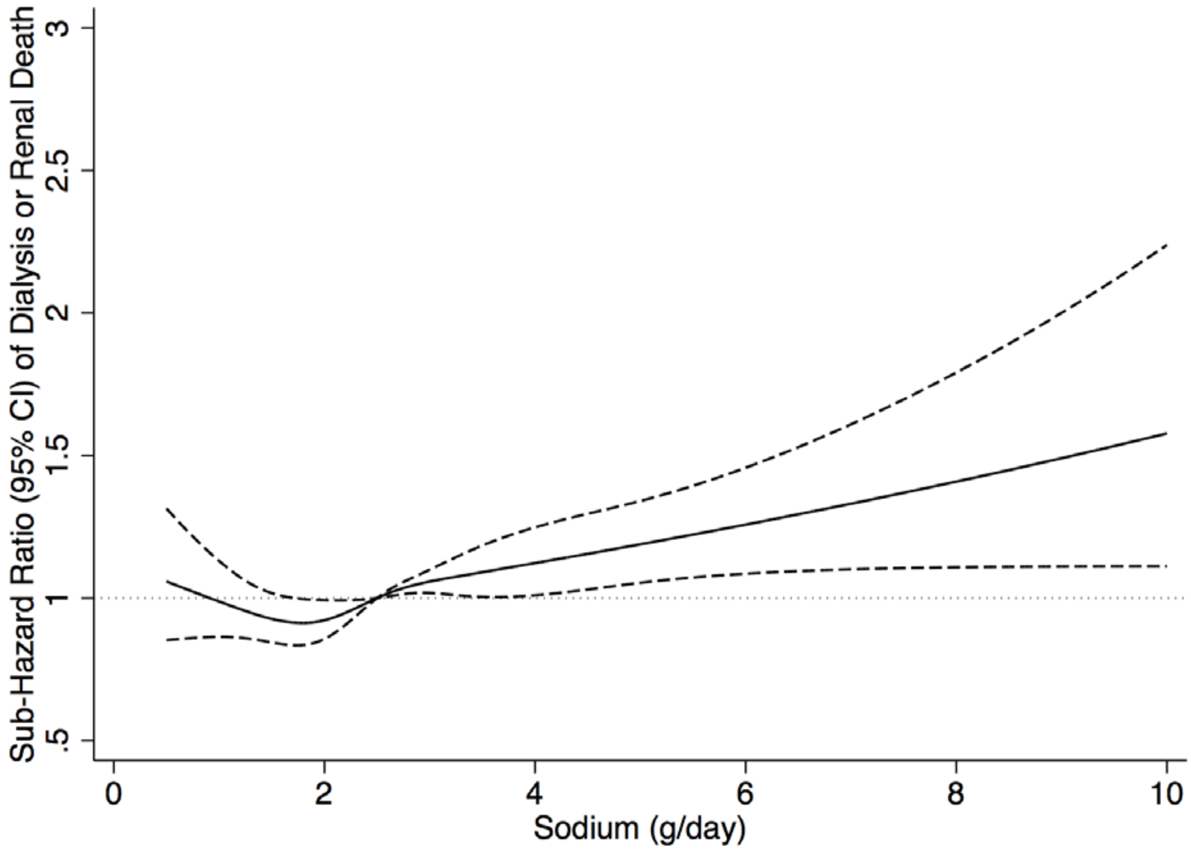


Figure 4.9 Sodium Intake and Composite Outcome

Reference point = median sodium intake (2.5g/day); adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke. Wald test for non-linearity, p=0.0027, indicating the relationship is non-linear.

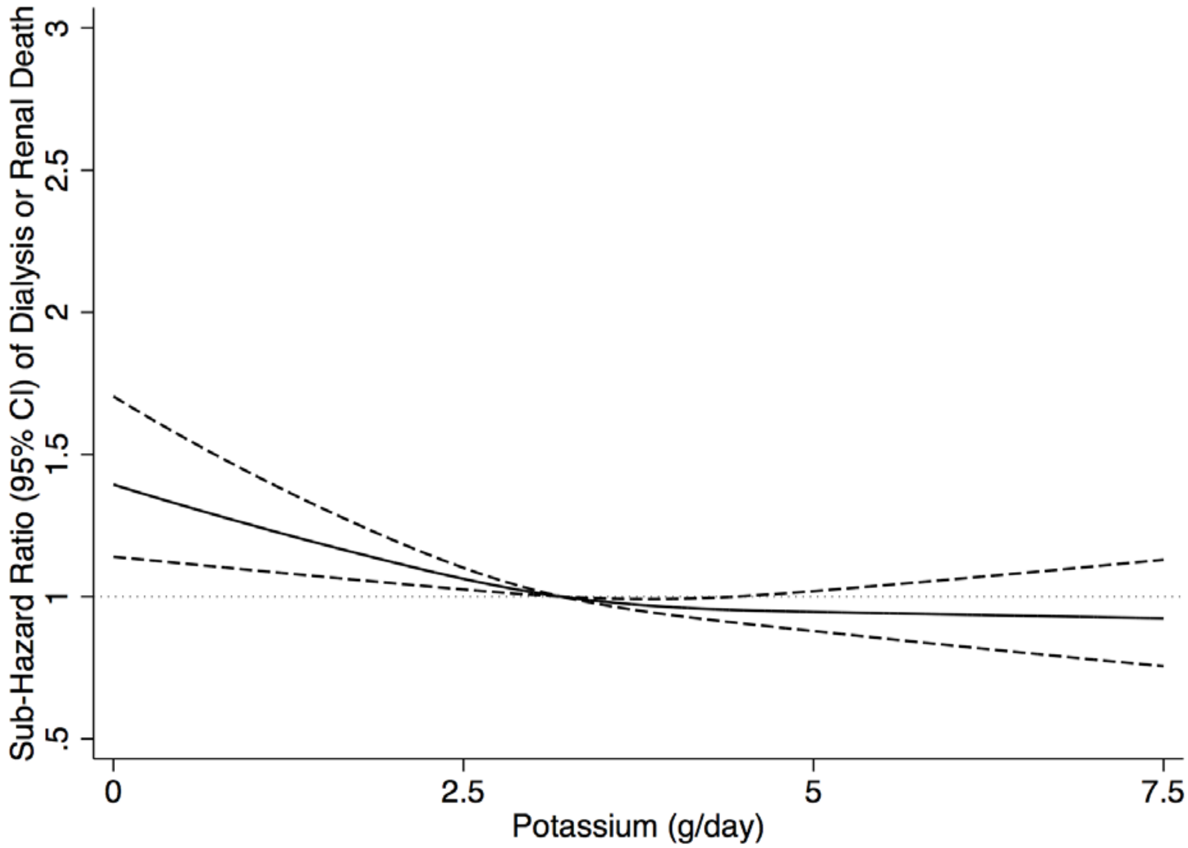


Figure 4.10 Potassium Intake and Composite Outcome

Reference point = median potassium intake (3.2g/day); adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke. Wald test for non-linearity, p=0.0225, indicating the relationship is non-linear.

Table 4.5 Sodium and Potassium Intakes and composite outcome (death due to a renal cause and dialysis)

*Adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke; [~]Fully adjusted model also adjusted for potassium intake; [$]Fully adjusted model also adjusted for sodium intake.

	Range	Composite Outcome (SHR, 95% CI)		Renal Death (SHR, 95% CI)		Dialysis (RRR, 95% CI)	
		Unadjusted	**Adjusted***	**Unadjusted**	**Adjusted***	**Unadjusted**	**Adjusted***
Sodium (g/day)[~]							
Q1	<1.7	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)
Q2	1.7-2.3	0.97 (0.89-1.07)	1.01 (0.91-1.12)	0.96 (0.87-1.07)	1.00 (0.89-1.12)	0.96 (0.77-1.20)	0.95 (0.75-1.22)
Q3	2.3-2.8	1.04 (0.95-1.14)	1.05 (0.94-1.17)	1.01 (0.92-1.12)	1.03 (0.92-1.16)	1.17 (0.95-1.45)	1.13 (0.89-1.44)
Q4	2.8-3.6	1.12 (1.02-1.22)	1.10 (0.99-1.24)	1.07 (0.97-1.18)	1.07 (0.95-1.21)	1.25 (1.01-1.54)	1.11 (0.87-1.43)
Q5	>3.6	1.29 (1.18-1.40)	1.17 (1.02-1.33)	1.25 (1.13-1.37)	1.15 (0.99-1.33)	1.55 (1.26-1.89)	1.16 (0.86-1.56)
p for trend		<0.001	0.1314	<0.0001	0.2879	<0.001	0.201
Potassium (g/day)[$]							
Q1	<2.3	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)	1.0 (Ref)
Q2	2.3-2.9	0.88 (0.80-0.96)	0.87 (0.79-0.96)	0.85 (0.77-0.94)	0.84 (0.75-0.93)	0.92 (0.75-1.13)	0.97 (0.77-1.22)
Q3	2.9-3.5	0.93 (0.85-1.02)	0.92 (0.83-1.01)	0.92 (0.84-1.02)	0.90 (0.81-1.00)	0.92 (0.75-1.14)	0.93 (0.74-1.18)
Q4	3.5-4.3	0.88 (0.80-0.96)	0.83 (0.75-0.93)	0.84 (0.77-0.93)	0.80 (0.71-0.90)	0.98 (0.79-1.20)	0.95 (0.74-1.21)
Q5	>4.3	1.03 (0.95-1.12)	0.83 (0.73-0.95)	0.99 (0.90-1.08)	0.78 (0.67-0.90)	1.31 (1.08-1.60)	1.10 (0.83-1.48)
p for trend		<0.001	0.0112	<0.0001	0.0011	0.004	0.722

4.3.4 Interactions

There were no statistically or clinically significant interactions between the dietary variables (including interactions between diet scores and sodium or potassium intake) and between the dietary variables and categories of age (≤63years vs. >63 years), gender, BMI (≤30 vs. >30) or diabetes for the composite outcome.

4.3.5 Sensitivity Analyses

The exclusion of dialysis (n=236 events) or death due to a renal cause (n=355 events) occurring in the first four years of follow-up (1995-1999) did not materially alter the findings for the composite (Table 4.6 & 4.7). The inclusion of hypertension as a covariate in the fully adjusted model (available in 44.0%, (n=239,593)) reduced the number of composite events by 46.0% to 2,229 events, but did not materially modify the observed associations (Table 4.8 & 4.9) for the AHEI, HEI, MDS and RFS. In particular, the highest quintile of sodium intake (sHR 1.29 (95% CI 1.08-1.54)) continued to be associated with an increased hazard for the composite. The exclusion of diabetes and CVD from the fully adjusted models also did not materially alter the observed associations for the composite.

Table 4.6 Diet indexes and composite outcome, excluding events in the first four years

*Adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke

	Range	Composite Outcome (SHR, 95% CI)	
		Unadjusted	**Adjusted***
Alternative Healthy Eating Index[164]			
Q1	12.8-44.1	1.00 (Ref)	1.00 (Ref)
Q2	44.1-49.7	0.91 (0.83-0.99)	0.92 (0.84-1.01)
Q3	49.7-54.7	0.85 (0.77-0.93)	0.85 (0.77-0.93)
Q4	54.7-60.7	0.77 (0.70-0.84)	0.81 (0.73-0.89)
Q5	>60.7	0.60 (0.55-0.67)	0.69 (0.62-0.77)
p for trend		<0.001	<0.001
Healthy Eating Index 2010[165]			
Q1	18.2-57.0	1.00 (Ref)	1.00 (Ref)
Q2	57.0-64.4	0.98 (0.90-1.07)	0.98 (0.89-1.09)
Q3	64.4-69.9	0.88 (0.80-0.96)	0.88 (0.80-0.98)
Q4	69.9-73.3	0.87 (0.79-0.95)	0.89 (0.80-0.98)
Q5	>73.3	0.74 (0.67-0.82)	0.79 (0.71-0.88)
p for trend		<0.001	<0.001
Mediterranean Diet Score[166,167]			
Q1	0-3	1.00 (Ref)	1.00 (Ref)
Q2	4	0.99 (0.92-1.07)	0.99 (0.91-1.08)
Q3	5	0.85 (0.78-0.92)	0.90 (0.82-0.99)
Q4	6	0.75 (0.68-0.83)	0.82 (0.74-0.92)
Q5	>6	0.71 (0.63-0.80)	0.81 (0.71-0.93)
p for trend		<0.001	<0.001
Recommended Food Score[168]			
Q1	0	1.00 (Ref)	1.00 (Ref)
Q2	1	0.99 (0.91-1.07)	0.95 (0.88-1.04)
Q3	2	0.94 (0.86-1.03)	0.90 (0.82-1.00)
Q4	3	1.03 (0.92-1.14)	0.94 (0.83-1.05)
Q5	>3	1.00 (0.91-1.11)	0.91 (0.81-1.02)
p for trend		0.6385	0.2524

Table 4.7 Sodium and potassium Intakes and composite outcome, excluding events in the first four years

*Adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease and stroke; [~]Fully adjusted model also adjusted for potassium intake; [$]Fully adjusted model also adjusted for sodium intake.

	Range	Composite Outcome (SHR, 95% CI)	
		Unadjusted	**Adjusted***
Sodium (g/day) [~]			
Q1	<1.7	1.00 (Ref)	1.00 (Ref)
Q2	1.7-2.3	0.99 (0.89-1.09)	1.03 (0.92-1.15)
Q3	2.3-2.8	1.05 (0.95-1.16)	1.05 (0.94-1.18)
Q4	2.8-3.6	1.11 (1.01-1.22)	1.09 (0.97-1.23)
Q5	<3.6	1.30 (1.19-1.43)	1.17 (1.01-1.35)
p for trend		<0.001	0.2783
Potassium (g/day) [$]			
Q1	<2.3	1.00 (Ref)	1.00 (Ref)
Q2	2.3-2.9	0.87 (0.79-0.96)	0.85 (0.76-0.94)
Q3	2.9-3.5	0.92 (0.84-1.01)	0.89 (0.80-0.99)
Q4	3.5-4.3	0.89 (0.81-0.99)	0.83 (0.74-0.93)
Q5	>4.3	1.01 (0.93-1.11)	0.79 (0.68-0.91)
p for trend		0.0034	0.005

Table 4.8 Diet indexes and composite outcome including adjustment for hypertension

*Adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease, stroke and hypertension

	Range	Composite Outcome (SHR, 95% CI)	
		Unadjusted	**Adjusted***
Alternative Healthy Eating Index[164]			
Q1	12.8-44.1	1.00 (Ref)	1.00 (Ref)
Q2	44.1-49.7	0.90 (0.80-1.00)	0.90 (0.80-1.02)
Q3	49.7-54.7	0.83 (0.74-0.93)	0.83 (0.74-0.94)
Q4	54.7-60.7	0.75 (0.67-0.84)	0.81 (0.71-0.92)
Q5	>60.7	0.57 (0.50-0.64)	0.68 (0.59-0.77)
p for trend		<0.001	<0.001
Healthy Eating Index 2010[165]			
Q1	18.2-57.0	1.00 (Ref)	1.00 (Ref)
Q2	57.0-64.4	0.97 (0.86-1.08)	0.95 (0.84-1.07)
Q3	64.4-69.9	0.85 (0.75-0.95)	0.87 (0.76-0.98)
Q4	69.9-73.3	0.87 (0.78-0.98)	0.90 (0.80-1.03)
Q5	>73.3	0.72 (0.63-0.81)	0.79 (0.69-0.91)
p for trend		<0.001	0.001
Mediterranean Diet Score[166,167]			
Q1	0-3	1.00 (Ref)	1.00 (Ref)
Q2	4	0.99 (0.90-1.10)	0.99 (0.89-1.11)
Q3	5	0.87 (0.78-0.97)	0.91 (0.81-1.02)
Q4	6	0.74 (0.65-0.84)	0.82 (0.72-0.94)
Q5	>6	0.67 (0.57-0.78)	0.79 (0.67-0.93)
p for trend		<0.001	<0.001
Recommended Food Score[168]			
Q1	0	1.00 (Ref)	1.00 (Ref)
Q2	1	0.94 (0.85-1.04)	0.93 (0.83-1.03)
Q3	2	0.91 (0.81-1.01)	0.92 (0.81-1.04)
Q4	3	0.98 (0.85-1.12)	0.92 (0.79-1.06)
Q5	>3	1.00 (0.88-1.14)	0.98 (0.85-1.13)
p for trend		0.4345	0.5240

Table 4.9 Sodium and potassium intakes and composite outcome including adjustment for hypertension

*Adjusted for age, sex, BMI, smoking, education, race, physical activity, diabetes, heart disease, stroke and hypertension; [~]Fully adjusted model also adjusted for potassium intake; [$]Fully adjusted model also adjusted for sodium intake.

	Range	**Composite Outcome (SHR, 95% CI)**	
		Unadjusted	**Adjusted***
Sodium (g/day) [~]			
Q1	<1.7	1.00 (Ref)	1.00 (Ref)
Q2	1.7-2.3	1.03 (0.90-1.16)	1.07 (0.94-1.23)
Q3	2.3-2.8	1.00 (0.88-1.14)	1.05 (0.90-1.21)
Q4	2.8-3.6	1.21 (0.99-1.27)	1.15 (0.99-1.33)
Q5	<3.6	1.37 (1.22-1.54)	1.29 (1.08-1.54)
p for trend		<0.001	0.0455
Potassium (g/day) [$]			
Q1	<2.3	1.00 (Ref)	1.00 (Ref)
Q2	2.3-2.9	0.90 (0.80-1.02)	0.91 (0.80-1.04)
Q3	2.9-3.5	0.95 (0.84-1.07)	0.96 (0.83-1.09)
Q4	3.5-4.3	0.92 (0.81-1.04)	0.91 (0.79-1.05)
Q5	>4.3	1.02 (0.90-1.14)	0.89 (0.74-1.06)
p for trend		0.2262	0.6017

4.4 Discussion

4.4.1 Summary of Findings

In this study, a significant association between a healthier diet, namely higher scores on validated diet quality scales and a reduced risk of the composite outcome of death due to a renal cause and dialysis, was observed. The greatest magnitude of effect was seen with AHEI, followed by HEI and MDS with no association with RFS, highlighting that not all measures of diet quality are associated with reduced hazards of renal outcomes. Diets that avoid high sodium intake and low potassium intake were also associated with a lower risk of the composite outcome.

4.4.2 Healthy Eating and Outcomes

Healthy eating patterns are reported to decrease the risk of vascular disease[9,38] and mortality[168,174-176] but, to date, no other longitudinal study has explored the association with clinical renal outcomes in a general population. While small observational studies report that a healthy diet is associated with reduced renal outcomes (including proteinuria[154,155], CKD[145,177] and decline in eGFR[154]), no large prospective cohort studies have evaluated the association with clinical renal events. Low-fat diets have been associated with improvements in eGFR over two years[178], the association with high-protein diets is inconsistent[179-181] and the Mediterranean Diet has been associated with improved eGFR at one year but not change in proteinuria[182].

The findings of this study have important implications for dietary guideline recommendations for the prevention of major renal events. First, simple measures of diet quality, known to be associated with reduced cardiovascular risk, are associated with reduced risk of renal events. The data suggest that AHEI has the strongest association with renal outcomes, although HEI and MDS were also significantly associated with reduced risk. In contrast, the data support avoided the use of RFS, as no significant association was observed. Therefore, these findings have implications for selecting optimal approaches to measuring overall dietary quality in populations. Poor quality diets are associated with hyperglycaemia, dyslipidaemia, hepatokine production (e.g. Fetuin A), adipokine production (e.g. leptin) and visceral obesity[183], which may ultimately lead to renal disease through metabolic syndrome[184], RAAS activation, insulin resistance, decreased insulin sensitivity and atherosclerosis[185]. The consumption of a “healthy diet” may modify these mechanisms and potentially reduce the risk of clinical outcomes. As foods and nutrients are not consumed in isolation, these findings further enforce the importance of an overall healthy diet, rather than an isolated focus on particular ‘healthy’ foods or food groups, which do not capture the synergy between foods and nutrients.

Second, this study questions current recommendations for low potassium intake among those with CKD without hyperkalemia. Higher potassium intake has been associated with reductions in blood pressure and cardiovascular disease (especially stroke[126]) and Chapter 3 reports reduced odds of renal outcomes, but not in those with lower eGFR (30-45ml/min/1.73m^2)[186]. While potassium intake may ‘trigger’ the need for dialysis (as decline eGFR is associated with a decreased ability to excrete potassium hyperkalemia), it appears protective in the general population.

Third, there was convincing evidence that high sodium intake was associated with an increased risk of renal events; the increased risk was observed only in the 20% of the population with the highest intake, and argues against a population-wide recommendation[39] to reduce sodium intake, as it is uncertain if reductions in blood pressure[21] directly translate into reduced risk of clinical endpoints. Similar to the J-shaped association observed between sodium intake and CV death[25,26,114], the association between sodium intake and renal outcomes was non-linear. We confirm the association between high sodium intake and increased risk of renal outcomes, consistent with a recent systematic review (Chapter 2)[117].

Although multiple methods can be used to characterize a diet as healthy, no consensus exists as to which method is optimal[177]. The apparent inconsistency in these findings (i.e. comparing the findings for RFS to the other scores) likely reflects differences in the components of each score (Table 4.1)[187]. The AHEI, HEI and MDS include both ‘healthy’ (such as grains, vegetables and fruits) and ‘unhealthy’ (such as oils, fat, alcohol, added sugar) components and likely give a better overall assessment of diet quality than the RFS, which includes only ‘healthy’ components (grains, vegetables, fruit, milk and meat/fish/legumes). In addition, although diet was measured at baseline only, changes in diet over time would be likely to attenuate the results towards the null and more likely to dilute rather than strengthen observed associations[38]. Although the criterion standard for sodium and potassium intake is repeated 24-hour urine collection, FFQ alone was used in this study, which may not capture salt added during/after cooking, resulting in systematic bias (i.e. underassessment). However, it is estimated that >85% of sodium intake in the US comes from non-discretionary sources[188]. Importantly, as FFQ is expected to underestimate sodium intake, the absolute level of sodium intake where the risk increases may be higher than we observed here, should the reference standard (24 hour urine collection for sodium) be employed. Finally, although hypertension was not included in our primary models (as it may be on the causal pathway of the association between diet and renal outcomes), sensitivity analyses including hypertension did not materially modify the observed associations.

4.4.3 Strengths

The main strength of this study is the large sample size, duration of follow-up (>7.7 million person-years follow-up available for mortality) and the inclusion of major renal events (i.e. death due to renal cause or dialysis) unlike previous studies that rely on surrogate outcomes. Renal disease as a contributing cause of death was included as patients with CKD frequently die from other causes before requiring dialysis[189]. Participants completed a detailed, previously validated[159], dietary assessment. Despite a potential to underestimate energy intake, and overestimate fruit, vegetable or dairy[190], diet quality indexes are less subject to measurement error as the ratios of reported foods are still likely to reflect actual consumption[191]. Moreover, while these scales may not accurately capture actual intake, their application in clinical practice can be guided by these findings. Finally, to account for potential survival bias, as those with a healthier diet may have lived longer and had a longer period in which to require dialysis, a competing risks approach was taken to these analyses.

4.4.4 Limitations

This study has a number of limitations. First, the observational design precludes conclusions of causation. Second, a baseline measure of kidney function (eGFR or proteinuria) or presence of CKD was not available, but this is expected to bias towards the null, as a proportion of the non-renal event group will be participants with established CKD, but without renal events. To further address reverse causation, participants with renal events during the first four years of follow-up were excluded, as it is expected to reduce the proportion of patients with advanced CKD at baseline, who would be more likely to receive dietary intervention or to experience outcome events early during follow-up. Third, dialysis events could not be classified as acute or chronic (and therefore distinguish between acute kidney injury (AKI) and ESRD), as events were self-reported. However, the majority of dialysis treatments are required for ESRD, making it unlikely that associations are heavily influenced by AKI. As diabetes and CVD may be on the causal pathway of association between diet and renal outcomes, they were excluded from the fully adjusted model in sensitivity analyses, but associations were materially unchanged. For the dialysis outcome, these results likely underrepresent the true incidence of dialysis, which may result in misclassification bias, as some participants in the control group would have received dialysis. However, this is expected to bias estimates towards the null, rather than increase the risk of type I error. Fourth, this study was performed exclusively in the USA and the findings may not be generalizable to other countries. Finally, healthier diet scores may reflect a healthy lifestyle in general[192], a contention supported by these results as those in the highest quintile of diet scores had lower BMI and were less likely to smoke or have hypertension. Although these known confounders were included in the primary model, the possibility of residual confounding cannot be excluded.

4.4.5 Conclusion

In summary, these findings extend the known benefits of healthy eating, suggesting that a healthy diet may protect from renal outcomes. Importantly, not all diet quality scores were associated with reduced risk, highlighting the importance of a comprehensive dietary assessment, and not just focusing on the intake of “healthy foods” alone. As dietary modification is a low-cost, simple intervention, it offers the potential to significantly reduce the global burden from CKD.

Chapter 5: Chronic kidney disease and functional impairment

Smyth A, Glynn LG, Murphy AW, Mulqueen J, Canavan M, Reddan DN, O'Donnell MJ. Mild chronic kidney disease and functional impairment in community-dwelling older adults. Age and Ageing 2013; 42: 488-494. Reproduced with permission (Appendix 4).

5.1 Introduction

The preservation of functional independence is a key determinant of 'successful ageing' and in subjective surveys of older adults, functional independence is reported to be more important than absence of disease[51]. Subclinical cardiovascular and cerebrovascular disease, especially covert stroke, has been associated with functional decline, with loss of independence and the ability to perform routine activities of daily living[52]. While the association between ESRD and functional impairment has been established[54], the relationship between earlier stages of CKD and functional impairment has been understudied. Such information would inform the epidemiology of CKD and the conduct of future clinical research on potential interventions for CKD prevention that include patient important outcomes.

In this study, the association between CKD and functional impairment will be explored in a representative sample of community-dwelling older adults.

5.2 Methods

5.2.1 Population

The *C*ardiovascu*lar* Multimorbid*ity* in Primary Care (CLARITY) study is a cross-sectional study of patients in the West of Ireland. Patients were recruited from primary care centres included in the Western Research and Education Network (WestREN), a University-affiliated primary care research network, previously reported to be representative of the Irish national general practice profile[193]. All general practices in the NUI Galway undergraduate database in June 2009 were invited to join WestREN, a dedicated website was established and a questionnaire on practice characteristics was completed in September 2009. Of 88 practices in the database, 71 were recruited to the network, who cared for both general medical service (GMS) and private patients. A geographical balance of primary care centres was identified and practices that used the same practice software programme were invited to take part in CLARITY. From a total of 71 practices, 17 were invited and 65% (n=11) took part in the study. Within each primary care centre, all patients aged ≥50 years that had two or more consultations in the previous 24 months were considered eligible. This approach was used to include only active patients and to ensure one-off visitors and patients who had moved away from the primary care centre were not included.

Ethical approval for the CLARITY study was obtained from the Irish College of General Practitioners (ICGP) and the work was supported by the Project Grant Scheme of the Health Research Board (HRB) of Ireland.

5.2.2 Data Collection

The following data were collected in all patients: demographics (age, gender and race), place of residence (including nursing home), hospital medical care (outpatient attendance and inpatient admissions in the previous two years), smoking status, past medical history (including medication use) and results of laboratory investigations including blood glucose, glycosylated haemoglobin (HbA1c), serum creatinine, cholesterol and triglyceride levels.

Smoking status was defined as current smoker, former smoker or non-smoker. Hypertension was defined as either documented hypertension in medical records or a systolic blood pressure ≥140mmHg or diastolic blood pressure ≥90mmHg. A prior history of coronary artery disease was defined as a history of angina, myocardial infarction, percutaneous coronary intervention or coronary artery bypass graft. Chronic kidney disease was defined as either documented CKD in medical records or estimated GFR (eGFR)<60ml/min/1.73m^2 using the MDRD formula[11]. To determine if there were significant differences in the prevalence of CKD by eGFR formula, eGFR was also calculated using the CKD-EPI formula[12].

5.2.3 Functional Assessment

A subset of patients received a postal, standardized, self-reported health questionnaire to measure perceived health, well-being and functional status (Appendix 5). Of the 9,698 patients recruited, 2,212 were excluded for the following reasons: patients who died since study initiation, moved away from the area and those deemed unsuitable (due to dementia or terminal illness) by their primary care physician. Of the 7,486 participants that received the questionnaire, 47% (n=3,499) returned it; only those who returned a completed questionnaire were included in this study. Compared to the overall CLARITY cohort, patients who responded to the questionnaire were older (mean age 66.3±10.3 vs. 65.2±10.6, p<0.001) and more likely to be female (36.1% of females vs. 34.2% of males, p=0.045) but were as likely to have serum creatinine measured as non-responders (p=0.734).

Instrumental activities of daily living (IADL) and basic activities of daily living (BADL) were measured using modified items from the Lawton and Barthel scales respectively (Appendix 4)[194,195]. Patients were asked about difficulty (including the need for assistance and/or dependence on another person) with mobility, self-care (washing and dressing) and usual activities (work, housework, leisure activities). They were also asked to report pain, discomfort, anxiety, depression, measures of well-being, smoking status, age on leaving formal education, marital status, employment status, alcohol use and if they had a fall within the preceding year.

The primary outcome measure was a composite of any impairment in IADL or BADL, with BADL defined as self-care tasks including washing, dressing and mobility and IADL as activities including work, housework and leisure activities. Secondary outcomes included impairment in BADL, IADL or fall requiring hospital admission within the preceding year.

5.2.4 Statistical Analysis

Continuous variables are expressed as mean (SD) and compared using t-test or Mann Whitney test, as appropriate. Categorical variables are expressed as proportions and compared using the Chi square test.

Binary logistic regression was used to determine the independent association between CKD and functional impairment in activities of daily living. A multivariable model was developed *a priori* including variables that may confound the association between eGFR and functional impairment (including age, gender, vascular risk factors, previous vascular events and prescribed treatments). The final model included the following predictor variables: CKD, age, gender, age on leaving formal education, hypertension, history of coronary artery disease, congestive cardiac failure, stroke or transient ischaemic attack, peripheral vascular disease, diabetes mellitus, smoking status, lipid lowering therapy, beta blocker therapy, renin-angiotensin aldosterone blocker therapy, antithrombotic therapy and tertile of LDL:HDL ratio. Odds ratios (OR) with 95% confidence intervals were calculated for each variable. Tests for effect modification for gender, age (<75 versus ≥75) and cardiovascular disease were performed using the Wald test for interaction. A two-sided p value of <0.05 was considered statistically significant and analyses were performed using SPSS for Mac Version 18.0

5.3 Results

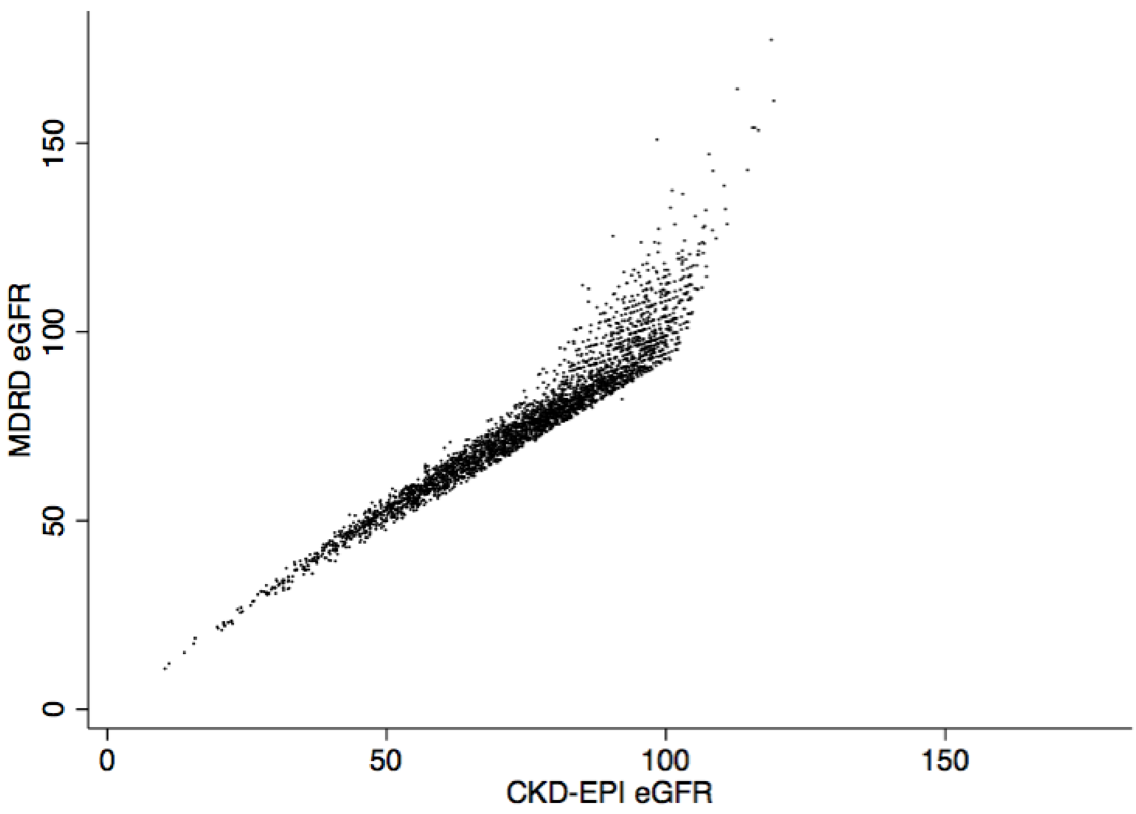
In total, 3,499 subjects, who returned the standardized questionnaire, were included in these analyses. Mean age was 66.2±10.3 years (95% confidence interval 65.9 to 66.6 years) and 45.6% were male. Serum creatinine measurement was available in 90.5% (n=3,165) of participants.

Mean MDRD eGFR was 76.6 ±19.1ml/min/1.73m^2 (95% confidence interval 75.9 to 77.2ml/min/1.73m^2). Mean CKD-EPI eGFR was 74.4±17.8ml/min/1.73m^2 (95% confidence interval 73.8 to 75.1ml/min/1.73m^2). Estimates of eGFR using MDRD and CKD-EPI were highly correlated (r=0.9595, p<0.01, Figure 5.1).

Based on medical records alone, a diagnosis of CKD was documented in 3.9% (n=138) of subjects, but when the eGFR-based definition (eGFR<60ml/min/1.73m^2) was applied, CKD was present in 18.0% (n=630). Mean eGFR was 83.1±14.8ml/min/1.73m^2 in those without CKD and 50.2±9.2ml/min/1.73m^2 in those with CKD (p<0.001). Using the CKD-EPI formulae, CKD was present in 19.6% (n=686). There was 96.6% agreement for the classification of a participant as having CKD between the two formulae (κ=0.8974).

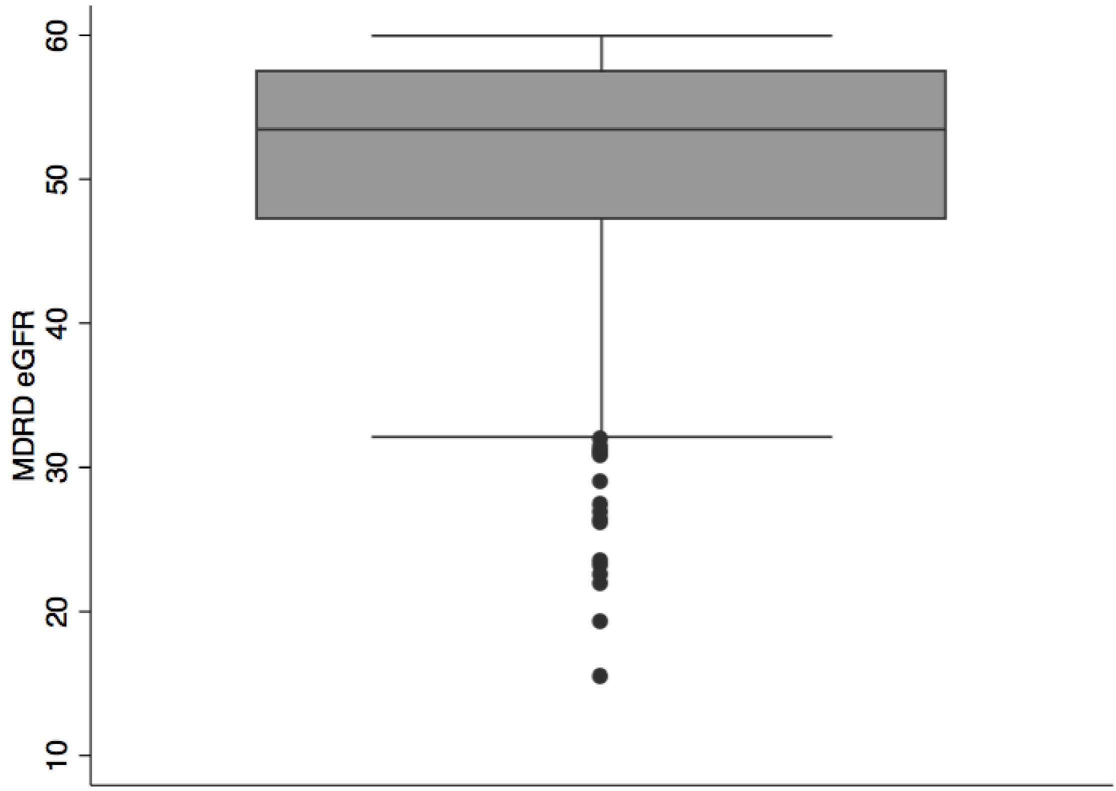
Compared to those without CKD, patients with CKD were more likely to be older, female and to have left formal education at younger ages than patients without CKD. Patients with CKD were also more likely to have coronary artery disease, heart failure, stroke, transient ischaemic attack, peripheral vascular disease, diabetes mellitus and hypertension (Table 5.1). The median MDRD eGFR in those with uncrecognised CKD was 53.5ml/min/1.73m^2 (IQR 47.2-57.4ml/min/1.73m^2) (Figure 5.2).

Figure 5.1 Scatterplot of eGFR using MDRD & CKD-EPI formulae



Correlation coefficient r=0.9596, p<0.001

Figure 5.2 Boxplot of MDRD eGFR in unrecognised CKD



Unrecognised CKD defined as MDRD eGFR<60ml/min/1.73m^2 and no documentation of a diagnosis of CKD in medical notes

Table 5.1 Parameters by CKD Status

*Coronary artery disease = composite for angina, myocardial infarction, percutaneous coronary intervention or coronary artery bypass graft; ^Hypertension = documented hypertension in medical records or a systolic blood pressure of ≥140mmHg or diastolic blood pressure of ≥90mmHg; ~Antithrombotic therapy = composite of either antiplatelet or warfarin therapy; $RAAS = renin angiotensin aldosterone system; reproduced with permission (Appendix 4).

Parameter	All (n=3499)	CKD (n=630)	No CKD (n=2869)	p-value
Mean Age (SD)	66.2 (10.3)	74.0 (9.9)	64.4 (9.5)	<0.001
Male Gender	1546/3390 (45.6%)	253/630 (40.2%)	1293/2760 (46.8%)	0.002
White	3387/3390 (99.9%)	629/630 (99.8%)	2758/2760 (99.9%)	0.465
Mean Age on Leaving Formal Education (SD)	17.3 (4.6)	16.2 (3.2)	17.5 (4.8)	<0.001
Past Medical History				
Coronary Artery Disease*	398/3392 (11.7%)	124/630 (19.7%)	274/2762 (9.9%)	<0.001
Congestive Cardiac Failure	84/3390 (2.5%)	50/630 (7.9%)	34/2760 (1.2%)	<0.001
Stroke or TIA	163/3392 (4.8%)	61/630 (9.7%)	102/2762 (3.7%)	<0.001
Peripheral Vascular Disease	104/3385 (3.1%)	25/629 (4.0%)	79/2756 (2.9%)	0.146
Risk Factors				
Diabetes Mellitus	413/3382 (12.2%)	117/627 (18.7%)	296/2755 (10.7%)	<0.001
Smoking Status				
Current Smoker	473/3479 (13.6%)	56/627 (8.9%)	417/2852 (14.6%)	0.001
Former Smoker	1406/3479 (40.4%)	263/627 (42.0%)	1143/2852 (40.1%)	0.001
Non-Smoker	1600/3479 (46.0%)	308/627 (49.1%)	1292/2852 (45.3%)	0.001
Hypertension^	1712/3499 (48.9%)	404/630 (64.1%)	1308/2869 (45.6%)	<0.001
Medications				
Antithrombotic Therapy~	835/3499 (23.9%)	214/630 (34.0%)	621/2869 (21.6%)	<0.001
Antiplatelet Therapy	744/3499 (21.3%)	182/630 (28.9%)	562/2869 (19.6%)	<0.001
Warfarin Therapy	108/3499 (3.1%)	41/630 (6.5%)	67/2869 (2.3%)	<0.001
Beta blocker Therapy	451/3499 (12.9%)	129/630 (20.5%)	322/2869 (11.2%)	<0.001
RAAS Blockade$	831/3499 (23.8%)	181/630 (28.7%)	650/2869 (22.7%)	0.001
Lipid Lowering Therapy	1053/3499 (30.1%)	206/630 (32.7%)	847/2869 (29.5%)	0.116
Mean Systolic Blood Pressure (SD)	133.4 (16.9)	135.4 (18.1)	132.9 (16.5)	0.001
Mean Diastolic Blood Pressure (SD)	78.0 (9.4)	76.9 (9.4)	78.3 (9.4)	0.001
Median Medications (Range)	4 (45)	3 (45)	6 (30)	<0.001
Mean Total Cholesterol mmol/L (SD)	5.0 (1.0)	4.7 (1.1)	5.0 (1.0)	<0.001
Mean Triglyceride mmol/L (SD)	1.4 (0.8)	1.5 (0.7)	1.3 (0.8)	0.006
Mean HDL mmol/L (SD)	1.5 (0.5)	1.4 (0.4)	1.5 (0.5)	<0.001
Mean LDL mmol/L (SD)	2.9 (0.9)	2.7 (0.9)	3.0 (0.9)	<0.001
Mean LDL:HDL Ratio (SD)	2.2 (0.9)	2.1 (0.9)	2.2 (0.9)	0.016

5.3.1 Chronic Kidney Disease and Activities of Daily Living

Functional impairment was reported by 40.4% (n=1,413) of included patients, with a higher proportion of impairment reported by patients with CKD (55.6%) than those without CKD (37.1%) ($p<0.001$). There was no difference in the proportion of patients reporting functional impairment between patients with eGFR<30ml/min/1.73m^2 (54.5%, 12/22) and patients with eGFR 30-60ml/min/1.83m^2 (55.6%, 338/608) ($p=0.895$). Dependence on a person for any ADL was reported by 17.3% (n=607), including 21.9% (n=138) of patients with CKD and 16.3% (n=469) of patients without CKD ($p=0.001$)

Unadjusted analyses shows that CKD was associated with an increased risk of any functional impairment (OR 2.12, 1.78-2.53), impairment in IADL (OR 2.02, 1.70-2.41) and BADL (OR 2.25, 1.88-2.68) compared to those without CKD. CKD was also associated with an increased risk of fall requiring hospital admission (OR 1.77, 1.12-2.81). Multivariable adjusted analyses shows that CKD was independently associated with an increased risk of any functional impairment (OR 1.43, 1.15-1.78), impairment in IADL (OR 1.43, 1.15-1.78) and impairment in BADL (OR 1.39, 1.11-1.75). The association between CKD and falls requiring hospital admission was no longer significant (Table 5.2). Results were similar when the CKD-EPI formula was used to define CKD (Table 5.3).

On subgroup analyses, age, sex and previous history of cardiovascular disease did not materially modify the association between CKD and functional impairment, showing no evidence of interaction (Table 5.4).

Table 5.2 Functional Outcomes and CKD Status using MDRD Formula

CKD defined as CKD documented in medical notes or MDRD eGFR<60ml/min/1.73m^2.

*Model adjusted for age, gender, age on leaving formal education, hypertension, coronary artery disease, congestive cardiac failure, stroke or transient ischaemic attack, peripheral vascular disease, diabetes mellitus, smoking status, lipid lowering therapy, beta blocker therapy, renin-angiotensin aldosterone blocker therapy, antithrombotic therapy and tertile of LDL:HDL ratio; reproduced with permission (Appendix 4).

Parameter	All (n=3499)	CKD (n=630)	No CKD (n=2,869)	p-value	CKD Unadjusted	CKD Adjusted*
Composite (Any Functional Impairment)	1413/3499 (40.4%)	350/630 (55.6%)	1063/2869 (37.1%)	<0.001	2.12 (1.78, 2.53)	1.43 (1.15, 1.78)
Impairment in IADL	1240/3499 (35.4%)	310/630 (49.2%)	930/2869 (32.4%)	<0.001	2.02 (1.70, 2.41)	1.43 (1.15, 1.78)
Impairment in BADL	1029/3499 (29.4%)	279/630 (44.3%)	750/2869 (26.1%)	<0.001	2.25 (1.88, 2.68)	1.39 (1.11, 1.75)
Impaired Mobility	1004/3499 (28.7%)	272/630 (41.9%)	732/2869 (25.5%)	<0.001	2.22 (1.86, 2.65)	1.36 (1.09, 1.71)
Impaired Ability to Provide Self-Care	344/3499 (9.8%)	107/630 (17.0%)	237/2869 (8.3%)	<0.001	2.27 (1.78, 2.91)	1.28 (0.92, 1.77)
Fall requiring Admission	94/3499 (2.7%)	26/630 (4.1%)	68/2869 (2.4%)	0.014	1.77 (1.12, 2.81)	1.21 (0.69, 2.13)

Table 5.3 Functional Outcomes and CKD Status using CKD-EPI Formula

CKD defined as CKD documented in medical notes or CKD-EPI eGFR<60ml/min/1.73m^2.

*Model adjusted for age, gender, age on leaving formal education, hypertension, coronary artery disease, congestive cardiac failure, stroke or transient ischaemic attack, peripheral vascular disease, diabetes mellitus, smoking status, lipid lowering therapy, beta blocker therapy, renin-angiotensin aldosterone blocker therapy, antithrombotic therapy and tertile of LDL:HDL ratio; reproduced with permission (Appendix 4).

Parameter	All (n=3499)	CKD (n=686)	No CKD (n=2813)	p-value	CKD Unadjusted	CKD Adjusted*
Composite (Any Functional Impairment)	1413/3499 (40.4%)	384/686 (56.0%)	1,029/2813 (36.6%)	<0.001	2.14 (1.81, 2.54)	1.34 (1.08, 1.67)
Impairment in IADL	1240/3499 (35.4%)	340/686 (49.6%)	900/2813 (32.0%)	<0.001	2.03 (1.71, 2.41)	1.35 (1.09, 1.69)
Impairment in BADL	1029/3499 (29.4%)	306/686 (44.6%)	723/2813 (25.7%)	<0.001	2.32 (1.94, 2.76)	1.29 (1.02, 1.62)
Impaired Mobility	1004/3499 (28.7%)	298/686 (43.4%)	706/2813 (25.1%)	<0.001	2.29 (1.92, 2.73)	1.27 (1.01, 1.60)
Impaired Ability to Provide Self-Care	344/3499 (9.8%)	118/686 (17.2%)	226/2813 (8.0%)	<0.001	2.45 (1.91, 3.13)	1.29 (0.93, 1.79)
Fall requiring Admission	94/3499 (2.7%)	30/686 (4.4%)	64/2813 (2.3%)	0.005	1.89 (1.21, 2.95)	1.33 (0.76, 2.32)

Table 5.4 Subgroup Analyses by Gender, Age and CVD

Multivariable adjusted logistic regression for the association between CKD and Composite for Any Functional Impairment, using Wald* test for interaction. Cardiovascular disease defined as presence of coronary artery disease or congestive heart failure or stroke. Models adjusted for age, gender, age on leaving formal education, hypertension, coronary artery disease, congestive cardiac failure, stroke or transient ischaemic attack, peripheral vascular disease, diabetes mellitus, smoking status, lipid lowering therapy, beta blocker therapy, renin-angiotensin aldosterone blocker therapy, antithrombotic therapy and tertile of LDL:HDL ratio; reproduced with permission (Appendix 4).

Parameter	Odds Ratio (95% Confidence Interval)	p-value*
Gender		
Male	1.43 (1.02, 2.02)	0.911
Female	1.43 (1.07, 1.91)	
Age		
<75 years	1.32 (0.996, 1.761)	0.716
>=75 years	1.52 (1.06, 2.20)	
Cardiovascular Disease˜		
Yes	1.37 (0.84, 2.25)	0.860
No	1.47 (1.15, 1.89)	

5.4 Discussion

5.4.1 Summary of Findings

This study shows that CKD was associated with an increased risk of functional impairment, independent of age, gender, co-morbidities, traditional vascular risk factors and cardiovascular events. Although the association with functional impairment is established in patients with ESRD[54] and those with moderate-severe CKD (mean of GFR 25ml/min/1.73m^2[196] and GFR 37ml/min/1.73m^2[197]), this study shows that this association extends to patients with milder CKD, with a mean GFR of 50ml/min/1.73m^2. The study also highlights the burden of functional impairment in patients with mild-moderate CKD, reported in one-fifth of participants.

5.4.2 Comparison with Previous Studies

A number of observational studies have reported the relationship between CKD and functional impairment. These include a small cross-sectional study (n=50) of patients with advancing CKD expected to require dialysis[197], cross-sectional analyses of the MDRD clinical trial (n=900) of patients with CKD[198] and cross-sectional analysis of the Heart and Estrogen/Progestin Replacement Study (HERS) (n=2,761) of menopausal women, half of whom had CKD[199]. Other studies included exclusively elderly populations including a prospective cohort study of patients without CKD at baseline[200] and a cross-sectional study (n=2,431) from NHANES[201]. An Australian population-based self-administered questionnaire of 10,525 adults aged 25 years and older, 11.2% of which had CKD, reported CKD was associated with functional impairment after multivariable adjustment[202]. However, the sample included in this study is representative of a community-dwelling population, with a prevalence of CKD closer to the 15.1% reported by the United States Renal Data System (USRDS)[203].

5.4.3 Under-Recognition of CKD

In line with previous reports, this study highlights a substantial under recognition of CKD in primary care[204], as only 21.9% of patients who fulfilled diagnostic criteria for CKD had a documented diagnosis in the medical notes. In a previous analysis of this cohort, increasing age and female gender were identified as independent risk factors for unrecognized CKD, independent of the formula used to calculate eGFR (MDRD vs. CKD-EPI)[152]. Although one would expect patients with established CKD to be prescribed more medication than those without CKD, this study reports the opposite. This likely reflects the high proportion of patients with unrecognised CKD, and therefore, not prescribed medications to reduce progression of renal disease and complications, as the median eGFR of patients with unrecognised CKD was 53ml/min/1.73m^2, significantly higher than the level of eGFR where the metabolic complications of CKD are more common. CKD was defined as eGFR <60ml/min/1.73m^2, in line with the NKF definition, where eGFR measurements considered reliable. However, higher measurements of eGFR may be more biased than lower eGFR[205].

5.4.4 Underlying Mechanisms

Impaired renal function is associated with accelerated atherosclerosis and endothelial dysfunction, which may lead to vascular disease and/or progressive renal disease[206]. In addition, traditional vascular risk factors may be more prevalent in patients with CKD, which may increase the risk of both clinical and subclinical vascular disease, especially covert stroke, which may result in functional impairment[207]. Even after controlling for previous cardiovascular disease and vascular risk factors, CKD remained significantly associated with functional impairment and suggests that abnormal renal function may contribute to covert cardiovascular disease. This finding is in line with previous studies that report a high incidence of subclinical cerebrovascular disease in patients with CKD[208].

Mechanisms potentially contributing to functional impairment in patients with CKD include anaemia[209], hyperphosphataemia[210], secondary hyperparathyroidism[211], acidosis [212] and protein-energy malnutrition[213]. Vitamin D deficiency, common in CKD patients, is associated with lower muscle strength, increased body sway, falls, sarcopenia and disability in older men and women[214]. The metabolic disturbance associated with CKD may cause myopathy and neuropathy [215], resulting in reductions exercise tolerance, independence and the ability to perform ADL. However, the patients included in this study, on average, had milder CKD where these metabolic complications are less likely to occur, and when present tend be mild in severity, compared to patients with more advanced CKD. In this study, these parameters were not measured, nor the use of non-steroidal anti-inflammatory drugs (NSAID), which may have contributed to reverse causation, in that patients with functional impairment may require the use of higher amounts of NSAIDs, which may increase the risk of CKD[216].

Sarcopenia, the loss of muscle strength and muscle mass with aging, is reported to increase the risk for functional limitation[217]. While studies have reported the association between renal impairment and frailty[218] and quality of life, no study has included a detailed assessment of function that includes instrumental and basic activities of daily living in community-dwelling patients with moderate CKD. In addition, a better understanding of the underlying mechanisms contributing to functional impairment in patients with CKD is needed to identify potential therapeutic targets to prevent functional decline.

5.4.5 Strengths

The main strength of this study is the high proportion of subjects with measured eGFR (90.5%). Secondly, unlike the majority of previous studies, the sample is representative of a community-based population.

5.4.6 Limitations

Similar to other studies addressing this research question, the main limitation of this study is that it is observational in nature, and cannot establish causation. However, few clinical trials, including those of hypertension, report functional status as clinical outcomes[53]. In addition, clinical trials that include interventions to improve the functional status of patients with CKD should be informed by observational studies. Secondly, only 47% of patients who received the questionnaire completed it, potentially introducing responder bias. Similarly, women were more likely to respond, which may partly explain the higher proportion of women with CKD, unlike previous studies. Although patients with CKD were younger on leaving formal education, there was no interaction between gender and age leaving formal education (p=0.408). Similarly, although responders were older than non-responders (mean difference 1.1 years), it is not likely to represent a clinically meaningful difference. In addition, patients with CKD were, on average, almost 10 years older than those without CKD. This is not surprising as age is a recognised risk factor for decline in renal function and age is used when estimating eGFR using the MDRD and CKD-EPI formulae from serum creatinine. Despite these differences, CKD continued to be associated with functional impairment, after multivariable adjustment. Thirdly, primary care physicians determined the suitability of sending questionnaires to participants, which may have introduced selection bias. Despite this, there was no significant difference in the proportion with measurements of serum creatinine (and calculation of eGFR) used to diagnose CKD. Finally, although the fully adjusted models included multiple risk factors for CKD and CVD, residual confounding cannot be excluded.

5.4.7 Conclusion

In conclusion, this study suggests that CKD is significantly underdiagnosed and is associated with functional impairment in community-dwelling adults. The association remained significant after adjustment for age, gender, vascular risk factors and previous cardiovascular events. As CKD was significantly under-recognised in this cohort, future studies examining if these patients develop progressive renal impairment over time are also required, to inform further study of targeted interventions to reduce the burden of CKD.

Chapter 6: Sodium InTake In Chronic Kidney Disease (STICK): A Randomised Controlled Trial

Trial protocol funded by Health Research Board of Ireland Health Research Awards Scheme 2014 (Grant Reference: HRA-POR-2014-686)

Table 6.1 STICK Study Synposis

Title	Sodium InTake In Chronic Kidney Disease (STICK): A Randomised Controlled Trial
Version	2.0 (September 24th 2014)
Trial Registration	To be registered at http://clinicaltrials.gov
Funding Sources	Health Research Board (HRA-POR-2014-686)
Study Sponsor	HRB Clinical Research Facility Galway
Public Queries	HRB Clinical Research Facility Galway
Scientific Queries	Dr Andrew Smyth, HRB Clinical Research Facility Galway
Public Title	Salt and Kidney Disease
Countries involved	Republic of Ireland
Health Condition	Chronic kidney disease
Interventions	Control: Usual Care including treatment of hypertension, comorbidities, optimising the metabolic profile and other dietary guidelines Intervention: Usual Care plus Low Sodium Intake (<2.3g/day) achieved through lifestyle and behavioural changes
Key Inclusion Criteria	(i) Age >17 years; (ii) MDRD eGFR 30-60ml/min/1.73m^2 on ≥2 occasions ≥3 months apart; (iii) Most recent eGFR measurement within 3 months; (iv) CKD attributed to diabetes, hypertension, vascular disease, obstructive uropathy, polycystic kidney disease or a combination;(v) SBP 110-140mmHg;(vi) No change in anti-hypertensive medications in previous three months; (vii) Moderate sodium intake (2.3-4.6g/day); (viii) Signed informed consent
Key Exclusion Criteria	(i) Other glomerular disease; (ii) Acute Kidney Injury within three months; (iii) Current or previous dialysis or renal transplantation; (iv) Severe heart failure (LVEF ≤20%); (v) High-dose loop or thiazide diuretic therapy; (vi) Regular NSAID use; (vii) Current or recent use of immunosuppressive drugs; (viii) Prescribed high-salt diet
Study Type	A Phase IIb, randomised, two-group, parallel, open-label, exploratory, single-centre controlled trial with a 1:1 allocation ratio
Enrollment	Planned to start 2015
Sample Size	99 participants per group (total 224 allowing for dropout & crossover)
Recruitment Status	Planned to start 2015
Primary Outcome	Change in creatinine clearance at two years
Secondary Outcomes	Proteinuria, eGFR, need for dialysis/transplantation, risk category for prognosis of CKD, 24-hour blood pressure, functional status, 24-hour urine sodium
Protocol Contributors	Dr Andrew Smyth and Prof Martin O'Donnell (Co-Principal Investigators), Dr Donal Reddan and Prof Matthew Griffin (Co Investigators), Dr Mahshid Dehghan (Collaborator) and Dr John Newell (Biotatistician).

Based on the World Health Organisation Trial Registration Data Set (Version 1.2.1)[219]

6.1. Background and Rationale

6.1.1. Burden of Chronic Kidney Disease

It is estimated that CKD affects over 300,000 people in Ireland[7] and is associated with increased risks of premature death, CVD, hospitalization, cognitive and functional impairment, as well as considerable health care costs[3-6]. Over 1.8 million people worldwide receive dialysis for ESRD[3-6,220], including approximately 4,000 people in Ireland[7]. There is a need to identify low-cost, effective, generalizable and simple interventions to reduce the burden of CKD (both the prevention of CKD and reducing CKD progression)[221].

6.1.2. Sodium Intake and Hypertension

Hypertension is a major risk for CKD, making dietary factors that reduce BP an important target for the prevention and treatment of CKD. INTERSALT reported a significant association between increasing sodium intake and BP, but the association was largely confined to populations with moderate to high intake, and no significant association was found where intake was low to moderate[222]. Meta-analyses of observational studies estimated a mean BP reduction of 10/5mmHg with a 2.3g (100mmol) reduction in daily sodium intake[23,24]. Meta-analyses of trials evaluating the effect of low vs. moderate intake on BP report inconsistent findings, with one study reporting an overall non-significant mean BP reduction of 1.1/0.6mmHg for those with low sodium intake[21], while a second reported a significant mean BP reduction of 5.0/2.7mmHg[22]. The most prominent clinical trials, with the most influence on current guidelines, are the Dietary Approaches to Stop Hypertension (DASH) diet[22] and The Trials of Hypertension Prevention (TOHP)[223], which reported BP reductions with reduced sodium intake. At population level, even modest reductions in BP may have large effects on CKD progression, and potential effects on clinical and subclinical cardiovascular disease, which may impact an individual's functional status.

6.1.3. Sodium Intake and RAAS

While low sodium intake (compared to moderate intake) results in a modest reduction in blood pressure, it may also activate RAAS[28,29], which is associated with glomerular fibrosis, increased intraglomerular pressure and proteinuria[30-32]. Moreover, the use of medications that block RAAS (ACE-I and ARB) are known to slow CKD progression[33,124,125]. In contrast, high sodium intake, but not moderate, may blunt the effect of RAAS blockade[71]. Therefore, there is conflicting mechanistic evidence as low sodium intake may reduce BP (which should be protective) but a commensurate activation in RAAS (which should have an adverse effect on renal function).

6.1.4 Current Evidence to Support Low Sodium Intake in CKD

Current guidelines recommend a daily sodium intake of <2.3g/day[18], but are based on studies of patients without CKD (Figure 2.2). In Ireland, the average daily sodium intake is 3.5-4g/day[224]. A recent, systematic review (Chapter 2) reported that high intake (>4.6g/day) was associated with an increased risk of adverse renal outcomes but no difference between moderate and low intakes[117]. Subsequent to that systematic review, the findings of additional observational studies exploring this research question have been published. Analyses of the Modification of Diet in Renal Disease (MDRD) study (n=840) reported no association between urinary sodium and “kidney failure”[123]; analyses of ONTARGET and TRANSCEND (Chapter 3) reported no association between sodium intake and renal outcomes, although the slow overall rate of decline in renal function precludes definitive conclusions[186]. The NIH Diet and Health Study (Chapter 4) reported an increased risk of renal outcomes with high sodium intake.

Clinical trials in patients with heart failure, treated with fluid restriction and high-dose furosemide, reported higher serum creatinine measurements with low sodium intake (1.8g/day). Two other similar clinical trials, from the same group, reported similar findings[225,226]; however, the findings of these trials have recently been called into question which resulted in the retraction of a meta-analysis of sodium trials in patients with heart failure[120]. One clinical trial has evaluated the effect of low sodium intake in patients with CKD; a six-week crossover study (n=20) reported a reduction in blood pressure and proteinuria with sodium restriction, but no long-term renal outcome data has been reported[87]. A Chinese cluster RCT of sodium reduction reported decreased proteinuria with reduced sodium intake over 18-month follow-up[88].

Overall available, but limited, evidence supports an association between high sodium intake and adverse renal outcomes. However, the association between low sodium intake (vs. moderate) and renal outcomes is uncertain, with inconsistent findings from cohort studies and small clinical trials of short follow-up. There is an urgent need to clarify the long-term efficacy and safety of low sodium intake in patients with CKD, echoed by a recent review of the evidence by the Institute of Medicine, who concluded "no evidence for benefit and some evidence suggesting risk of adverse health outcomes associated with sodium intake levels in ranges approximately 1500 to 2300 mg/day among those with diabetes, kidney disease or CVD"[39].

6.1.5 Modifiability of Sodium Intake

Dietary sodium intake is modifiable, as evidenced by numerous RCTs of standardized educational interventions that successful reduced dietary sodium intake to low intake range including TONE[20] and TOHP[98]. Two recent small clinical trials of sodium reduction in patients with CKD reported mean sodium reduction of 1.2g/day[95] and mean intake of 1.8g/day[87] with educational interventions. In addition, a cluster randomised controlled trial in China (n=120 villages, n=1,903 participants) reported a reduction in sodium of 0.8g/day[88]. It is clear from these trials that a sodium reduction intervention strategy is feasible. The intervention in this protocol is based on the intervention successfully applied in TONE[20]. This Phase IIb clinical trial will evaluate the role of low sodium intake (vs. moderate) on renal outcomes, and will provide preliminary information, to guide a future Phase III definitive clinical trial (if indicated) with critical relevance to the 300,000 people with CKD in Ireland.

6.1.6 Clinical Equipoise

Clinical trials that challenge current management guidelines are difficult. However, such trials are of considerable importance, and can have major implications for the management of patients and populations. For patients consuming high sodium diets, there is convincing evidence that high sodium intake is associated with reduced GFR[71,72] and increased proteinuria[71,74,227]. For patients with hypertension and moderate-high intake, most clinicians would recommend low sodium intake. However, in patients with non-severe CKD without hypertension, there is uncertainty as to whether low sodium intake is associated with more benefit than harm, and therefore needs interrogation with an RCT. Ideally, such an RCT would focus on patient important outcomes including dialysis, mortality or functional status. However, smaller studies are first needed to inform the design of a large trial. Although phase 1 of the LowSALT CKD study reported significant reductions in BP, extracellular fluid volume and proteinuria in patients with moderate-to-severe CKD (GFR 15-59ml/min/1.73m^2) after a six week intervention[87], data on longer term reductions in sodium intake and renal outcomes are lacking.

6.1.7 Impact of Trial

Effective, simple, inexpensive interventions are essential for population-based strategies that prevent progression of CKD. However, it is critical that these interventions are proven to be effective. Although patients with CKD and normal BP are routinely recommended low sodium diets, there is no convincing evidence that they benefit from it (vs. moderate intake). In addition, recent evidence has raised concerns that low sodium diet may cause harm, further incentivising the need for an RCT. Thus, the results of this trial will have important implications for CKD management.

6.2 Research Question

In adult patients with non-severe CKD and moderate sodium intake, is an educational intervention to reduce sodium intake to low levels, compared to moderate intake, associated with better renal outcomes over two years?

6.3 Study Objectives

6.3.1 Primary Research Question

In normotensive patients with non-severe CKD, is low sodium intake, compared to moderate sodium intake, associated with a reduction in the rate of decline in creatinine clearance, over two years?

6.3.2 Secondary Research Questions

(i) To determine the distribution of sodium intake in patients with CKD (by screening patients for recruitment into the trial by measuring sodium intake). This data will be used to determine the proportion exceeding current guidelines and to determine key modifiable sources of excess sodium intake in the population with CKD.

(ii) In normotensive patients with non-severe CKD, is low sodium intake, compared to moderate sodium intake, associated with a reduction in the rate of decline in eGFR (using the MDRD [11] and CKD-EPI formulae[12]), over two years?

(iii) In normotensive patients with non-severe CKD, is low sodium intake, compared to moderate sodium intake, associated with a reduction in the rate of requirement for renal replacement therapy (dialysis or transplantation), over two years?

(iv) In normotensive patients with non-severe CKD, is low sodium intake, compared to moderate sodium intake, associated with a reduction in 24 hour urinary protein excretion, over two years?

(v) In normotensive patients with non-severe CKD, is low sodium intake, compared to moderate sodium intake, associated with a reduction in the rate of change in risk category for prognosis of CKD (Table 1.1, KDIGO 2012 classification system[18]), over two years?

(vi) In normotensive patients with non-severe CKD, is low sodium intake, compared to moderate sodium intake, associated with a greater reduction in 24-hour ambulatory blood pressure over two years?

6.4 Study Design

A Phase IIb, randomised, two-group, parallel, open-label, exploratory, single-centre (Galway University Hospitals [GUH]) controlled trial with an allocation ratio of 1:1. The study sponsor and funder had no role in the study design of this trial.

6.4.1 Sampling Frame & Clinical Setting

The sampling frame includes patients with CKD attending the nephrology outpatient clinic at GUH. All attending patients will be screened using the eligibility criteria (section 6.5). Established non-severe CKD is defined by eGFR 30-60ml/min/1.73m^2 measured using the MDRD formula[11], reported by the local laboratory from IDMS traceable serum creatinine measurements. Patients meeting the eGFR criterion will be invited to participate and informed consent obtained (by a research nurse) before further screening with a food frequency questionnaire to identify only those patients with moderate sodium intake (2.3-4.6g/day). All patients enrolled into the trial will continue to receive standard care (section 6.7.2). Eligible participants will be randomised to (a) Low Sodium Intake Target (<2.3g/day) or (b) Moderate Sodium Intake (2.3-4.6g/day) (Figure 6.1).

6.4.2 Population

This study is a Phase IIb trial of a secondary prevention population of patients with established CKD, who have relatively stable disease, but based on previous robust research have a predictable rate of progression of disease (i.e. known rate of change of creatinine clearance with routine care). All patients with CKD are routinely followed-up every three to six months where they undergo clinical and laboratory assessment including measurement of eGFR and proteinuria. Therefore, the proposed clinical trial is aligned with routine care.

6.4.3 Feasibility

Previous studies report that the majority of patients with CKD do not adhere to current sodium guidelines (Figure 2.2) and consume “excess” sodium[228-230]. As the average intake in Ireland is 3.5-4.0g/day[231], it is likely the majority of patients consume moderate sodium (2.3-4.6g/day). In addition to patients already attending CKD clinics at GUH, the CLARITY study (Chapter 5) of patients aged ≥50 years, living in the catchment area of GUH, identified 1,319 patients with CKD[152]. Therefore, a large pool of potential participants is available. A recently published trial of low vs. high sodium intake in a population with CKD had a high rate of screen failures – 391 of 506 excluded participants did not meet inclusion criteria, the majority of which did not meet the criteria for controlled blood pressure, defined as 130/70mmHg, followed by inability to adhere to study protocol[87]. In this study, the systolic blood pressure eligibility criterion (see section 6.5.1) is less stringent and does not include diastolic blood pressure as an eligibility criterion. As such, a lower rate of screen failures is anticipated. This trial will inform the feasibility of recruitment of patients with CKD; should recruitment be lower than expected at the clinical site (GUH), two additional sites in Ireland (Cork and Limerick) will be formally approached to recruit patients.

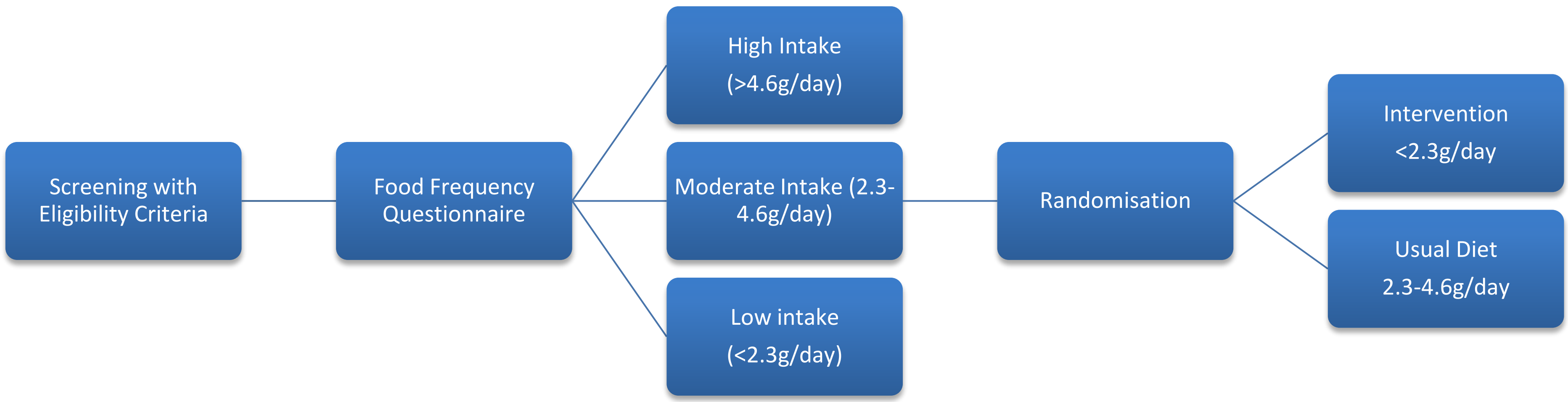


Figure 6.1 Schematic of Randomisation Process

6.5 Eligibility Criteria

6.5.1 Inclusion Criteria

(i) Age >17 years

(ii) MDRD eGFR[11] of 30-60ml/min/1.73m^2 on ≥2 occasions ≥3 months apart

(iii) Most recent MDRD eGFR measurement within 3 months

(iv) CKD attributed to diabetes, hypertension, vascular disease, obstructive uropathy, polycystic kidney disease or a combination (which does <u>not</u> have to be biopsy proven)

(v) SBP 110-140mmHg recorded three times, five minutes apart using a standardised methodology using an automated, calibrated BP monitor

(vi) No change in anti-hypertensive medications, including dose changes, in the previous three months. Changes in anti-hypertensive medications are permitted during the trial, as part of routine care.

(vii) Moderate sodium intake at baseline, defined as an estimated daily sodium intake of 2.3-4.6g/day estimated from FFQ completed during the screening visit (section 6.7.1)

(viii) Signed written informed consent

6.5.2 Exclusion Criteria

(i) Glomerular disease due to post-infectious glomerulonephritis, IgA nephropathy, thin basement membrane disease, Henoch-Schonlein Purpura, proliferative glomerulonephritis, membranous nephropathy (including lupus), rapidly progressive glomerulonephritis, minimal change disease, focal segmental glomerulosclerosis or amyloidosis, as patients with these glomerulonephritides are characteristically different from the general CKD population, where hypertension, diabetes and vascular disease are the predominant underlying causes of CKD.

(ii) Acute Kidney Injury in the previous three months (doubling of baseline serum creatinine <u>OR</u> rapidly declining MDRD eGFR over the preceding six months, defined as a decline in MDRD eGFR of ≥10ml/min/1.73m^2 from the previous measurement)

(iii) Current or previous chronic dialysis (≥1 per week for ≥3 months)

(iv) Previous renal transplantation

(v) Severe heart failure (LVEF ≤20%)

(vi) High-dose loop or thiazide diuretic therapy (exceeding a total daily dose of furosemide 80mg, bumetanide 2mg, hydrochlorothiazide 50mg, bendroflumethiazide 2.5mg, indapamide >2.5mg or metolazone >2.5mg)

(vii) Current use of regular non-steroidal anti-inflammatory drugs (more than once weekly)

(viii) Current or recent use (within one month) of immunosuppressive drugs (tacrolimus, cyclosporine, azathioprine or mycophenolate mofetil) as these medications may stimulate the renal sodium chloride co-transporter and have additional effects on sodium balance by limiting urinary excretion[232]

(ix) Unable to follow educational advice in the opinion of the clinician or dietician

(x) Kidney pathology associated with salt-wasting (Bartter syndrome)

(xi) Prescribed high-salt diet

(xii) Prescribed sodium bicarbonate therapy

(xiii) Pregnancy or lactation

(xiv) Urinary incontinence (due to difficulties completing 24-hour urine collections)

(xv) Cognitive impairment defined as a known diagnosis of dementia, mild cognitive impairment or inability to provide informed consent due to cognitive impairment in the opinion of the investigator

(xvi) Body Mass Index (BMI) <20 or >35kg/m^2

6.6 Randomisation

A fixed probability randomisation strategy will be planned prior to the start of recruitment, where each participant has an equal probability of assignment to intervention or control and the probability remains constant throughout the course of the study. *Prevalent* participants are defined as those who have previously had a consultation with a registered dietician and are expected to have received previous dietary advice on low sodium intake, and, therefore may not be as 'responsive' to the educational intervention. However, it is important to emphasise that the intervention is more intense than usual dietary advice. *Incident* participants are defined as those who have not previously had a consultation with a registered dietician. Accordingly, stratification by incident/prevalent status will be factored into randomisation, with a target recruitment ratio of 60% incident and 40% prevalent.

6.6.1 Allocation Sequence Generation

The randomisation schedule will be constructed using a computer-generated list of pseudo-random numbers, stratified by incident/prevalent status. A centrally administered, computer-generated randomisation scheme will be used to randomly assign participants in a 1:1 ratio using randomly permuted blocks.

6.6.2 Allocation Concealment

Once a patient returns for the randomization visit, the research nurse will contact a central operator by telephone, who will additionally check eligibility criteria and confirm that the participant has provided consent, prior to assigning a unique study identification number to the participant and a number corresponding to an opaque envelope containing individual patient treatment assignments. A central record (at the HRB-CRFG) will be kept of all participants who have been randomised. The randomisation schedule will be securely maintained by the central randomisation facility throughout the trial. At trial end, treatment assignments will be compared to the randomisation list, to detect any incorrect assignments.

6.6.3 Blinding

This study is open-label as blinding of participants and the research dietician is not possible for practical reasons, as the intervention is entirely behavioural and participants and study staff (research nurse, registered dietician) will be aware of treatment assignment. This approach was chosen over supplementation with sodium tablets or matching placebo, as performed in other trials[87], as this approach is more reflective of routine clinical practice, similar to the approach taken in the recently reported cluster RCT in China that achieved a reduction in sodium of 0.8g/day[88]. However, partial blinding will be employed, as the investigator will be blinded for the purposes of outcome assessment and the statistician completing the final analyses will also be blinded to treatment allocation. In addition, although eGFR will be available to the research dietician at study visits, the final study outcome (creatinine clearance) will not be measured or calculated between randomisation and the final study visit. Clinicians will be instructed to not discuss which treatment group the patient has been allocated, during routine consultation. No unblinding procedure is required as the trial is open-label.

6.7 Intervention and Usual Care

6.7.1 Pre-Randomisation & Screening Visit

All eligible participants who provide written informed consent will be included in an initial screening period prior to randomisation. The screening period will consist of a baseline nutritional assessment, including food frequency questionnaire and a medical nutrition therapy session, delivered by a registered dietician, as outlined in usual care (section 6.7.2). The European Prospective Investigation of Cancer (EPIC) food frequency questionnaire, which has been validated in Ireland[233], will be used for screening purposes. Participants will also be asked about eating habits (Table 6.2). Patients will be asked specifically about loss of appetite and the dietician will assess for evidence of malnutrition. The key aim of the screening period is to randomise only those individuals who adhere to the initial follow-up visit (at one month) and are thus likely to adhere to the intervention and study visits. Although it is possible that participants will modify his/her dietary intake due to study entry independent of randomisation, it is unlikely that such changes will be sufficient to result in long-term modification of dietary intake, without additional counselling.

Table 6.2 Questions on Eating Habit

Do you think what you eat could be healthier?	Yes/No
Do you read food labels	Yes/No
If yes, what do you look for?	
Ingredients	Yes/No
Nutrients	Yes/No
Calories	Yes/No
Weight of food	Yes/No
Additives	Yes/No
Do you follow any diets?	
Vegetarian	Yes/No
Diabetic	Yes/No
Weight reducing	Yes/No
Vegan	Yes/No
Gluten free	Yes/No
Do you take vitamins, minerals or food supplements?	Yes/No

6.7.2 Usual Care

For all participants, standard care includes treatment of hypertension, underlying comorbidities and optimising the metabolic profile (which includes the treatment of anaemia, metabolic acidosis, hyperkalaemia and calcium-phosphate disorders). Although changes in anti-hypertensive medications are not permitted during the three months prior to screening and prior to randomisation, changes in these medications are permitted throughout the trial, as part of standard care. Medical nutrition therapy, provided by a registered dietician, is recommended to maintain adequate nutrition status, to prevent or minimise disease progression and to delay need for RRT[234]. No restriction on routine clinical care, including possible co-interventions, will be enforced during the trial. Usual care sessions will be delivered to all enrolled patients at all visits during the trial.

6.7.3 Nutritional Assessment

For prevalent patients (previously had a consultation with a registered dietician), the dietician will review notes from previous dietetic consultations and review key aspects of the patient history including medications (prescription and over-the-counter) and supplements during a 20-minute assessment. Key messages for the management of CKD, specifically protein, energy, potassium and phosphate intake will be re-enforced. For incident patients (not previously had a consultation with a registered dietician), the dietician will take a full history during a 45-minute assessment. Baseline anthropometric measurements including height, weight, BMI and waist circumference will be taken and biochemical laboratory parameters reviewed for all patients. The nutrition history will include food intake (macronutrients and micronutrients), physical activity, exercise, food availability, psychosocial nutrition (including readiness to change nutrition and lifestyle behaviours) and economic issues (i.e. availability) affecting access to food and nutrition. Comorbid conditions including diabetes, obesity and dyslipidaemia will be factored into the assessment. A particular emphasis will be placed on identifying the patient's dietary sources of sodium, especially those that are the most modifiable.

6.7.4 Nutritional Recommendations

Based on standard practice[234], an individualised nutrition prescription will be created for all patients (Table 6.3).

Table 6.3 Dietary Guidelines for Patients with CKD

Dietary Factor	**Guideline Recommendation**
Protein	0.6-0.8g/kg of body weight per day if not diabetic 0.8-0.95g/kg of body weight per day if diabetic
Energy	23-35kcal/kg of body weight per day guided by weight, age, gender, physical activity and metabolic stressors
Potassium	For those with hyperkalaemia (serum potassium ≥5.0mmol) an intake of <2.4g/day is recommended
Phosphate	800-1000mg/day
Calcium	<2000mg/day, which includes dietary calcium, calcium supplementation and calcium-based phosphate binders

6.7.5 Follow-Up Review

A registered dietician will review all patients at all scheduled visits. Each visit will consist of a review of the baseline assessment and current status to determine changes since the previous visit as well as a review of laboratory and anthropometric data. The dietician will enquire about dietary intake, and 24hour urine sodium collections will be reviewed at visits that include these measures. These collections are conducted primarily to measure sodium, but will also include urinary creatinine in order to verify that the urine collections are complete. However, as creatinine clearance (the primary outcome) is <u>not</u> required for the consultation with the dietician, it will not be calculated at the time of the study visit. The individualized dietary prescription will then be modified. During all visits, key messages on dietary management of CKD will be re-enforced. This process will be standardised for all patients during all visits, independent of treatment allocation. In particular, a review of changes in dietary habits and potentially beneficial or harmful dietary substitution effects (e.g. increased potassium intake) will be determined in all participants.

6.7.6 Sodium Intake in Control Group

Participants randomized to the control group (2.3-4.6g/day) will not receive any recommendation on sodium intake at randomization, as these participants have moderate intake, based on pre-screening. However, on follow-up, participants will be reviewed for changes in diet that are associated with high sodium intake (>4.6g/day), and if present, will be counselled on sodium reduction to the moderate intake range.

6.7.7 Sodium Reduction Education Intervention

In addition to usual care (section 6.7.2), those randomised to the low sodium intake target group will receive additional counselling on behavioural and environmental factors that promote sodium reduction after randomisation and at all post-randomisation visits. The target for sodium intake in this group is <100mmol/day (<2.3g/day), in line with current guideline recommendations. The research dietician will develop the intervention, based on standardised approaches to education interventions, prior to study start up. The key aspects of the proposed dietary intervention are based on the intervention successfully applied in the TONE[20]. This trial was a multi-centre factorial controlled trial of men and women aged 60-80years with BP<145/85mmHg on single agent treatment for hypertension (single combination preparations were also permitted). Participants were randomly assigned to sodium reduction or weight loss or sodium reduction and weight loss combined or usual care and withdrawal of antihypertensive medications as attempted after three months.

A registered dietician will administer the intervention to those randomized to the sodium reduction, on an individual basis, during an additional 15-minutes dedicated to sodium intake at randomisation and at all follow-up visits. The primary goal of the intervention in the initial six months is to provide participants with the knowledge and behaviour skills necessary to achieve and maintain the desired reduction in sodium intake. The second six months will focus on problem solving and prevention of relapse to the baseline level of sodium intake. The final year of the intervention will focus on maintenance of participant interest and to reengage those participants who were less successful in reducing sodium intake. In addition to in-person visits, standardised phone follow-up with a registered dietician for additional support and further re-enforcement of the sodium reduction strategy only, will also be included in the intervention and be provided at weeks one, two, four, six and ten to intensify and re-enforce key components of the intervention, provide support and assist with troubleshooting. The registered dietician will also provide participants with a pre-prepared written pamphlet, containing advice on reducing dietary sodium intake to <100mmol/day (<2.3g/day) and motivate participants to make and sustain long-term lifestyle changes targeting sodium reduction. In particular, the intervention for this trial will focus on:

(i) Identification of sodium content in foods
(ii) Advice on food shopping, label reading, modification of recipes, menus
(iii) Sodium-specific dietary behaviour problem solving
(iv) Food-tasting & encouragement to use less salt and more spices
(v) Local shopping-guide for sodium-related products
(vi) Take-home samples of low sodium food
(vii) Use of a food diary for self-monitoring of sodium intake

6.7.8 Feasibility and Efficacy of Intervention

Dietary sodium intake is modifiable, as was shown in clinical trials that emplyed standardized educational interventions to successfully reduce dietary sodium intake to the low intake range (including TONE[20] and TOHP[98]). As it is expected that most dietary sodium intake comes from non-discretionary sources[235], the educational intervention will target these sources. The trial will help determine the feasibility of sodium restriction to the level recommended by current guidelines, in patients with CKD, and also determine if reduction in intake to such levels is sustainable. A Vanguard phase (first 40 patients, 20 in each group) will be included to determine the efficacy of the intervention in reducing sodium intake by ensuring that a difference between the low and moderate intake groups has been established.

6.8 Contamination

A major threat to the validity of this trial is contamination, whereby patients or clinicians may change their behaviour due to the conduct of the trial. For example, patients in low-sodium intake group may discuss their care with patients in the usual care group, and this may result in patients in the usual care group adopting a low-sodium intake regimen. As nephrology outpatient clinics take place twice weekly (Monday and Wednesday morning) at GUH, to minimise contamination, patients will be invited to attend pre-planned clinic days for follow-up, based on study allocation. This will also help to reduce potential contamination by the registered dietician administering the follow-up visits as all visits on specified clinic days will be with patients in either the usual care (i.e. no specific sodium counselling unless intake has risen to high levels) or intervention arms (sodium counselling), but not both on the same day. Clinicians and other treating healthcare professionals will be instructed to not discuss dietary factors with participants, as the registered dietician will address all dietary recommendations. At study end, we will re-administer the food frequency questionnaire and complete 24-hour urine collections to determine change in sodium intake in control and intervention groups.

6.9 Follow-Up Schedule

With the exception of the interval between the pre-randomisation (T_{-1}) and randomisation (T_0) visits, all other scheduled visits will occur three monthly (+/- two weeks) throughout the study period (Figure 6.2) (Table 6.4). Patients who attend the first follow up visit at month one (T_0) will be randomised to intervention or usual care. Only those randomised to sodium reduction will receive additional counselling on sodium intake (section 6.7.7), which will again be administered to this group only during months three (T_1), six (T_2), nine (T_3), twelve (T_4), fifteen (T_5), eighteen (T_6) and twenty-one (T_7) as well as telephone follow-up at weeks one, two, four, six and ten. Study-specific measurements will be carried out at randomisation (T_0), 12 months (T_4) and the final visit at 24 months (T_8) (Table 6.3). Patients will continue to be passively followed up (including recording of clinical measurements such as eGFR and ACR taken at routine clinical follow-up visits), without further counselling on the intervention, until the end of the study period.

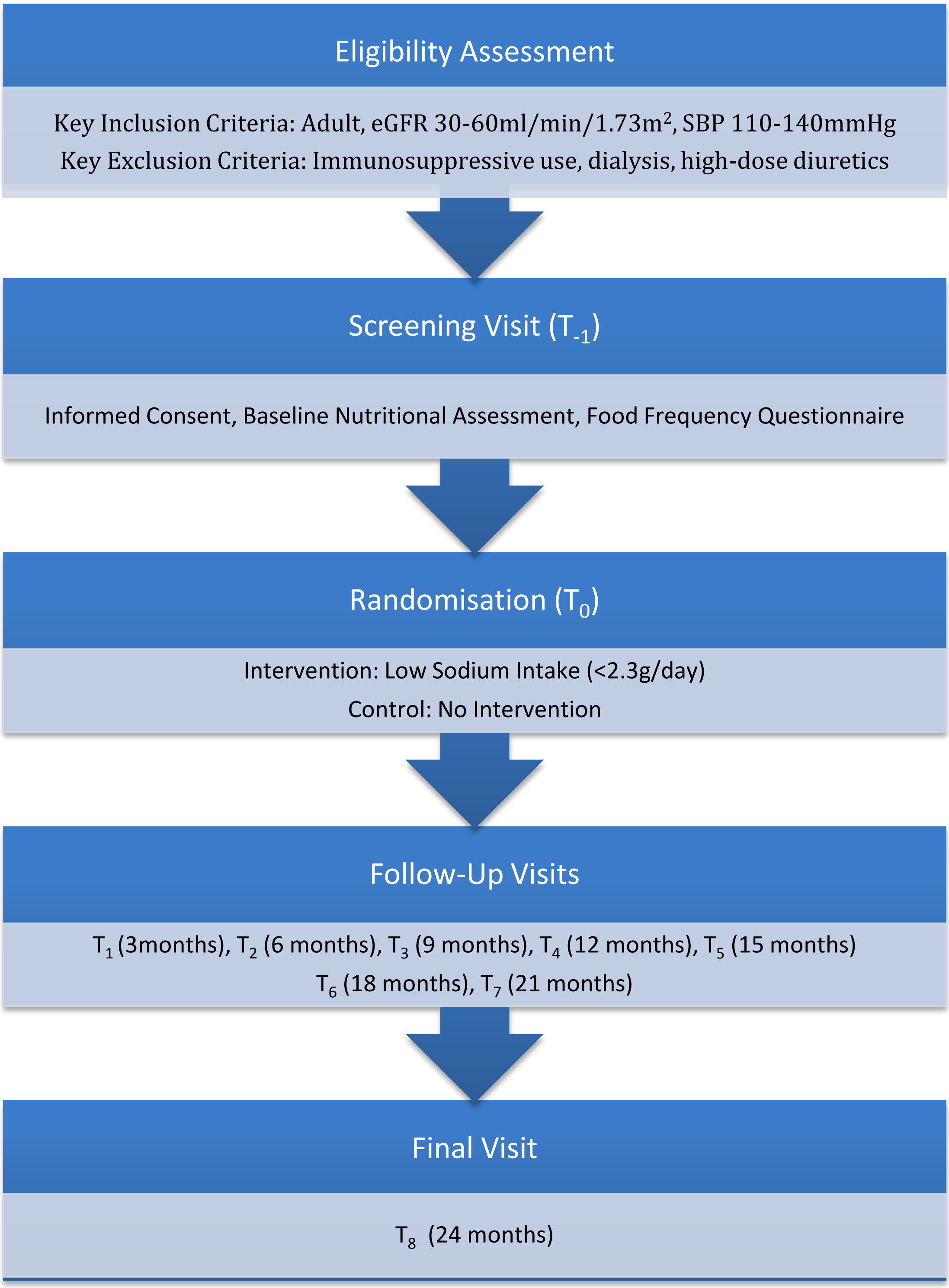


Figure 6.2 Study Flow Chart

Table 6.4 Schedule of Events

[a]24-hour urinary collections at T_0 and T_8 to be completed one week apart; [b]Passively followed serum creatinine, eGFR , UACR will be recorded from routine clinical visits; [c]Telephone support for intervention arm will be provided at weeks one, two, four, six and ten.

	T_{-1}	T_0	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
Timing Window	-1 month	Time Zero	3 Months	6 Months	9 Months	12 Months	15 Months	18 Months	21 Months	24 Months
Eligibility Criteria Review	X									
Informed Consent	X									
Demographics	X									X
Comorbidities	X									X
Office Systolic & Diastolic BP	X	X	X	X	X	X	X	X	X	X
Medications	X					X				X
Anthropometrics (Height & Weight)	X					X				X
Food Frequency Questionnaire	X									X
Randomisation		X								
24-hour Ambulatory Blood Pressure		X								X
Serum sodium, potassium, urea		X								X
Serum creatinine[b] & eGFR[11]		X	X	X	X	X	X	X	X	X
Urine Albumin: Creatinine Ratio		X	X	X	X	X	X	X	X	X
HbA1c		X								
24-hour urine collection[c] for creatinine & protein		XX[a]								XX[a]
24-hour urine collection for sodium		XX[a]	X	X				X		XX[a]
Dietitian Review ± Counselling		X	X	X	X	X	X	X	X	X
Telephone support (Intervention)		X[c]								
Adherence		X	X	X	X	X	X	X	X	X
Outcomes		X	X	X	X	X	X	X	X	X

6.9.1 Standardisation of Intervention & Follow-Up

Patients in both groups will attend for standardised usual care at the same visit frequency. There will be a standardised operations manual for usual care and also for the application of the intervention in the low sodium group to ensure that all sessions are conducted in the same way for every participant.

6.9.2 Methods to Maximise Participant Adherence

To improve compliance with the intervention and with follow-up procedures, only individuals who are adherent with the initial follow-up visit (i.e. return for the randomisation visit, T_0) will be included, thereby we will self-select potential participants who are more likely to adhere with a behavioural intervention. In addition, all follow-up visits will be scheduled at a convenient time for patients with adequate study staff to minimise wait times. To further improve adherence to the intervention, the initial period will include standardised phone follow-up with a dietician (weeks one, two, four, six and ten) to provide additional advice and support, answer questions and reinforce key messages. A written pamphlet will also be provided to participants. Assuming that the majority of ingested sodium is excreted on the same day, the current gold standard measurement to estimate dietary sodium intake is 24-hour urine collection. Each participant will complete four 24-hour urine collections for sodium at four visits over the course of the study (Table 6.4) to measure adherence.

To reduce loss to follow-up, patients who fail to return for a follow-up visit will be contacted by phone and/or mail to arrange an alternative convenient appointment. Participants will be asked to provide contact details for two nominated close acquaintances to act as additional contact persons, if necessary. The cost of transport to and from the study site will be reimbursed for the randomisation visit (which occurs outside the interval for routine care) to further encourage adherence. If participants are unable to make the final follow-up visit to the study site, a home visit will be arranged, where possible, to obtain final outcome data. If the patient refuses or is unable to attend for follow-up visits, every effort will be made to establish the reason for nonattendance.

6.10 Measurements

6.10.1 Screening

At the pre-randomisation (T_{-1}) visit, patients will be screened using a food frequency questionnaire administered by the registered dietician. Although FFQ is not the accepted gold standard to estimate dietary sodium intake, this approach was chosen to improve the feasibility of screening, and the FFQ data will be used for screening purposes only (i.e. food frequency questionnaire is not part of any outcome measure). The gold standard measurement, 24-hour urine collection, will be performed at the randomization visit and follow-up visits to quantify sodium intake at baseline and final visit, as well as to measure adherence to the intervention throughout the study.

6.10.2 Baseline Characteristics

Data will be collected at pre-randomisation (T_{-1}) for relevant participant demographic and clinical characteristics (Table 6.4) and entered into an electronic case report form (eCRF) including:

(i) Demographics: date of birth, age, sex, ethnicity, employment status

(ii) Cause of CKD, including if diagnosis is presumed or confirmed by renal biopsy

(iii) Co-morbidities: diabetes mellitus, hypertension, congestive heart failure, dyslipidaemia, obstructive uropathy (including prostate cancer and benign prostatic hypertrophy), musculoskeleteal disease (including osteoarthritis, rheumatoid arthritis and chronic back pain), cardiovascular disease (including myocardial infarction, stroke, congestive heart failure), peripheral vascular disease, current smoking status (never vs. former vs. current smoker [within last six months], number of cigarettes smoked per day), alcohol intake and physical activity

(iv) Office, automated systolic and diastolic blood pressures measured using calibrated, identical machines on all participants

(v) Current medications: antihypertensive agents, diuretics, lipid lowering therapies, analgesics, non-steroidal anti-inflammatory drugs, treatments for diabetes mellitus (oral hypoglycaemic agents, insulin or other therapies)

(vi) Functional status (section 6.13)

(vii) Anthropometrics including height (using a wall-mounted non-stretchable standard tape measure) and weight (using a standard automated weighing scale) used to calculate body mass index

(viii) Food frequency questionnaire

6.10.3 Measurements at Randomisation

Participants who return for the randomisation visit will compete 24-hour ambulatory blood pressure monitoring and laboratory measurements. IDMS traceable serum creatinine[18] and laboratory reported eGFR (estimated from the 4-variable MDRD formula[11]) will be recorded. Laboratory samples will be analysed at the GUH laboratory using standardised storage, handling and analytical procedures. Results will be automatically entered into the GUH database and will be electronically imported into the trial database. Laboratory measurements taken at randomisation will include:

(i) Serum sodium, potassium, urea, creatinine and laboratory reported eGFR (MDRD)

(ii) HbA1c

(iii) Urine albumin:creatinine ratio (ACR)

(iv) 24-hour urine collection (two collections, completed one week apart) for sodium, creatinine and protein

6.10.4 Measurements at Follow-Up Visits

At all follow-up visits, participants will be asked about adherence to the educational intervention and dietary recommendations, changes in dietary pattern, and nutrient intake, and data on clinical outcomes (section 6.12) will be recorded. If serum creatinine, eGFR and ACR were completed as part of routine clinical care, these values will be recorded. 24-hour urine collections for sodium excretion (mmol/day) will be completed in the 24-hour period prior to the follow-up visits at three months, six months and eighteen months to assess adherence. These collections will include urinary creatinine in order to determine if the urine collections are complete, but creatinine clearance will <u>not</u> be calculated from these measurements (i.e. reserved for the randomisation visit and final visit).

6.10.5 Measurements at Final Visit

At the final study visit (two years after randomisation), participants will be asked about adherence to the intervention and dietary recommendations, changes in dietary pattern, nutrient intake, clinical events, current medications, assessment of functional status (section 6.13) and complete a 24-hour ambulatory blood pressure monitor. Laboratory measurements will include:

(i) 24-hour urine collection (two collections, completed one week apart) for sodium, creatinine and protein
(ii) Urine albumin:creatinine ratio (ACR)
(iii) Serum sodium, potassium, urea, creatinine and laboratory reported eGFR (MDRD)

6.10.6 Measurements after Final Visit

After completion of the final study visit, study staff will record available eGFR and ACR measured as part of routine clinical care, until close out of the study.

6.10.7 Confounding Variables

Potential confounding variables will be measured and controlled for in final analysis, including age, gender, RAAS blockade, diuretic therapy and changes in overall diet pattern (specifically changes in potassium intake).

6.11 Criteria for Permanent Withdrawal of Intervention

(i) Symptomatic orthostatic hypotension, defined as dizziness on standing, blurring of vision, weakness or syncope in the presence of SBP≤100mmHg
(ii) SBP≤100mmHg on ≥2 occasions
(iii) Diagnosis of glomerular disease as listed in section 6.5.2
(iv) Diagnosis of ESRD, initiation of dialysis or occurrence of renal transplantation
(v) Prescription of salt or sodium bicarbonate therapy or treating physician recommends against sodium restriction
(vi) Pregnancy

6.12 Study Outcomes

6.12.1 Primary Outcome

Change in 24-hour urinary creatinine clearance from baseline to final follow-up, calculated from 24-hour urine collections (with corresponding serum creatinine measurement) at randomisation and final visit (T_8). This approach is adopted rather than eGFR to improve the reliability and validity of the surrogate outcome measure. Two collections are completed at each time-point to determine 'usual' sodium excretion. However, for safety monitoring, eGFR at baseline, 3 months and 24 months, as well as collecting all eGFR measurements taken as part of routine clinical care, will be recorded.

6.12.2 Secondary Outcomes

(i) Change in eGFR (MDRD[11] formula) from baseline to final visit
(ii) Change in eGFR (CKD-EPI[12] formula) from baseline to final visit
(iii) Rates of requirement for RRT (dialysis or transplantation)
(iv) Change in 24-hour urinary protein from baseline to two-year visit
(v) Increase in risk category for prognosis of CKD (Table 1.1)
(vi) Change in functional status from baseline to two-year visit
(vii) Change in 24-hour urine sodium from baseline to two-year visit
(viii) Change in 24-hour ambulatory blood pressure completed at baseline and final visit (two years)

6.12.3 Outcome Events of Interest

(i) Hypotension defined as SBP≤100mmHg
(ii) Cardiovascular Events including:
   a) Myocardial infarction defined as rise in troponin and one of: ischaemic signs or symptoms, ECG changes (Q waves, ST elevation, ST depression), coronary artery intervention or new cardiac wall motion abnormality on echocardiography
   b) Stroke defined as new focal neurological deficit thought to be vascular in origin with signs or symptoms lasting >24hours or leading to death
   c) Cardiac revascularization procedures including percutaneous coronary intervention (PCI) and coronary artery bypass graft (CABG) surgery

6.13 Functional Status

Functional decline is the final common pathway of many chronic conditions, by capturing the overall impact of comorbidities, and may result in patient important outcomes include loss of independence, falls or admission to nursing home[236]. The work presented in Chapter 5 shows an association between functional impairment and CKD. However, there is no clear consensus on the most appropriate method of assessing functional status in clinical trials. As functional status has multiple components (including physical and cognitive), five tools will be used to measure function at randomisation and the final study visit.

6.13.1 Basic Activities of Daily Living

The self-reported Katz Index of IADL[237], which includes bathing, dressing, toileting, transferring, continence and feeding, will be used to assess participants' basic activities of daily living (Table 6.5).

6.13.2 Instrumental Activities of Daily Living

The self-reported Frenchay Activities Index[238], which includes preparing meals, washing, housework, shopping, social occasions, hobbies, driving/travel, gardening, reading, household maintenance and work will be used to assess participants' instrumental activities of daily living (Table 6.6).

6.13.3 Cognitive Function

Multiple methods will be used to assess participants' cognitive function including the digit symbol substitution test[239] (Figure 6.3), the Montreal Cognitive Assessment (MoCA)[240] (Figure 6.4) and the trail making test part B[241,242] (Figure 6.5).

Table 6.5 Katz Index of Independence in Basic Activities of Daily Living

Participants are scored yes/no for independence in each of six functions, with a score of six indicating full function, four indicating moderate impairment and two or less indicating severe functional impairment.

Activities	**Independence** (1 Point for No supervision, direction or personal assistance)	**Dependence** (0 Points With supervision, direction, personal assistance or total care
Bathing Points:___	Bathes self completely or needs help in bathing only one part of the body (back, genital area or disabled extremity)	Needs help with bathing more than one part of the body, getting in or out of the tub or shower. Requires total bathing
Dressing Points:___	Gets clothes from closets and drawers and puts on cloths and outer garments, complete with fasteners. May have help typing shoes	Needs help with dressing self or needs to be completely dressed
Toileting Points:___	Goes to toilet, gets on and off, arranges clothes, cleans genital area without help	Needs help transferring to the toilet, cleaning self or uses bedpan or commode
Transferring Points:___	Moves in and out of bed or chair unassisted. Mechanical transferring aides acceptable	Needs help in moving from bed to chair or requires a complete transfer
Continence Points:___	Exercises complete self control over urination and defecation	Is partially or totally incontinent of bowel or bladder
Feeding Points:___	Gets food from plate into mouth without help. Preparation of food may be done by another person	Needs partial or total help with feeding or requires parenteral feeding
Total Points:____		

Table 6.6 Frenchay Activities Index

Concentrate on the patient's actual frequency of the activity over the *recent* past.

<u>**Question**</u>	<u>**Response**</u>
In the last three months how often have you undertaken:	
1. Preparing main meals *Needs to play role*	0 = Never 1 = Less than once per week 2 = 1-2 times per week 3 = Most days
2. Washing up after meals *Must do all or share equally*	
3. Washing clothes *Organisation of washing & drying clothes*	0 = Never 1 = 1-2 times in three months 2 = 3-12 times in six months 3 = At least weekly
4. Light housework *Dusting, polishing, ironing, tidying small objects*	
5. Heavy housework *Changing beds, cleaning floors/windows, vacuuming*	
6. Local shopping *Organizing & buying groceries, not just push a cart*	
7. Social occasions (*Must take active part)*	
8. Walking outside for >15 minutes	
9. Actively pursuing hobby *Requires active participation & thought*	
10. Driving car / going on bus *Must drive or take bus independently*	
In the last six months how often have you undertaken:	
11. Travel outing / car ride *To place for pleasure, not for a routine social outing, must involve some organization and decision making*	0 = Never 1 = 1-2 times in six months 2 = 3-12 times in six months 3 = At least weekly
12. Gardening *Light: occasional weeding/sweeping* *Moderate: regular weeding/raking/pruning* *Heavy: all necessary work*	0 = Never 1 = Light 2 = Moderate 3 = Heavy / All necessary
13. Household maintenance *Light: repairing small items/lamp light bulb or pug* *Moderate: spring cleaning, hanging a picture, routine car maintenance* *Heavy: painting, decorating, most necessary household/car maintenance*	
14. Reading books *Full length books, not periodicals/magazines*	0 = None 1 = 1 in six months 2 = Less than one in two weeks 3 = More than one every two weeks
15. Gainful work *Work for which the patient is paid, not voluntary, time averaged over six months*	0 = None 1 = Up to 10 hours/week 2 = 10-30 hours/week 3 = Over 30 hours/week
Total Score	

1	2	3	4	5	6	7	8	9
—	⊥	⊐	└	⊔	o	Λ	x	=

Sample Items

2	1	3	7	2	4	8	2	1	3	2	1	4	2	3	5	2	3	1	4

5	6	3	1	4	1	5	4	2	7	6	3	5	7	2	8	5	4	6	3

7	2	8	1	9	5	8	4	7	3	6	2	5	1	9	2	8	3	7	4

6	5	9	4	8	3	7	2	6	1	5	4	6	3	7	9	2	8	1	7

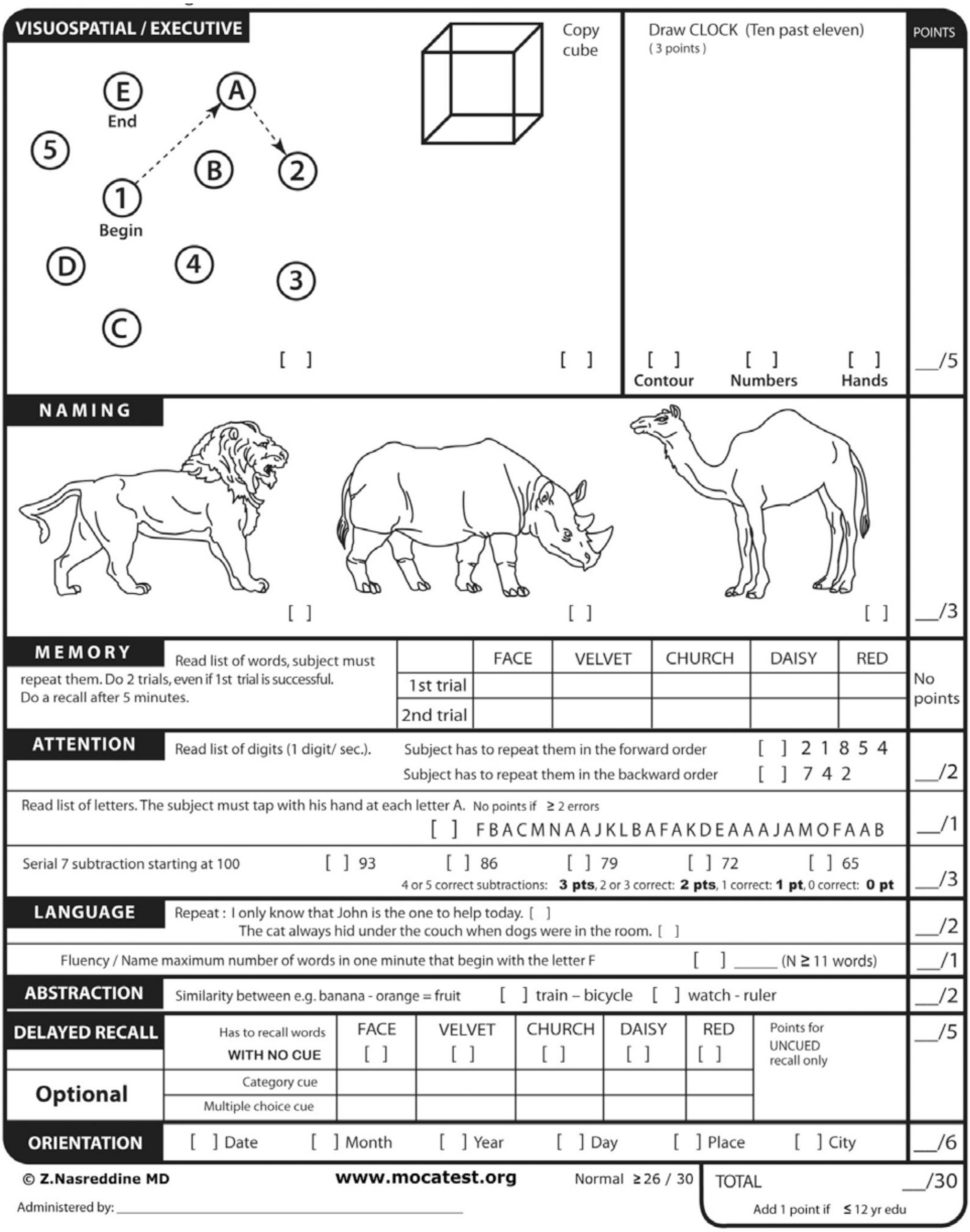
9	4	6	8	5	9	7	1	8	5	2	9	4	8	6	3	7	9	8	6

2	7	3	6	5	1	9	8	4	5	7	3	1	4	8	7	9	1	4	5

7	1	8	2	9	3	6	7	2	8	5	2	3	1	4	8	4	2	7	6

Figure 6.3 Digit Symbol Substitution Test

Study staff will explain the test asks the participant to copy symbols, specific for each digit and demonstrate by coping the symbol for the first two boxes of the sample items. The participant then completes the next five (as far as the heavy link). Once complete, the participant is instructed to complete as many squares as possible without skipping any over a timed period of two minutes. Final score is the number of correct boxes completed within the time period.



VISUOSPATIAL / EXECUTIVE

[ ]

Copy cube [ ]

Draw CLOCK (Ten past eleven) (3 points)

[ ] Contour [ ] Numbers [ ] Hands

POINTS __/5

NAMING

[ ] [ ] [ ] __/3

MEMORY Read list of words, subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.

	FACE	VELVET	CHURCH	DAISY	RED
1st trial					
2nd trial					

No points

ATTENTION Read list of digits (1 digit/ sec.). Subject has to repeat them in the forward order [ ] 2 1 8 5 4
Subject has to repeat them in the backward order [ ] 7 4 2 __/2

Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors
[ ] FBACMNAAJKLBAFAKDEAAAJAMOFAAB __/1

Serial 7 subtraction starting at 100 [ ] 93 [ ] 86 [ ] 79 [ ] 72 [ ] 65
4 or 5 correct subtractions: **3 pts**, 2 or 3 correct: **2 pts**, 1 correct: **1 pt**, 0 correct: **0 pt** __/3

LANGUAGE Repeat : I only know that John is the one to help today. [ ]
The cat always hid under the couch when dogs were in the room. [ ] __/2

Fluency / Name maximum number of words in one minute that begin with the letter F [ ] _____ (N ≥ 11 words) __/1

ABSTRACTION Similarity between e.g. banana - orange = fruit [ ] train – bicycle [ ] watch - ruler __/2

DELAYED RECALL	Has to recall words WITH NO CUE	FACE [ ]	VELVET [ ]	CHURCH [ ]	DAISY [ ]	RED [ ]	Points for UNCUED recall only	__/5
Optional	Category cue							
	Multiple choice cue							

ORIENTATION [ ] Date [ ] Month [ ] Year [ ] Day [ ] Place [ ] City __/6

© Z.Nasreddine MD www.mocatest.org Normal ≥26 / 30 TOTAL __/30
Administered by: ____________ Add 1 point if ≤ 12 yr edu

Figure 6.4 Montreal Cognitive Assessment

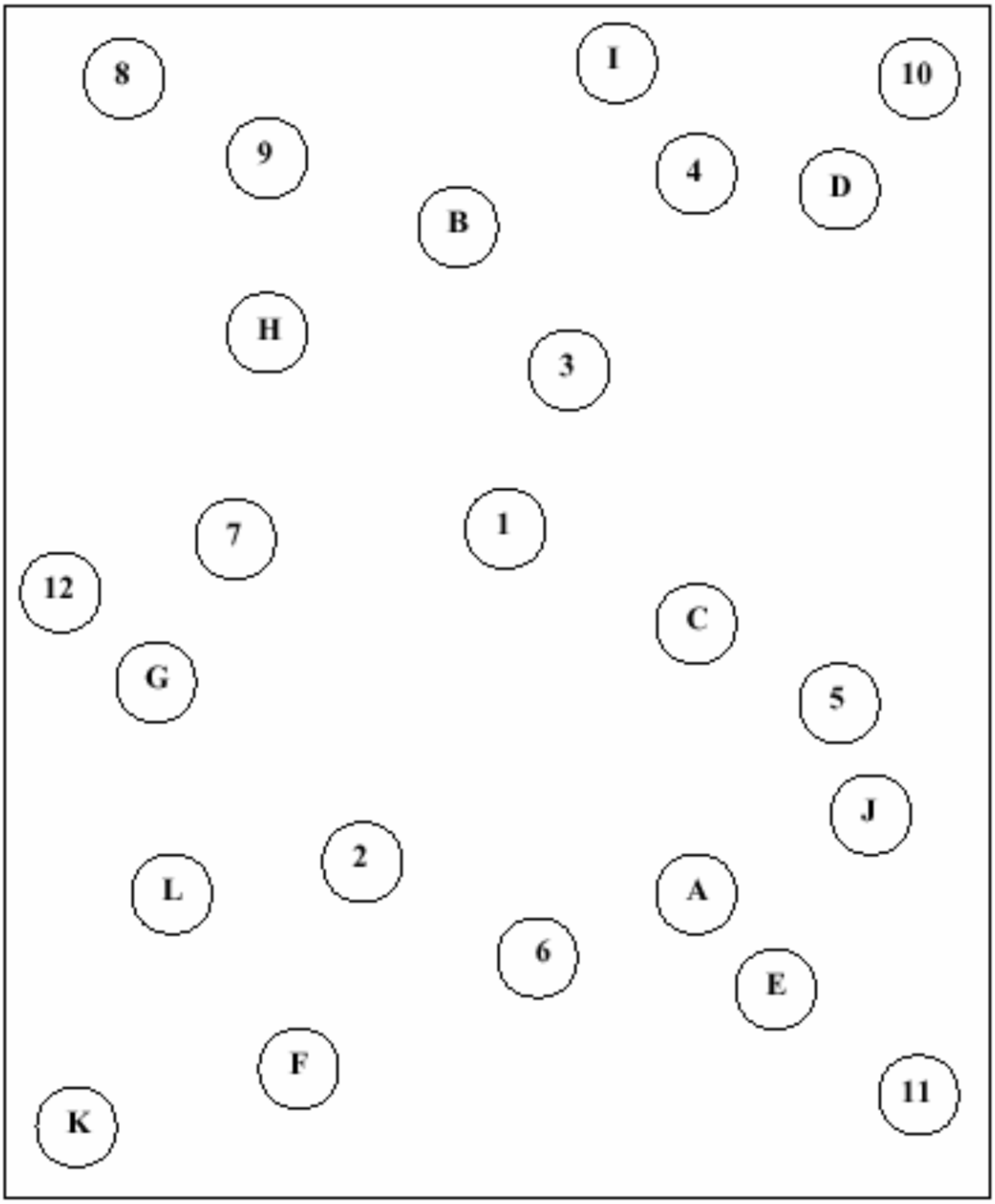


Figure 6.5 Trail Making Test Part B

Participant to be instructed to draw lines to connect the 25 circles in an ascending pattern, alternating between the numbers and the letters (i.e. 1-A-2-B-C), as quickly as possible without lifting the pen/pencil from the paper and without skipping any circles. The time taken for the patient to connect the complete trail is recorded.

6.14 Statistical Analysis

6.14.1 Sample Size Considerations

The primary outcome is mean change in 24-hour creatinine clearance between the two study groups. A two-sample t-test will be used to detect a difference in the mean change between groups. Over two year follow-up, an estimated mean decline in creatinine clearance of 8±5ml/min is expected. The estimated minimum clinically meaningful effect size is a 25% relative reduction in decline of creatinine clearance, in that the group randomised to low sodium intake will have less decline in creatinine clearance than those in the moderate intake group. Based on a two-sided alpha of 0.05 and power of 90%, a per group sample size of 99 participants will be required (Figure 6.6). Assuming a dropout rate of 5% and a net crossover/non-adherence of 5% in favour of the control group (i.e. participants in the low sodium intake arm who continue with moderate sodium intake), a total sample size of 224 participants is required. This sample size will have 74% power to detect a 20% difference in eGFR decline, should the treatment effect be <25% (Figure 6.7).

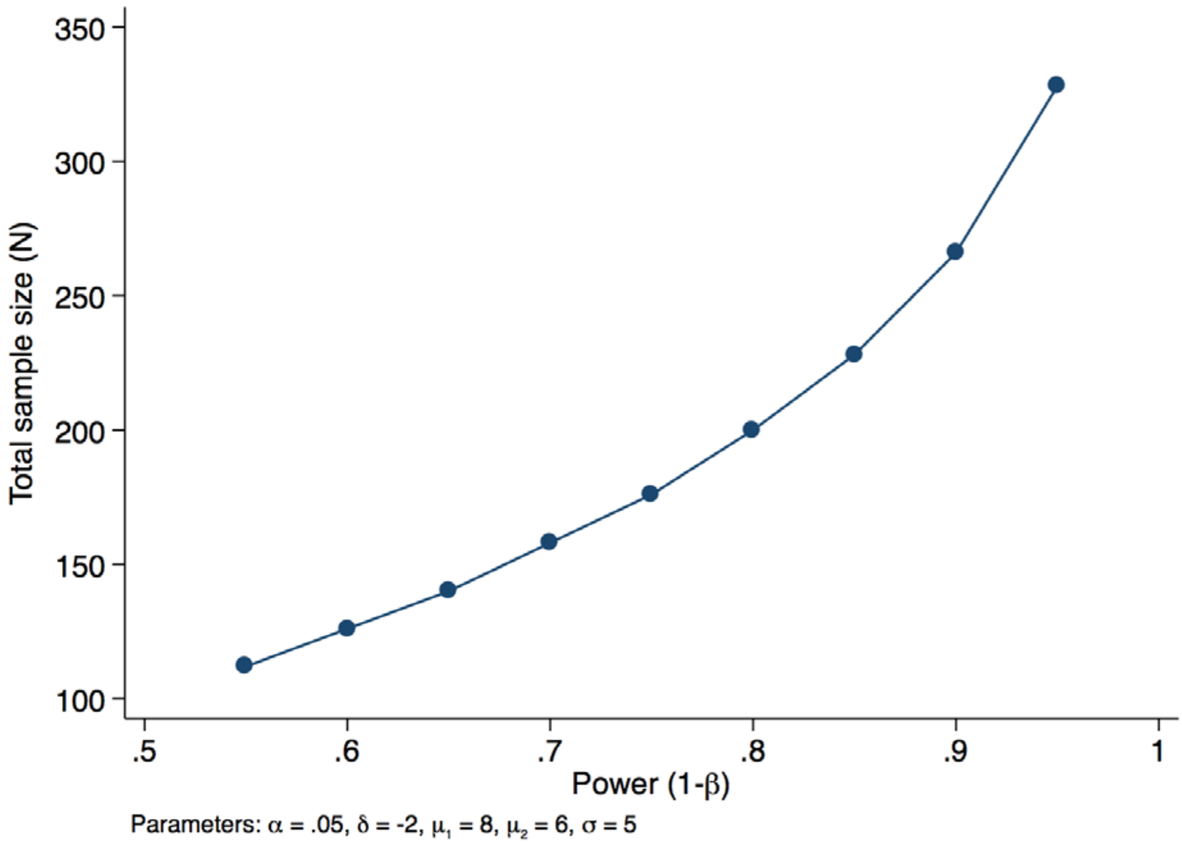


Figure 6.6 Sample Size Calculation

Assuming an effect size of 25%, this plot shows the total sample size required (y-axis) by level of power (x-axis)

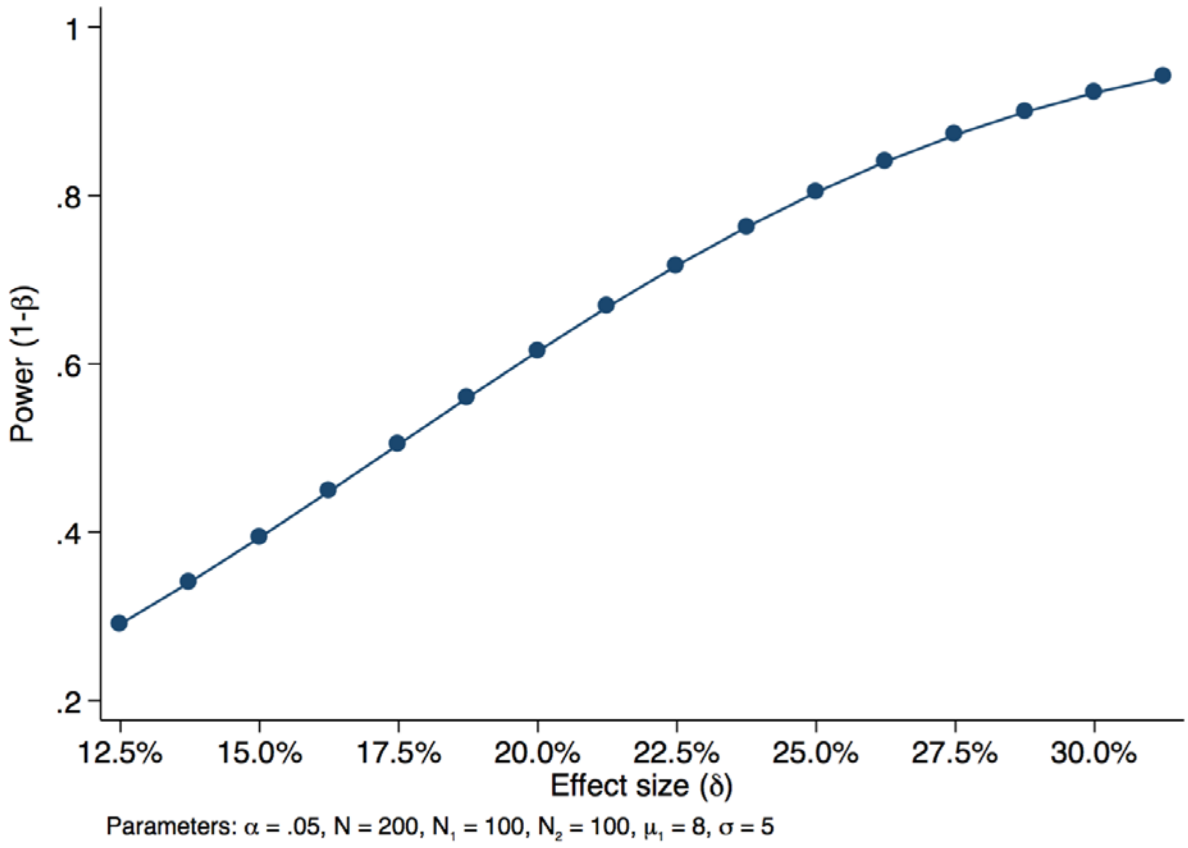


Figure 6.7 Influence of Effect Size on Statistical Power

Assuming a total sample size of 200, this plot shows the statistical power (y-axis) by differences in effect size (x-axis)

6.14.2 Descriptive Statistics & Baseline Characteristics

Descriptive statistics will be used to describe the baseline characteristics of the study population, the flow of trial participants and the level of missing data for both predictor and outcome variables. All losses to follow-up and dropouts will be accounted for and reasons documented. Categorical variables will be described using frequencies and percentages. Histograms and boxplots will be used to evaluate the distribution of continuous variables and to identify any outliers or potential errors in the data, with follow-up verification from CRFs. If the population distribution of any variables does not appear to be normal, logarithmic transformation to approximate normality will be carried out. For continuous variables, the mean and standard deviation will be reported and if not normally distributed, the median and interquartile range will be reported. All tests of significance will be two-sided and conducted at an alpha of 0.05 for statistical significance. The intention-to-treat dataset will be used for the analysis of all outcomes. A secondary per-protocol analysis will also be carried out for all outcomes, with post randomization exclusions due to ineligibility, non-compliance, loss to follow-up and missing data.

6.14.3 Statistical Analysis of Primary Outcome

A two sample paired t-test will be used to compare mean change 24-hour creatinine clearance from baseline (T_0) to the final visit (T_8) and 95% CI reported to indicate precision. If there are any chance imbalances in the baseline distribution of important covariates between treatment groups, an adjusted secondary analysis will also be performed with the intention to treat dataset and the relevant covariates, in a linear mixed model for a continuous response.

6.14.4 Statistical Analysis of Secondary Outcomes

For continuous outcome variables (eGFR, ACR, BP, scores of functional status and 24hour urine sodium), a two sample paired t-test will be used to compare changes from T_0 to T_8, including 95% CI. For categorical outcomes (need for renal replacement therapy), a Chi-square or Fisher's exact test will be used, depending on the number of outcome events after checking underlying statistical assumptions.

6.14.5 Subgroup & Sensitivity Analyses

The primary outcome will be analysed by subgroups based on incident/prevalent (i.e. previously saw a dietician vs. did not previously see a dietician), age (<70 versus ≥70 years), gender, smoking status (never smoker vs. ever smoker), cause of CKD (diabetes vs. hypertension vs. other), and use of RAAS blockade. Statistical tests of interaction (Wald) will be performed for all subgroup analyses. A sensitivity analysis will be performed to assess the effect of participant reported compliance on treatment effect.

6.14.6 Enrollment and Disposition

All subjects screened, randomized and followed up will be presented as will the level of missing data for predictor and outcome variables.

6.14.7 Missing Data

As missing data may compromise inferences from clinical trials[243], multiple approaches are required to minimize the effects from missing data. A key strategy is to optimize study design and conduct to minimize the amount of missing data[243], as statistical approaches to dealing with missing data rely on unverifiable assumptions on the reasons for missing data.

First, although multiple treatments for CKD exist, some degree of decline in renal function is inevitable and the potential additive benefit from the intervention will be an incentive for participants to stay in the trial. Second, by including a run-in period and only randomizing those that adhere to the run-in protocol, this will reduce the dropout rate during the trial. Third, by integrating with routine clinical care, the trial will not be a significant burden to patients. Fourth, due to the high likelihood of decline in renal function, a complete dataset for covariates is anticipated as well as outcome data for 90% of participants, preserving the ability to analyse endpoints using the intention-to-treat approach. Fifth, the informed consent form will distinguish between withdrawing from the treatment allocation and withdrawal from the study, allowing the collection of outcome data on patients that discontinue the intervention. Sixth, this small, feasibility proof-of concept trial, which will be used to inform the conduct and design of a larger trial, if indicated. Seventh, the telephone contact between the registered dietician and the participants in the intervention arm during the initial study period will re-enforce key components of the intervention, increasing the likelihood of adherence and minimizing the cross over, as well as enhancing participant engagement in the study.

If participants drop out, the reason(s) for withdrawal will be sought and used to test the assumption that data is missing at random. The features of this trial seek to minimize missing data but, if required, complete case analysis will be used to handle missing data. The sample size was increased to allow for dropouts and crossovers. Although this approach accounts for a reduction in study precision due to missing data, it is possible that missing data will differ substantially from the observed data and not be missing completely at random. The extent of missing data will be explored by comparing the characteristics of those with and without missing data, to understand the reasons for missing data. Last observation carried forward, which assumes that outcomes of participants do not change when data is missing, will not be used as it does not propagate imputation uncertainty resulting in inappropriate low p values and standard errors[244]. As such, a sensitivity analysis using multiple imputation will also be carried out if there is >10% missing data for covariates, to assess the robustness of findings to plausible alternative assumptions concerning missing data[245]. This approach generates multiple datasets with the observed and different plausible values imputed for missing data points, which are then analysed and the results combined, generating a statistically plausible set of values[246].

6.14.8 Data Safety & Monitoring Board

To ensure the safety of participants, two interim analyses will be performed and reviewed by the DSMB when 25% (A_1) and 50% (A_2) of participants have completed one year (T_4) of follow-up. The DSMB will be chaired by a clinician scientist with experience in clinical trials in the area of nephrology who will appoint the remainder of the DSMB, which will include an odd number of members and, at a minimum, an ethicist and an independent, non-voting biostatistician, all of whom are knowledgeable about the question being studied. All members of the DSMB will provide written declarations of freedom from conflicts of interest.

The DSMB will be blinded and review outcomes whilst considering three stopping guidelines: (i) the proportion of serious adverse events for an unacceptable safety risk at A_1 and A_2; (ii) new external information that convincingly answers the primary research question or raises serious safety issues with either the intervention or usual care at A_1 and A_2; and (iii) overwhelming benefit from either the intervention or usual care approach when the mean change in MDRD eGFR (as creatinine clearance is not measured throughout the trial, only at randomisation and at the study visit) is compared using a two sample paired t-test resulting in a p-value of <0.001, at A_2 (50% completed 1-year follow-up), consistent with the Haybittle-Peto rule[247], with a p-value of <0.05 used for the final analyses. This approach was chosen because the use of a p value <0.05 as the stopping criterion may lead to the false conclusion that the difference is statistically significant when interim data are analysed multiple times, due to multiple testing[248]. Group sequential methods, such as the Pocock[249] methodology were not chosen because this method uses the same p-value throughout the study and carries a higher likelihood of type 1 error, with a higher chance of stopping the trial early. The O'Brien Fleming[248] methodology was not chosen as it spends little of the type 1 error early in the trial but requires adjustment of the p-value for the final analyses, which may make the trial methodology appear overly complex to the wider medical community. A formal 'stopping boundary' was not established because of the overall small sample size required for this trial.

Results of interim analysis will not be disclosed to investigators or study participants (unless the DSMB makes a formal recommendation to stop the trial is made) but will be published along with the DSMB recommendations on trial completion. The evidence derived from the interim analyses may not be conclusive and concealing the findings prevents any influence on study participants or investigators, by disturbing the assumption of clinical equipoise – i.e. it could make physicians hesitant to continue to recruit patients and/or adversely affect adherence[250].

6.15 Ethical Considerations

This study will be conducted in accordance with the International Conference on Harmonisation guidelines for Good Clinical Practice (ICH-GCP)[251] and the requirements of the research ethics committee, from whom ethical approval will be sought prior to study initiation. Informed consent forms, patient information leaflets and other study documentation (including case report forms) will be developed prior to submission to the research ethics committee.

6.15.1 Patient Confidentiality

There is a risk to patient confidentiality with this study, which could result in mental discomfort and distress to involved participants if breached. The risk is estimated to be low/moderate. In order to preserve patient confidentiality, all information obtained in this study will be handled with strict privacy and electronic data security standards. Unique subject identifiers will be assigned to each participant to prevent unauthorised identification of research participants. In addition, all primary data will be stored in the database devoid of any personal information or identifiers. All computers and laptops used to store the data will have password protection and encryption software in place.

6.15.2 Risk-Benefit Ratio

Although there are risks associated with this trial and interim analyses are planned, based on available evidence these risks are anticipated to be low with reasonable chance of benefit, as the intervention is to reduce sodium intake to that currently recommended by international guidelines. As such, there is a reasonable chance of benefit and the overall risk benefit ratio is favourable. The DSMB (section 6.14.8) will continue to monitor risks, and may recommend study discontinuation if the risk-benefit ratio changes significantly.

6.16 Limitations & Methods to Control Bias

First, this study is open-label as it is not possible to blind patients or the dietician. Partial blinding of study staff (data collection and entry), outcome assessment by the investigator and blinding of the statistician will be performed, as previously described.

Second, patients in the control group may receive recommendations on sodium reduction from non-trial sources, which may reduce the likelihood of seeing an effect from the intervention and bias toward the null. In addition, to minimise the risk of co-intervention and contamination (section 6.8), participants in different trial arms will attend separate clinic sessions, where possible, and participants and clinicians will be instructed not to discuss the education sessions with other participants or clinicians. Patient consent forms and information leaflets will not specifically describe sodium reduction, and will instead refer to general aspects of risk factor modification for progression of CKD.

Third, individuals who participate in studies are often more healthy than those who refuse to provide consent (volunteer bias) and the actual sample of individuals included in the trial (i.e. trial participants) may not be representative of the target population, limiting the generalizability of the results. A screening log will be compiled to determine if participants and non-participants differ significantly.

Fourth, only patients attending the nephrology clinic are included (referral bias). Primary care physicians refer the majority of their patients in whom they recognise CKD to the nephrology outpatient service, but there may be a significant proportion of unrecognised CKD in the community[152,204] (Chapter 5).

Fifth, the primary outcome measure is absolute reduction in creatinine clearance, which although clinically relevant, is a surrogate measure for progression to ESRD. Although it is likely that a small proportion of trial participants will progress to ESRD during the study, a much longer follow-up period would be required to demonstrate a treatment effect, even in a population with CKD. As shown in Chapter 4, over eight years of follow-up, only 0.2% of participants of a large, US prospective cohort study of older adults required dialysis. To date, no RCTs of sodium reduction with two-year follow-up report decline in creatinine clearance as the primary outcome and the findings of this trial can be used to inform larger, multi-centre trials.

Sixth, there is a risk of non-compliance with the intervention, a particular challenge in trials evaluating behavioural interventions. Multiple methods will be implemented to maximize adherence (section 6.9.2) and to determine the influence of adherence on outcomes a sensitivity analysis will be performed. Unlike clinical trials of medications, where pill counts are often used to measure adherence, this trial of a behavioural intervention will be assessed through patient self-report and 24hour urine collection for sodium (objective measure), during study conduct.

Seventh, to minimize within and between subject variability in the administration of the intervention, all specialists administering the intervention will be trained, certified and periodically observed. Finally, to maximize study quality, all components of the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT)[252] statement were considered during the drafting of this protocol (Table 6.7).

Table 6.7 Checklist for Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT)

Section	Checklist Item	Section
Administrative Information		
Title	Descriptive title identifying study design, population, interventions and trial acronym	Title Page
Trial Registration	Trial identifier and registry name	Table 6.1
	All items from WHO Trial Registration Data Set	
Protocol Version	Date and version identifier	
Funding	Sources and types of financial, material and other support	
Roles and Responsibilities	Names of protocol contributors	
	Name of trial sponsor	
	Role of study sponsor and funders, in any, in study design, collection, management, analysis, interpretation of data, writing	6.4
	Composition, roles and responsibilities of the coordinating centre, steering committee, adjudication, data management	6.17.2
Introduction		
Background and Rationale	Description of research question and justification for the trial	6.1-2
Objectives	Specific objectives or hypotheses	6.3
Trial Design	Description including type of trial, allocation ratio and framework	6.4
Methods: Participants, interventions and outcomes		
Study Setting	Study settings including countries where data will be collected	6.4.1
Eligibility criteria	Inclusions and exclusion criteria for participants	6.5
Interventions	Sufficient detail to allow replication	6.7
	Criteria for discontinuing or modifying allocated interventions	6.11
	Strategies to improve adherence to intervention protocols	6.9.2
	Relevant concomitant care and interventions	6.7
Outcomes	Primary and secondary outcomes including the specific measurement variable, analysis metric and time point	6.7
Participant Timeline	Time schedule of enrolment, interventions (including run-in), assessments and visits	6.9, Fig 6.2 Table 6.3
Sample Size	Estimated number, how determined and any assumptions	6.14.1
Recruitment	Strategies for achieving adequate enrolment	6.4.3
Methods: Assignment of Interventions (Allocation and Blinding)		
Sequence generation	Method of generating the allocation sequence and list of any factors for stratification	6.6.1
Concealment mechanism	Mechanism of implementing the allocation sequence, describing any steps to conceal until interventions are assigned	6.6.2
Implementation	Who will generate the allocation sequence, who will enroll and who will assign participants to interventions	6.6
Blinding	Who will be blinded and how	6.6.3

Table 6.7 (continued) Checklist for Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT)

Section	Checklist Item	Section
Methods: Data Collection, Management and Analysis		
Data Collection	Plans for assessment and collection of outcome, baseline and other data including processes to promote data quality	6.10, 6.17.4
	Plans to promote participant retention and complete follow-up	6.9
Data Management	Plans for data entry, coding, security and storage	6.17.4
Statistical Methods	Analysis of primary and secondary outcomes	6.14.3-4
	Methods for any additional analyses including subgroups	6.14.5
	Analysis population & statistical methods to handle missing data	6.14.7
Methods: Monitoring		
Data Monitoring	Data monitoring committee or processes	6.17.3
	Interim analyses and stopping guidelines	6.14.8
Harms	Plans for collecting, assessing, reporting and managing solicited and spontaneously reported adverse events	6.12.3
Auditing	Frequency and procedures for auditing trial conduct	6.17.3
Ethics and Dissemination		
Research ethics approval	Plans for seeking research ethics committee / institutional review board (REC/IRB) approval	6.15
Protocol Amendments	Plans for communicating important protocol modifications	6.19
Consent or Assent	Who will obtain informed consent/assent and how	6.7.1
Confidentiality	How personal information about potential and enrolled participants will be collected, shared and maintained in order to protect confidentiality before, during and after the trial	6.15.1
Declaration of Interests	Financial and other competing interests for PI	6.21
Access to data	Who will have access to the final trial dataset	6.20
Ancillary & Post-Trial Care	Provisions, if any, for ancillary and post-trial care, and for compensation to those who suffer harm from trial participation	6.22
Dissemination Policy	Plans to communicate trial results to participants, healthcare professionals, the public and other relevant groups	6.18
	Authorship eligibility guidelines	6.20
	Plans, for granting public access to the protocol, dataset & code	

6.17 Trial Administration

6.17.1 Site

This single-centre study will be conducted at the Nephrology Outpatient Clinic, GUH and the co-ordinating centre will be the HRB-CRFG. Completed CRFs and other paperwork relating to the trial, including the trial master file, will be held at the coordinating centre (HRB-CRFG).

6.17.2 Steering, Local Operations and Publication Committees

As this is a single site study, one committee will deal with steering, operations and publications and consist of Dr. Andrew Smyth and Prof. Martin O'Donnell (Co Principal Investigators), a Research Nutritionist/Dietician (to be appointed), Dr. Donal Reddan and Prof. Matthew Griffin (Consultant Nephrologists at GUH), Dr. Mahshid Dehghan (a Clinical Nutritionist), Dr. John Newell (Biostatistician) and Prof. Ivan Perry (Centre for Health and Wellbeing, University College Cork, who has considerable experience in developing and implementing dietary interventions). The steering committee will meet every three months to review progress of the clinical trial, identify problems and implement solutions. The local operations committee will meet every month to review progress, recruitment update, quality of data, etc. and will include the research nurse, research assistant, Prof O'Donnell and Dr Smyth. Specifically, in the first year the committee will closely review recruitment to ensure that all study participants are randomised by a specific date in order to complete two-years of follow-up by study end. In years two and three, the committee will focus on reviewing progress with patient follow-up and ensuring that the highest possible rate of follow-up is achieved. This process strives to minimise missing data.

6.17.3 Study Monitoring

Members of the steering committee will be responsible for the overall conduct and on-site monitoring of the study and for ensuring that all study procedures are compliant with ICH-GCP. Monitoring will be independent of the study funder and free from competing interests. Progress meetings will be held to monitor: (i) recruitment and adherence to procedures for informed consent; (ii) adherence to the protocol and any protocol amendments by reviewing a random subsample of patient consent forms and CRFs; (iv) adherence to the intervention and to follow-up visits; (v) quality of the data collected; (vi) verification that the data collected is valid and consistent on import into the study database. On-going training will occur to ensure that a good understanding of the protocol and of standardised operating procedures is maintained, to resolve problems and to promote staff commitment and enthusiasm for the study. As this is a single centre study on-site monitoring will be conducted. One internal audit by the Quality and Regulatory Affairs Manager from the HRB-CRFG will be performed during the conduct of the trial.

6.17.4 Data Collection & Quality Control

All data collection and study outcome measurements will be conducted locally and entered into eCRFs by the attending research nurse and transferred into the study database by trial staff. To minimise errors, two staff members will enter data independently and any discrepancies will be highlighted, with follow-up checks of the CRF to determine validity. Data to be extracted from CRFs will include: participant identification numbers, verification of eligibility criteria, verification of written informed consent, relevant participant demographic and clinical characteristics, current medications, physical parameters (e.g. BMI and BP), adherence with follow-up, patient reported adherence with the intervention and outcome events. Laboratory results will be electronically imported into the trial database. Trial staff entering data will be blinded to study group assignment and will remain blinded until all data has been collated, entered into the database and the full database has been blindly reviewed for errors. Blinding will help to avoid differential measurement bias or biased data entry.

To ensure high quality data is collected, staff will receive training on completing CRFs and extracting data from CRFs into the study database. Central statistical monitoring of the data using the following data entry edit checks will be employed, to improve the validity of the data entry process: (i) confirmation checks that participant identification numbers match dates of birth; (ii) plausible range checks (e.g. to require confirmation of values outside of the *a priori* specified range for variables such as systolic blood pressure, age, etc); (iii) checks for missing data; (iv) calendar checks (e.g. to ensure that follow-up visit occurs on a date after the randomization visit); (v) checks for unusual data patterns (digit preference, rounding or unexpected frequency distributions; (vi) checks of performance indicators such as the number of appointments per day and (vii) logic checks to ensure that values for data points don't contradict each other. Database integrity checks will be conducted weekly, to ensure data consistency.

6.17.5 Study Timeline

It is expected that 12 potentially eligible patients will be reviewed at the nephrology outpatient clinic per week, of which 50% are expected to participate in the trial. Therefore, a minimum of six patients per week will be recruited, with the recruitment phase completed in approximately 49 weeks. In addition to the main trial, patients will passively be followed up after completion of the final visit, where eGFR and ACR taken as part of standard clinical care will be recorded. No additional intervention or study visits will be required during this period.

6.18 Dissemination Strategy

It is anticipated that this research may demonstrate the benefits of a sodium reduction intervention on decline in renal function in patients with CKD. Study results will be presented locally, nationally and internationally by the study Co-PIs.

The primary study results will be submitted in manuscript form for publication to peer reviewed, internationally recognized, journals with a high impact factor. The focus will be on broad medical journals as well as those specific to CKD. The database created at HRB-CRFG will be available for further research studies.

6.19 Protocol Amendments

The investigator will not implement any deviation from or changes of the trial protocol without review and documented approval/favourable opinions from the REC, except to eliminate an immediate hazard to the trial participants. Changes will be recorded in writing, signed by the principal investigator and filed with the protocol. Approval from the REC must be received prior to implementation of changes.

6.20 Ownership of Data and Use of Study Results

The CRFG and Co-PIs have the ownership of all data and results collected during the trial and the Co-PIs have full rights to publication on data from this study, without restriction. For all publications and presentations from this work, authors will be required to meet make substantial contributions to conception or design, acquisition/analysis/interpretation of data, drafting or critically revising the manuscript and giving final approval of the version to be published, in line with the International Committee of Medical Journal Editors guidelines for authorship. The dataset will not be made publicly available; instead it will be maintained at HRB-CRFG and the Co-PIs must review and approve all requests for access to the data.

6.21 Declaration of Interests

The Co-PIs, collaborators and other protocol contributors have no financial or other competing interests to declare.

6.22 Ancillary & Post-Trial Care

Patients who meet the eligibility criteria and are included in this trial all have chronic kidney disease and will continue to receive standard care at the nephrology outpatient clinic at GUH on study completion.

Chapter 7: Overall Conclusions

Chronic kidney disease, including end-stage renal disease, is increasingly prevalent and associated with significant premature mortality and morbidity, including hospitalisation, cardiovascular disease and functional impairment. The prevention of CKD, as well as reducing the progression of CKD, is essential to improving health outcomes. Dietary intake is a physiological requirement, and its modification has previously been shown to modify the risk of mortality and cardiovascular disease. As cardiovascular disease and chronic kidney disease share many comment risk factors, and are inter-related, it is imperative that low-cost, simple, population-wide interventions are found to reduce the burden of CKD. In this thesis, I examined the association between dietary factors (including diet quality, dietary sodium intake and dietary potassium intake) and renal outcomes (including surrogate outcomes such as eGFR and proteinuria and clinical endpoints such as need for dialysis and death from renal cause). The data from these observational studies were used to develop a protocol for a randomised controlled trial testing the hypothesis that reduced sodium intake improves renal outcomes.

Chapter 2

I completed a systematic review of the literature exploring the association between dietary sodium intake and renal outcomes. For this chapter, I contributed actively to all components including: development of the research question, design of the electronic search strategy, electronic searching, abstract, title and full text review, data extraction, collation of results and writing of the manuscript. This systematic review, using standardised methodology, identified only seven eligible studies (cohort studies and clinical trials). Due to differences in study design, measurement of sodium intake and reported outcomes, a decision was made *a priori* not to perform meta-analysis or generate quantitative summary estimates. Despite differences in study design, size and duration of follow-up, a consistent association between high-sodium intake (>4.6g/day) and adverse renal outcomes was observed, but importantly, no difference between low (<2.3g/day, as recommended by guidelines (Figure 2.2)) and moderate intake (2.3-4.6g/day) was observed. The findings were most consistent when restricted to four studies that included only patients with chronic kidney disease. These results are similar to recent reports of the association between sodium intake and cardiovascular disease, and suggest that reducing sodium intake from high to low or moderate intake is associated with benefit. The findings echo a recent statement by the US Institute of Medicine[39]. Despite reductions in blood pressure, the existing literature does not support guidelines for patients with chronic kidney disease to restrict sodium intake to low levels. However, the evidence base is insufficient to make firm guideline recommendations highlighting the need for further studies in this area, including those discussed in Chapters 3, 4 and 6.

Chapter 3

Based on the findings of the systematic review (Chapter 2), I completed analyses of a prospective cohort study (comprised of the participants from two clinical trials, ONTARGET and TRANSCEND), exploring the association between sodium intake and renal outcomes. As sodium and potassium intakes are inter-related, this study also included analyses of the association between potassium intake and renal outcomes. In addition, as previous studies report high potassium intake is associated with reduced blood pressure and cardiovascular disease, the hypothesis was that high potassium intake is associated with improved renal outcomes, whilst acknowledging a potential adverse effect of hyperkalemia, particularly in those with reduced eGFR. For this chapter, I contributed actively to all components including: development of the research question, design of the statistical analysis plan, oversaw all statistical analyses (completed with statisticians Daniela Dunkler and Peggy Gao), collation and interpretation of statistical analyses and writing of the manuscript. In this large prospective cohort study, of patients with diabetes or at high cardiovascular risk, but with eGFR>30 and low rates of proteinuria, no association between sodium intake and multiple surrogate renal outcomes were observed, with trends towards increased odds in those with high intake. As such, this data does not support current recommendations to restrict sodium intake to low levels. For potassium intake, high intake was associated with reduced odds of renal outcomes, but there was a trend towards increased risk of hyperkalemia. Importantly, there was a loss of statistical significance in the association towards reduced odds of renal outcomes in those with lower eGFR or macroalbuminuria, suggesting that although high potassium intake is associated with reduced outcomes, this effect may be lost in those with significant renal disease. Further studies are required to explore the benefits and risks of increasing potassium intake. It is likely that the benefits may be confined to those without significant renal disease, as those patients have increased risks of hyperkalemia.

Chapter 4

To determine if the observed associations in Chapter 3 were restricted to a high cardiovascular risk population or restricted to surrogate outcomes, I explored the association between sodium and potassium intake, as well as diet quality, and clinical outcomes (dialysis or death from a renal cause) in a different population. Amongst the challenges of the ONTARGET/TRANSCEND cohort is that all participants completed active run-in (i.e. didn't experience significant decline in eGFR or develop hyperkalemia with the introduction of RAAS) and may be different to the general population and all participants were at high cardiovascular risk but relatively low renal risk. For this chapter, I contributed actively to all components including: development of the research question, study design, research proposal submission to the NIH for approval, data retrieval, design of the statistical analysis plan, completion of all statistical analyses (with oversight from biostatistician Dr John Newell), collation and interpretation of results and writing of the manuscript. These analyses of the NIH Diet and Health Study provide unique and novel insight into the association between diet quality and renal outcomes in a large cohort that completed a detailed dietary assessment, with a sufficient sample size and duration of follow-up to capture a sufficient number of clinical outcomes to complete robust analyses. In these analyses, higher sodium intake was associated with an increased risk of renal outcomes, while higher potassium intake was associated with a reduced risk. As foods and nutrients are not consumed in isolation, in this study I also explored the association between a healthy diet pattern using multiple scoring methods to characterize the diet. Higher diet quality, based on an extensive dietary assessment (such as the Alternate Healthy Eating Index, Healthy Eating Index and Mediterranean Diet Score), was associated with a reduced risk of clinical outcomes. However, dietary assessments based on the intake of desirable foods alone (including fruits and vegetables, such as those captured by the Recommended Food Score) without considering other aspects of the diet, was not associated with any reduction in risk of renal outcomes.

Chapter 5

Taken together, chapters 2, 3 and 4 suggest that a higher quality diet is associated with reduced risk of renal outcomes, although there continues to be an inconsistent association with dietary sodium. In addition, most clinical studies in nephrology focus on surrogate outcomes, rather than patient-important outcomes including dialysis, and death. In chapter 5, I explored the association between chronic kidney disease and functional impairment, another patient important outcome. For this chapter, I contributed actively to all components including: development of the research question, design of the statistical analysis plan, completion of all statistical analyses, collation and interpretation of results and writing of the manuscript. In this cross-sectional study of adults attending general practices in the West of Ireland, an association between chronic kidney disease and impairment in instrumental and basic activities of daily living was observed. These outcomes are particularly important to older adults, who value highly the preservation of functional impairment as part of 'successful ageing'. After adjustment for multiple confounders, increased odds of any functional impairment, impairment in IADL and impairment in BADL was observed in those with chronic kidney disease, as defined by eGFR<60ml/min/1.73m^2, with no significant differences by subgroups of age, gender and cardiovascular disease.

Chapter 6

Based on the previously presented work, I developed a protocol for a randomised controlled trial to test the hypothesis that reducing sodium intake, to levels recommended by current guidelines, improves renal outcomes. As no reported clinical trial has tested this hypothesis with follow-up greater than six months at the time of drafting this protocol and completion of this PhD thesis, the protocol was developed as a feasibility study planned for two-year follow-up. For this chapter, I contributed actively to all components including: development of the research question, study design, methodological approaches and drafting the protocol. I co-lead a successful funding application (with my PhD Supervisor, Professor Martin O'Donnell) to the Health Research Board of Ireland (HRA-POR-2014-686). I am the lead investigator for the STICK trial. As there will likely be an insufficient number of clinical endpoints such as diagnosis of end stage renal disease (requiring dialysis, transplantation or plan for conservative management) during this feasibility trial, the focus is on surrogate outcomes. However, to incorporate patient-important outcomes (including functional impairment, as demonstrated in Chapter 5), patients will complete assessments of functional status on study entry and at study end. It is anticipated that the findings of this trial will lead to a larger definitive Phase III trial and a longer follow-up period that will be powered to detect differences in clinical endpoints to definitively answer this research question.

Future Directions

In conclusion, dietary modification offers the potential to significantly reduce the burden from chronic kidney disease, including end-stage renal disease. There continues to be significant inconsistency in the association between sodium intake and renal outcomes, highlighted in this PhD thesis (Chapters 2, 3 and 4). In particular, as the results of the NIH Diet and Healthy Study (Chapter 4) suggests that the absolute level of sodium intake associated with decreased risk of clinical outcomes is higher than current guideline recommendations, this further contributes to the clinical equipoise. As such, it is anticipated that the randomised controlled trial (Chapter 6), will provide clarity to this clinical research question. Despite the inconsistencies for sodium intake, a consistent potential benefit from higher potassium intake on renal outcomes was observed in both prospective cohort studies. However, interventions to increase potassium intake ultimately require testing in randomised controlled trials, due to the potential risk of adverse events such as hyperkalemia. Few previous trials have explored this research question, but the methodologies that will be developed and refined during the conduct of the STICK trial (Chapter 6) could be used for other lifestyle interventions, such as increased potassium intake. Finally, the analyses of the association between higher diet quality and clinical renal endpoints is a novel finding and extends the previously reported benefits of healthy eating on cardiovascular outcomes. Dietary modification represents a low-cost, simple intervention that offers the potential to significantly reduce the global burden of chronic kidney disease, but further studies are required.

Appendices

Appendix 1: Outputs arising from this work

Presentations:

1. Smyth A, Dunkler D, Gao P, Teo KK, Yusuf S, O'Donnell M, Mann JFE, Clase CM. Relationship between estimated sodium and potassium excretion and subsequent renal outcomes. Poster presentations at (i) Canadian Society of Nephrology Meeting, Vancouver, April 2014 and (ii) American Society of Nephrology Meeting, Atlanta, November 2013 (Chapter 3)

2. Smyth A, Griffin M, Yusuf S, Mann JFE, Reddan DN, Canavan M, Newell J, O'Donnell M. Diet and major renal outcomes: the NIH-AARP Diet and Health study. Oral presentation at American Society of Nephrology Meeting, Philadelphia, November 2014 (Chapter 4)

3. Smyth A, Canavan M, Glynn LG, O'Donnell M, Reddan DN. Unrecognised chronic kidney disease: who is at risk? Poster presentation at American Society of Nephrology, San Diego, November 2012 (Chapter 5)

Publications

1. Smyth A, O'Donnell M, Yusuf S, Clase CM, Teo KK, Canavan M, Reddan DN, Mann JFE. Sodium intake & renal outcomes: a systematic review. American Journal of Hypertension. 2014; 27(10): 1277-84. PMID 24510182. (Chapter 2)

2. Smyth A, Dunkler D, Gao P, Teo KK, Yusuf S, O'Donnell M, Mann JFE, Clase CM. The relationship between estimated sodium and potassium excretion and subsequent renal outcomes. Kidney International 2014; 86(6): 1205-12. PMID 24918156. (Chapter 3)

3. Smyth A, Glynn LG, Murphy AM, Mulqueen J, Canavan M, Reddan DN, O'Donnell M. Mild chronic kidney disease and functional impairment in community-dwelling older adults. Age and Ageing. 2013; 42(4): 488-94. PMID 23438445. (Chapter 5)

Successful Funding Applications

1. O'Donnell M, Smyth A, Murphy AM, Griffin M, Newell J, Reddan D. Sodium intake in chronic kidney disease (STICK): a randomised controlled trial. Health Research Board Health Research Awards Scheme 2014. (Chapter 6)

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Title: The relationship between estimated sodium and potassium excretion and subsequent renal outcomes

Author: Andrew Smyth, Daniela Dunkler, Peggy Gao, Koon K Teo, Salim Yusuf et al.

Publication: Kidney International

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Appendix 5: CLARITY Questionnaire

OÉ Gaillimh
NUI Galway

The Clarity Study
Patient Questionnaire

This questionnaire will take about 5 minutes to complete. Please answer all questions if possible.

If you have difficulty reading or understanding the questions, please ask a family member, friend or carer to assist you.

The information that you provide is confidential and will only be viewed by the researcher. You don't have to write your name on the questionnaire.

When you have completed the questionnaire, please return it in the stamped, addressed envelope.

Thank you very much for your help.

Appendix 5 (continued): CLARITY Questionnaire

SECTION A: YOUR OWN HEALTH STATE TODAY

By placing a tick (✓) in one box in each group below, please indicate which statements best describe your own health **today**.

A1. Mobility	Please tick (✓) one answer
I have no problems in walking about	❑
I have some problems in walking about	❑
I am confined to bed	❑

A2. Self-Care	Please tick (✓) one answer
I have no problems with self-care	❑
I have some problems washing or dressing myself	❑
I am unable to wash or dress myself	❑

A3. Usual Activities (*e.g. work, study, housework, family or leisure activities*)

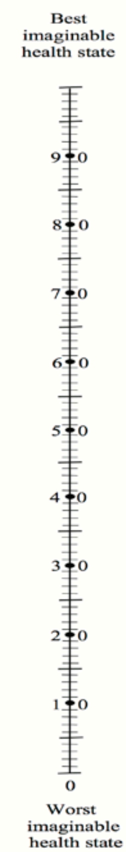
	Please tick (✓) one answer
I have no problems with performing my usual activities	❑
I have some problems with performing my usual activities	❑
I am unable to perform my usual activities	❑

A4. Pain/Discomfort	Please tick (✓) one answer
I have no pain or discomfort	❑
I have moderate pain or discomfort	❑
I have extreme pain or discomfort	❑

A5. Anxiety/Depression	Please tick (✓) one answer
I am not anxious or depressed	❑
I am moderately anxious or depressed	❑
I am extremely anxious or depressed	❑

To help people say how good or bad a health state is, we have drawn a scale (rather like a thermometer) on which the best state you can imagine is marked 100 and the worst state you can imagine is marked 0.

We would like you to **place an X on this scale to show how good or bad your own health is today**, in your opinion.



Appendix 5 (continued): CLARITY Questionnaire

SECTION B: WELL-BEING

Choose one response from the four given below to describe your current feelings. Please give your immediate response and do not think too long about your answers.

B1. I still enjoy the things I used to enjoy (Please tick (✓) one answer)
- Definitely as much ☐
- Not quite so much ☐
- Only a little ☐
- Hardly at all ☐

B2. I can laugh and see the funny side of things (Please tick (✓) one answer)
- As much as I always could ☐
- Not quite so much now ☐
- Definitely not so much now ☐
- Not at all ☐

B3. I feel cheerful (Please tick (✓) one answer)
- Never ☐
- Not often ☐
- Sometimes ☐
- Most of the time ☐

B4. I feel as if I am slowed down (Please tick (✓) one answer)
- Nearly all the time ☐
- Very often ☐
- Sometimes ☐
- Not at all ☐

B5. I have lost interest in my appearance (Please tick (✓) one answer)
- Definitely ☐
- I don't take quite as much care as I should ☐
- I may not take quite as much care ☐
- I take as much care as ever ☐

B6. I look forward with enjoyment to things (Please tick (✓) one answer)
- As much as I ever did ☐
- Rather less than I used to ☐
- Definitely less than I used to ☐
- Hardly at all ☐

B7. I can enjoy a good book or radio or television programme
(Please tick (✓) one answer)
- Often ☐
- Sometimes ☐
- Not Often ☐
- Very seldom ☐

Appendix 5 (continued): CLARITY Questionnaire

SECTION C: YOUR VIEWS ABOUT MANAGING ANY ILLNESS IN YOUR LIFE

We would like to know **how confident you are** in doing certain activities when you are feeling unwell. For each of the following questions, please **circle the number** that corresponds to your confidence level

For example in question D1, if you want to say that you are totally confident that you can keep the tiredness caused by illness from interfering with the things you want to do, you would circle the number 10.

C1. How confident are you that you can keep any tiredness caused by illness from interfering with things you want to do? (**Please circle one number only**)

Not at all confident	1	2	3	4	5	6	7	8	9	10	**Totally confident**

C2. How confident are you that you can keep any physical discomfort or pain caused by illness from interfering with the things you want to do? (**Please circle one number only**)

Not at all confident	1	2	3	4	5	6	7	8	9	10	**Totally confident**

C3. How confident are you that you can keep any emotional distress caused by illness from interfering with the things you want to do? (**Please circle one number only**)

Not at all confident	1	2	3	4	5	6	7	8	9	10	**Totally confident**

C4. How confident are you that you can keep any other symptoms or health problems you have from interfering with the things you want to do? (**Please circle one number only**)

Not at all confident	1	2	3	4	5	6	7	8	9	10	**Totally confident**

C5. How confident are you that you can do the different things needed to manage your health so as to reduce your need to see a doctor? (**Please circle one number only**)

Not at all confident	1	2	3	4	5	6	7	8	9	10	**Totally confident**

C6. How confident are you that you can do things other than just taking medication to reduce how much illness affects your everyday life? (**Please circle one number only**)

Not at all confident	1	2	3	4	5	6	7	8	9	10	**Totally confident**

Appendix 5 (continued): CLARITY Questionnaire

SECTION D: GENERAL INFORMATION

This part of the survey asks for your views about your general health.
If you are unsure about how to answer a question, please give the best answer you can.

D1. Which best describes you? (Please tick (✓) one box)

- I smoke every day ☐
- I smoke occasionally (less than once a day) ☐
- I used to smoke but I quit ☐
- I do not smoke but am exposed to other people's smoke on most days ☐
- I have never smoked ☐

D2. What age were you when you left formal education? [______] years

D3. What is your present marital status? (Please tick (✓) one box)

- Single/Never married ☐
- Married/Living with partner ☐
- Separated/Divorced ☐
- Widowed ☐

D4. What is your present employment status? (Please tick (✓) one box)

- Employed ☐
- Working in the home ☐
- Unemployed ☐
- Retired ☐
- Receiving disability allowance ☐

D5a. Which best describes your use of alcohol? (Please tick (✓) one box)

- Never ☐
- Former (used to drink but stopped) ☐
- Currently use alcohol ☐

D5b. At least once per month, I drink 5 or more drinks in a single day
(Please answer Yes or No) Yes ☐ No ☐

D6. Social support (Please answer Yes or No to each question)

- Do you live alone? Yes ☐ No ☐
- Do you drive a car? Yes ☐ No ☐

D7. Do you require help from another person for any of the following?
(Please answer Yes or No to each question)

- Shopping? Yes ☐ No ☐
- Laundry or preparing a meal? Yes ☐ No ☐
- Walking? Yes ☐ No ☐
- Dressing? Yes ☐ No ☐

D8. Do you use any of the following every day?
(Please answer Yes or No to each question)

- Walking stick ? Yes ☐ No ☐
- Walking frame? Yes ☐ No ☐
- Wheelchair? Yes ☐ No ☐

D9. **Have you had a fall in the last 12 months?** Yes ☐ No ☐
If yes did this result in overnight stay in hospital? Yes ☐ No ☐

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